

Chapter 4: Amino acids, Peptides and proteins

Contents:

- **Amino acids**
 - Properties
 - Structures of common amino acids
 - Chirality
 - Essential vs. non-essential
 - Isoelectric point
 - Titration curve
 - Synthesis
 - Reactions
- **Peptides**
 - Structure
 - Synthesis
 - Sequencing
- **Proteins**
 - Structure and properties

Why study amino acids, peptides and proteins?

- They are important primary metabolites
- Amino acids are the fundamental building blocks of proteins
- Many are pharmacologically active compounds
 - antibiotics
 - antitumor
 - allergens
 - anti-inflammatory agents
- Important components in the discussions of drugs (distribution, action, elimination of drugs)

Amino Acids, Peptides, and Proteins

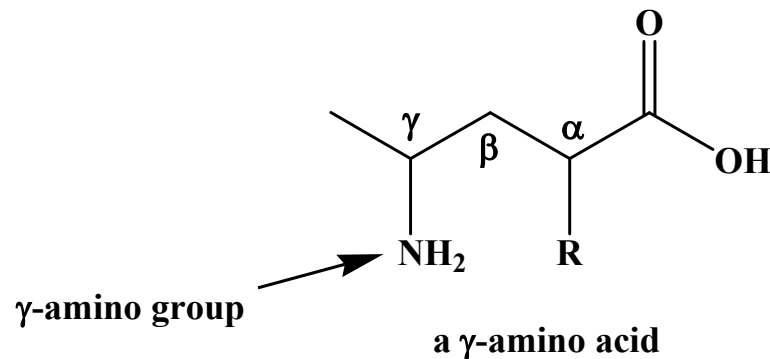
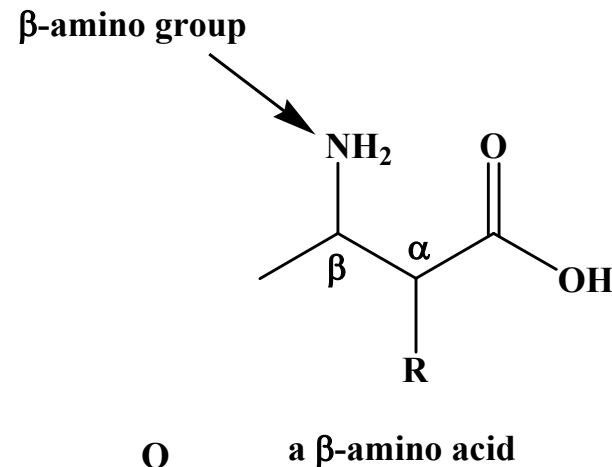
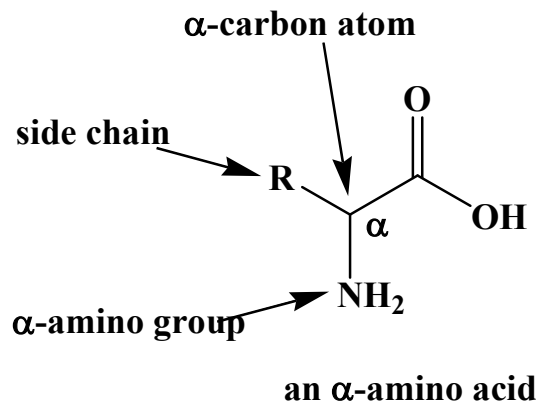
Amino acids: contain a basic amino group and an acidic carboxylic group

Peptides: are joined as amides between the -NH_2 of one amino acid and the $\text{-CO}_2\text{H}$ to the next amino acid (≤ 50 units)

Protein: large chains that have structural or catalytic functions in biology

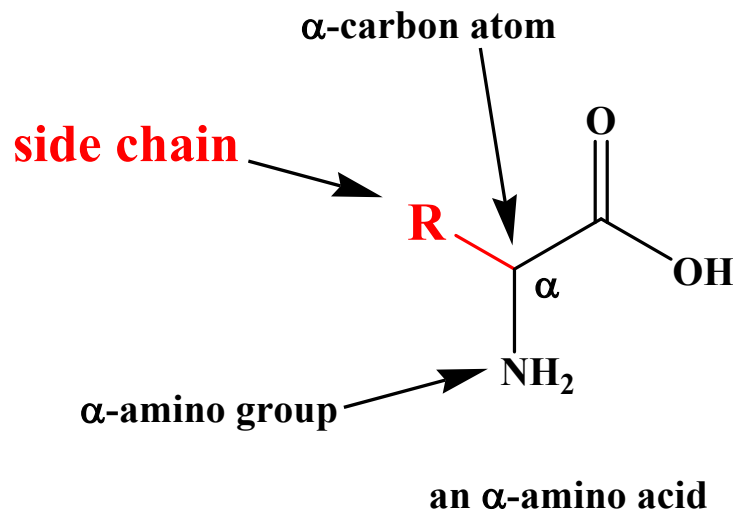
- **Amino acids**

- Derivatives of carboxylic acids with amine function
- Relative position of the amino group with respect to the carboxyl, :



- **The side chain (R):**

- Responsible for **structural and functional** diversity of proteins
- Determines the physicochemical properties amino acids and ultimately their polymers

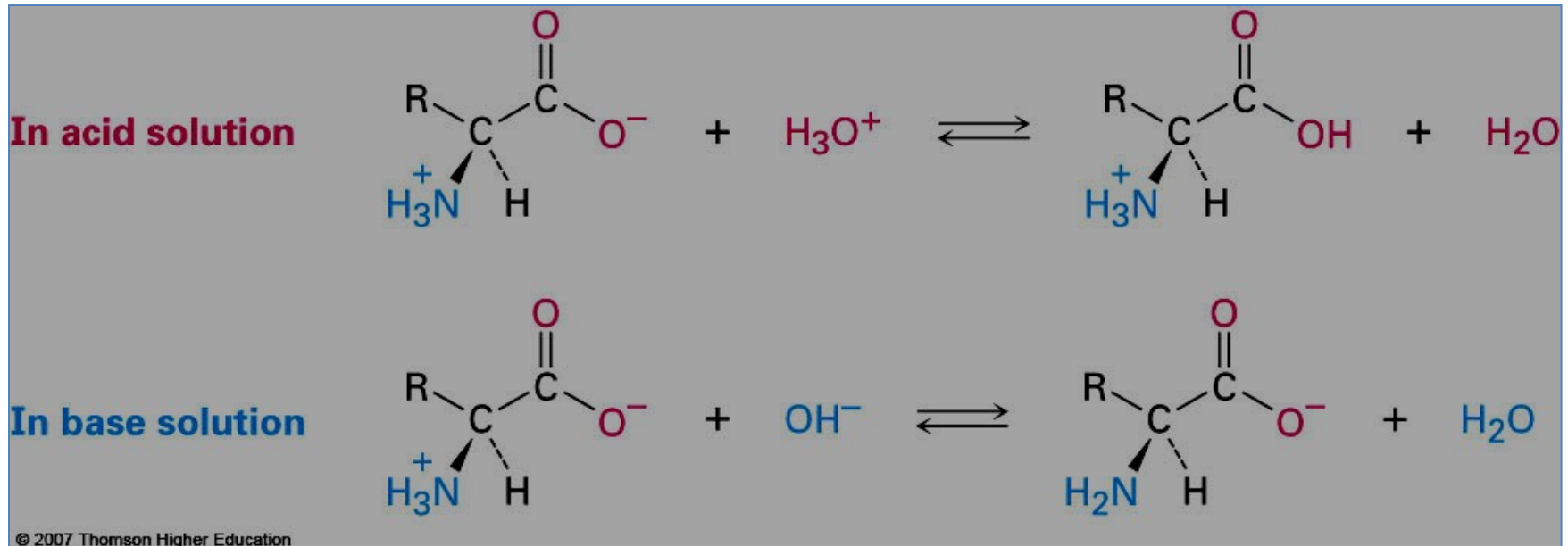


3.1 Amino Acids

In neutral solution, the COOH is ionized and the NH₂ is protonated

The resulting structures have “+” and “-” charges (a *dipolar ion*, or *zwitterion*)

They are like ionic salts in solution



I. Properties of Amino Acids

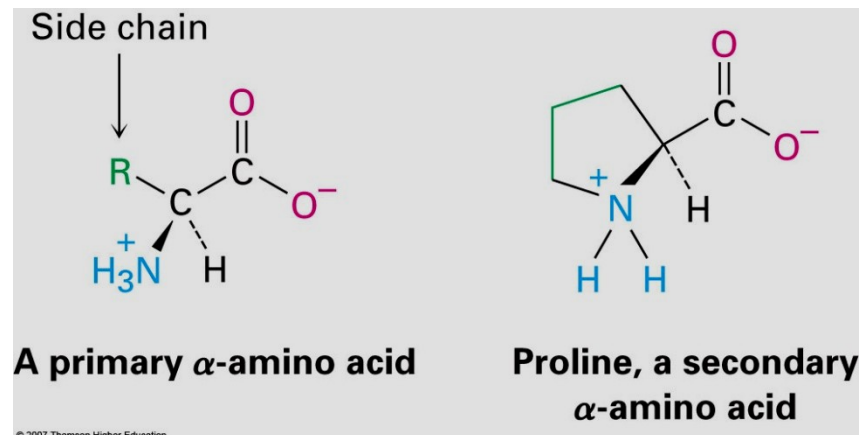
- High melting points, over 200°C, colorless crystalline substances.
- More soluble in water than in ether/other solvents.
- Less acidic than most carboxylic acids, less basic than most amines.
- High dielectric constant

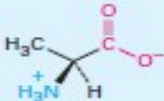
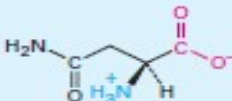
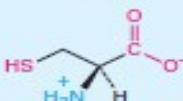
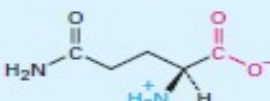
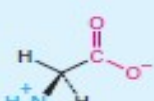
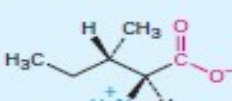
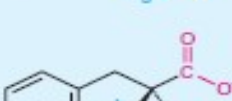
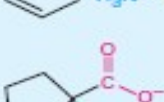
II. The Common Amino Acids

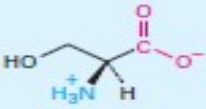
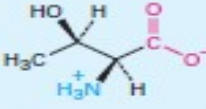
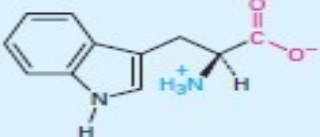
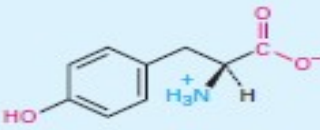
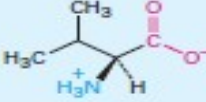
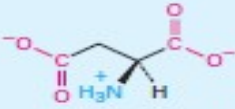
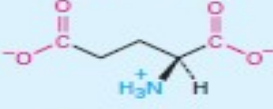
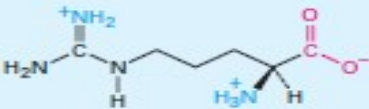
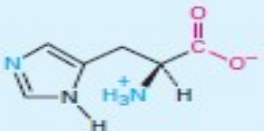
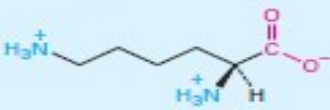
20 amino acids form amides in proteins

All are α -amino acids - the amino and carboxyl are connected to the same C

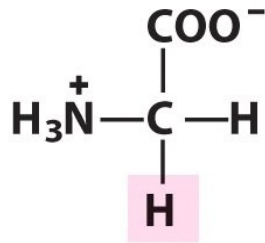
They differ by the other substituent attached to the α carbon, called the side chain, with H as the fourth substituent



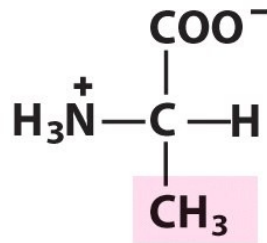
Name	Abbreviations	MW	Structure	pK _a α-CO ₂ H	pK _a α-NH ₃ ⁺	pK _a side chain	pI	
Neutral Amino Acids								
Alanine	Ala	A	89		2.34	9.69	—	6.01
Asparagine	Asn	N	132		2.02	8.80	—	5.41
Cysteine	Cys	C	121		1.96	10.28	8.18	5.07
Glutamine	Gln	Q	146		2.17	9.13	—	5.65
Glycine	Gly	G	75		2.34	9.60	—	5.97
Isoleucine	Ile	I	131		2.36	9.60	—	6.02
Leucine	Leu	L	131		2.36	9.60	—	5.98
Methionine	Met	M	149		2.28	9.21	—	5.74
Phenylalanine	Phe	F	165		1.83	9.13	—	5.48
Proline	Pro	P	115		1.99	10.60	—	6.30

Name	Abbreviations		MW	Structure	pK _a α-CO ₂ H	pK _a α-NH ₃ ⁺	pK _a side chain	pI
Neutral Amino Acids (continued)								
Serine	Ser	S	105		2.21	9.15	—	5.68
Threonine	Thr	T	119		2.09	9.10	—	5.60
Tryptophan	Trp	W	204		2.83	9.39	—	5.89
Tyrosine	Tyr	Y	181		2.20	9.11	10.07	5.66
Valine	Val	V	117		2.32	9.62	—	5.96
Acidic Amino Acids								
Aspartic acid	Asp	D	133		1.88	9.60	3.65	2.77
Glutamic acid	Glu	E	147		2.19	9.67	4.25	3.22
Basic Amino Acids								
Arginine	Arg	R	174		2.17	9.04	12.48	10.76
Histidine	His	H	155		1.82	9.17	6.00	7.59
Lysine	Lys	K	146		2.18	8.95	10.53	9.74

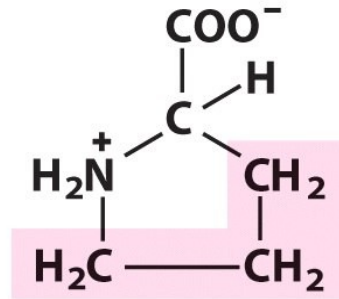
Nonpolar, aliphatic R groups



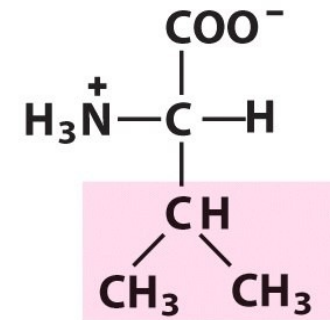
Glycine



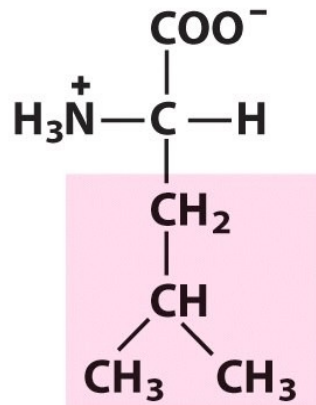
Alanine



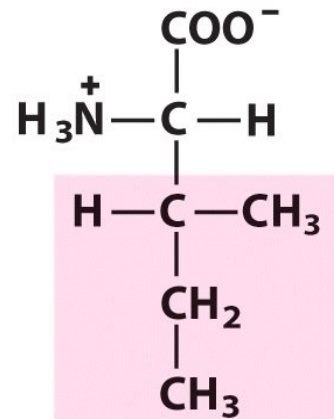
Proline



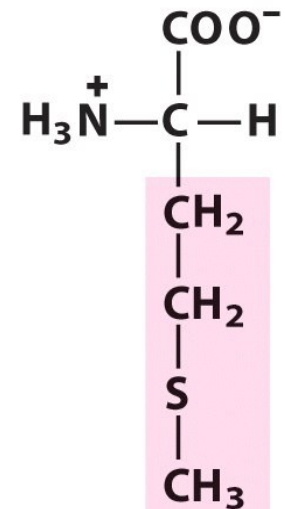
Valine



Leucine

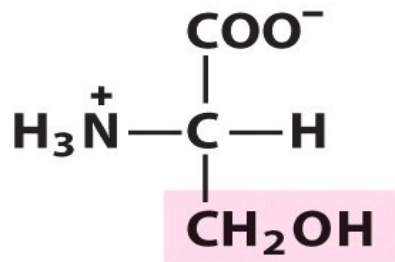


Isoleucine

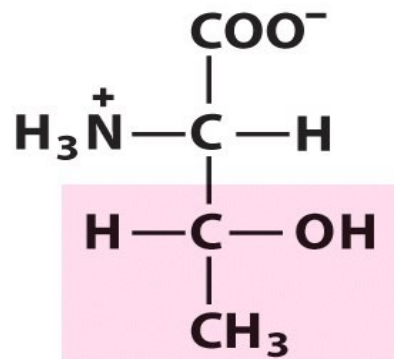


Methionine

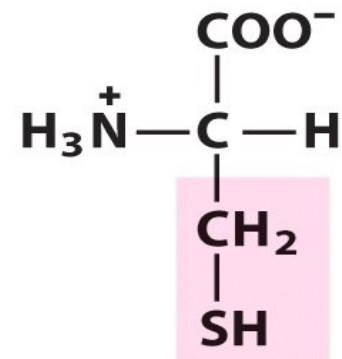
Polar, uncharged R groups



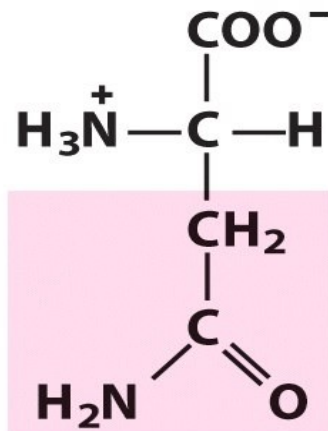
Serine



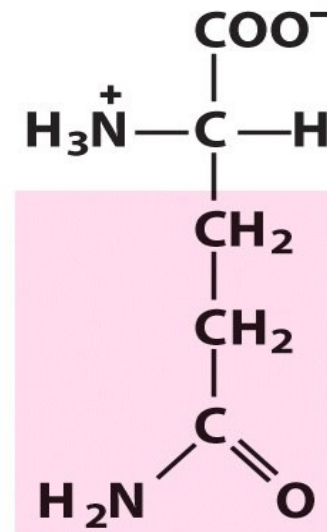
Threonine



Cysteine

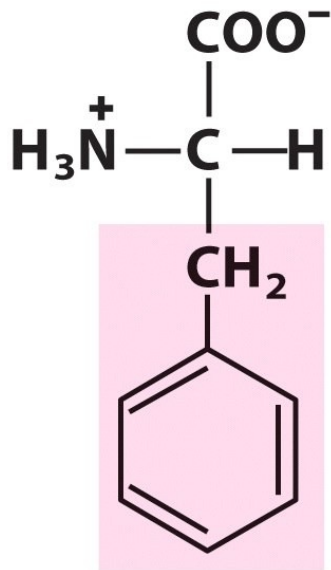


Asparagine

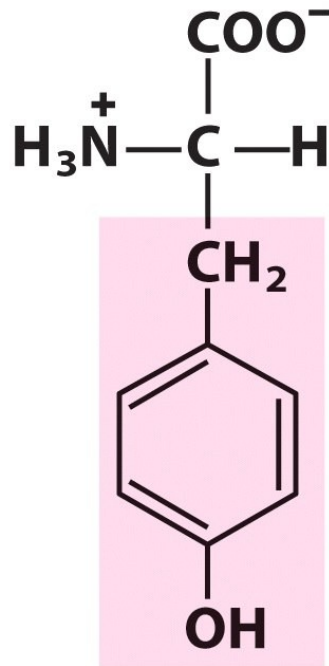


Glutamine

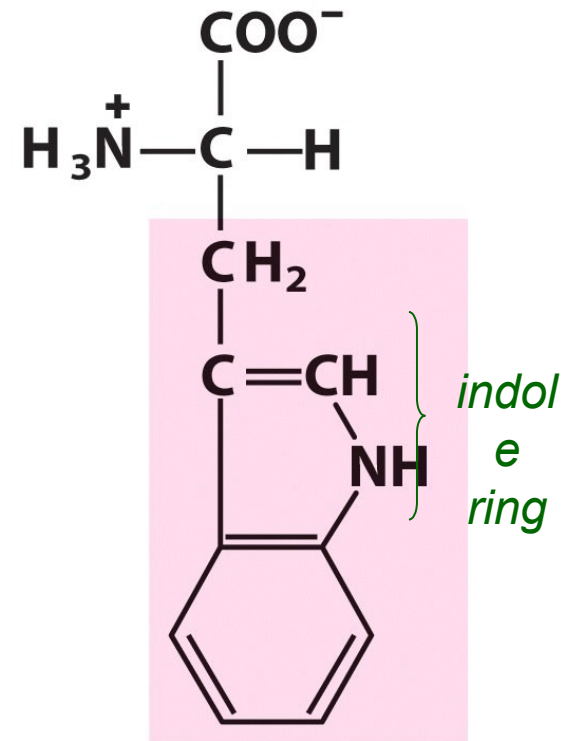
Aromatic R groups



Phenylalanine

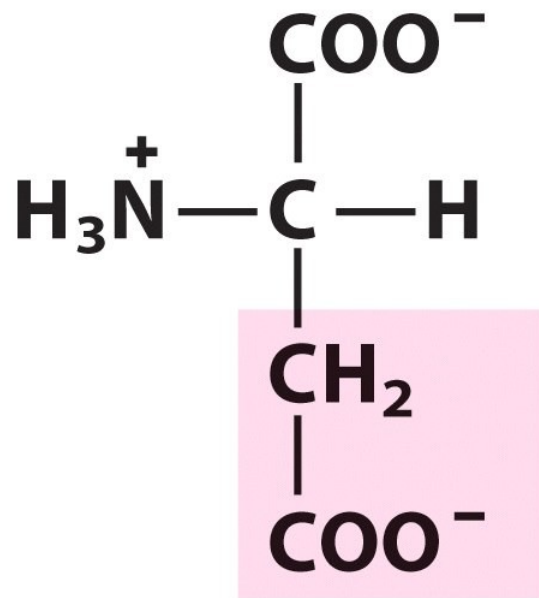


Tyrosine

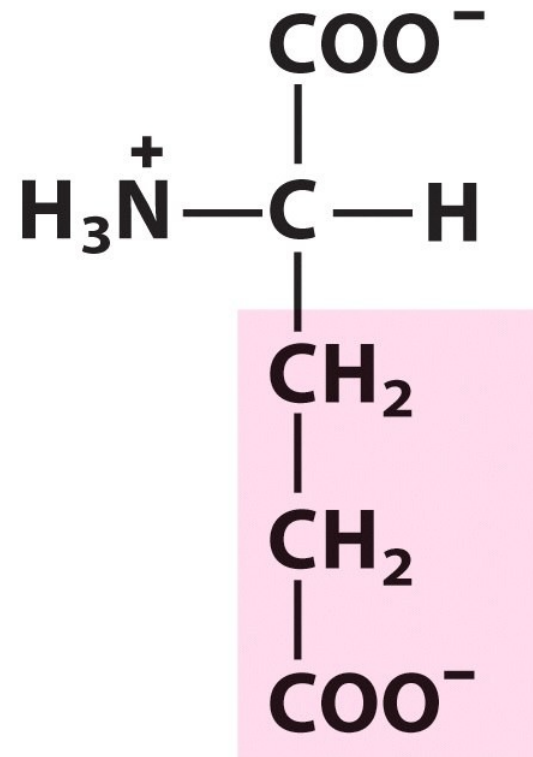


Tryptophan

Negatively charged R groups

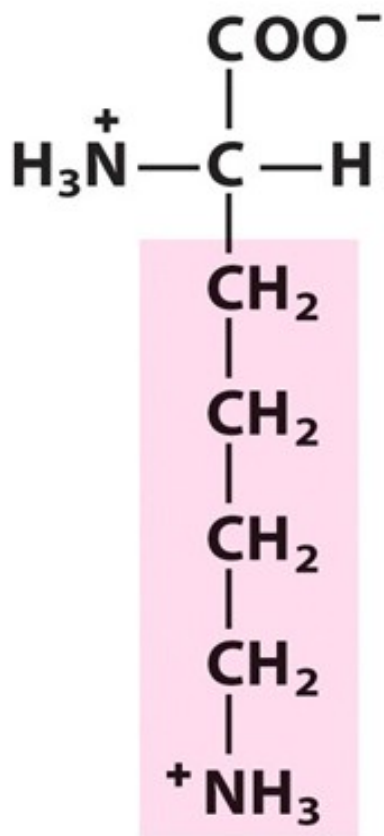


Aspartate

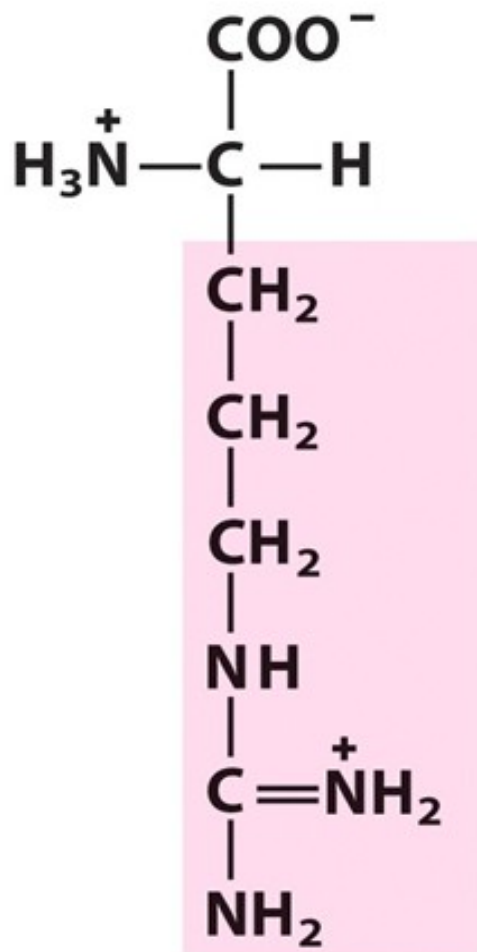


Glutamate

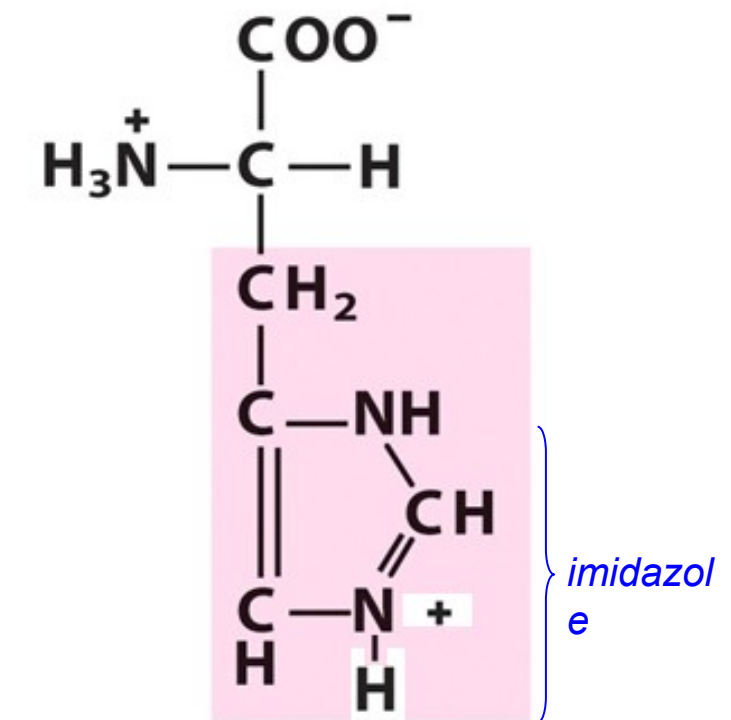
Positively Charged R groups



Lysine



Arginine



Histidine

Types of side chains

Neutral: Fifteen of the twenty have neutral side chains
Asp (Aspartate) and Glu (Glutamate) have a second COOH and are acidic

Lys (Lysine), Arg (Arginine), His (Histidine) have additional basic amino groups side chains (the N in tryptophan is a very weak base)

Cys (Cysteine), Ser (Serine), Tyr (Threonine) (OH and SH) are weak acids that are good nucleophiles

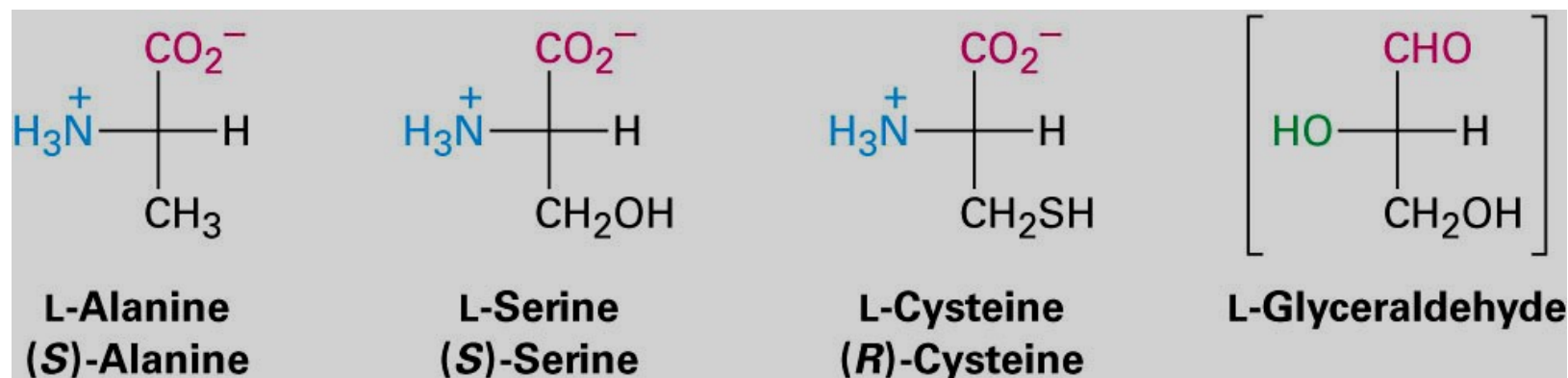
III. Chirality of Amino Acids

Glycine, 2-amino-acetic acid, is achiral

In all the others, the α carbons of the amino acids are centers of chirality

The stereochemical reference for amino acids is the Fischer projection of L-serine

Proteins are derived exclusively from L-amino acids



IV. Essential Amino Acids

All 20 of the amino acids are necessary for protein synthesis

Humans can synthesize only 10 of the 20

The other 10 must be obtained from food (Essential amino acid, listed below)

- Arginine (Arg)
- Threonine (Thr)
- Lysine (Lys)
- Valine (Val)
- Phenylalanine (Phe)
- Tryptophan (Trp)
- Methionine (Met)
- Histidine (His)
- Leucine (Leu)
- Isoleucine (Ile)

V. Isoelectric Points

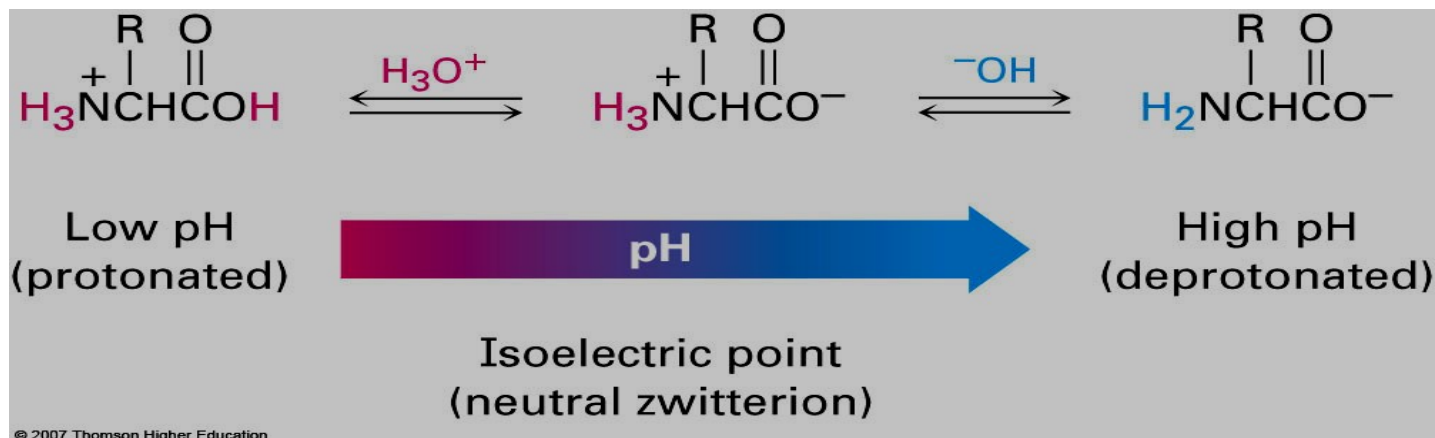
In acidic solution, the carboxylate and amine are in their conjugate acid forms, an overall cation

In basic solution, the groups are in their base forms, an overall anion

In neutral solution cation and anion forms are present

□ The isoelectric point (pI)

“is the pH at which the molecule has an average net charge of zero and therefore does not migrate in an electric field”

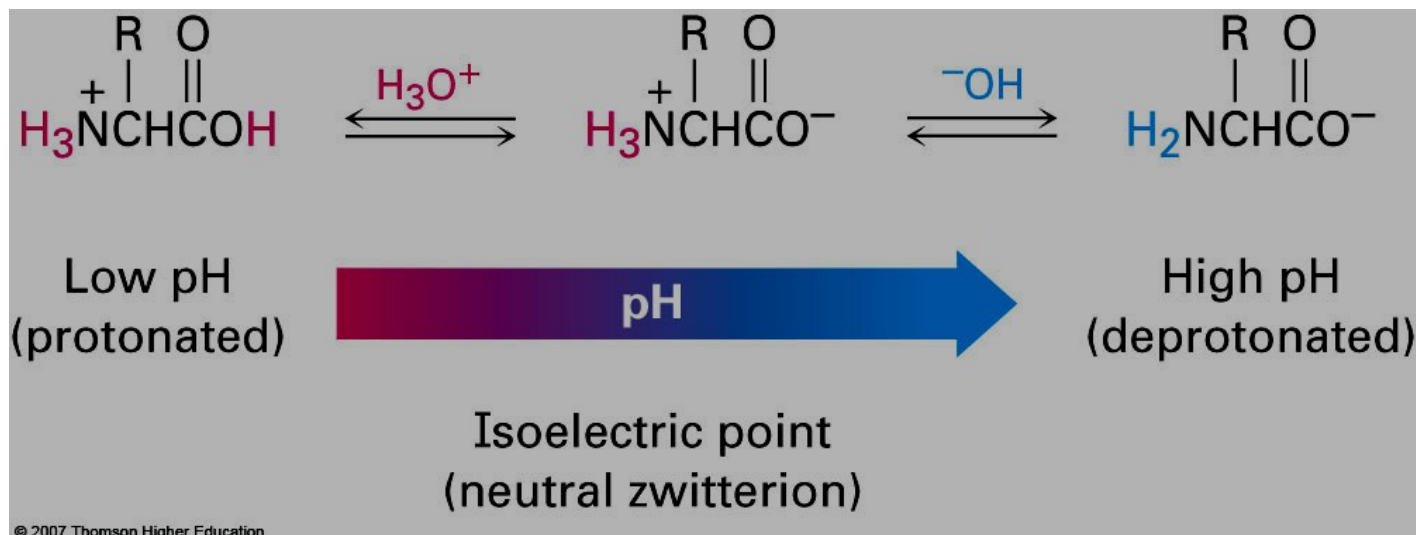


pI Depends on Side Chain

Acidic amino acids, isoelectric pH ~3.

Basic amino acids, isoelectric pH ~9.

Neutral amino acids, isoelectric pH is slightly acidic, 5-6.



VI. The Henderson Hasselbalch Equation

If pK_a values for an amino acid are known the fractions of each protonation state can be calculated

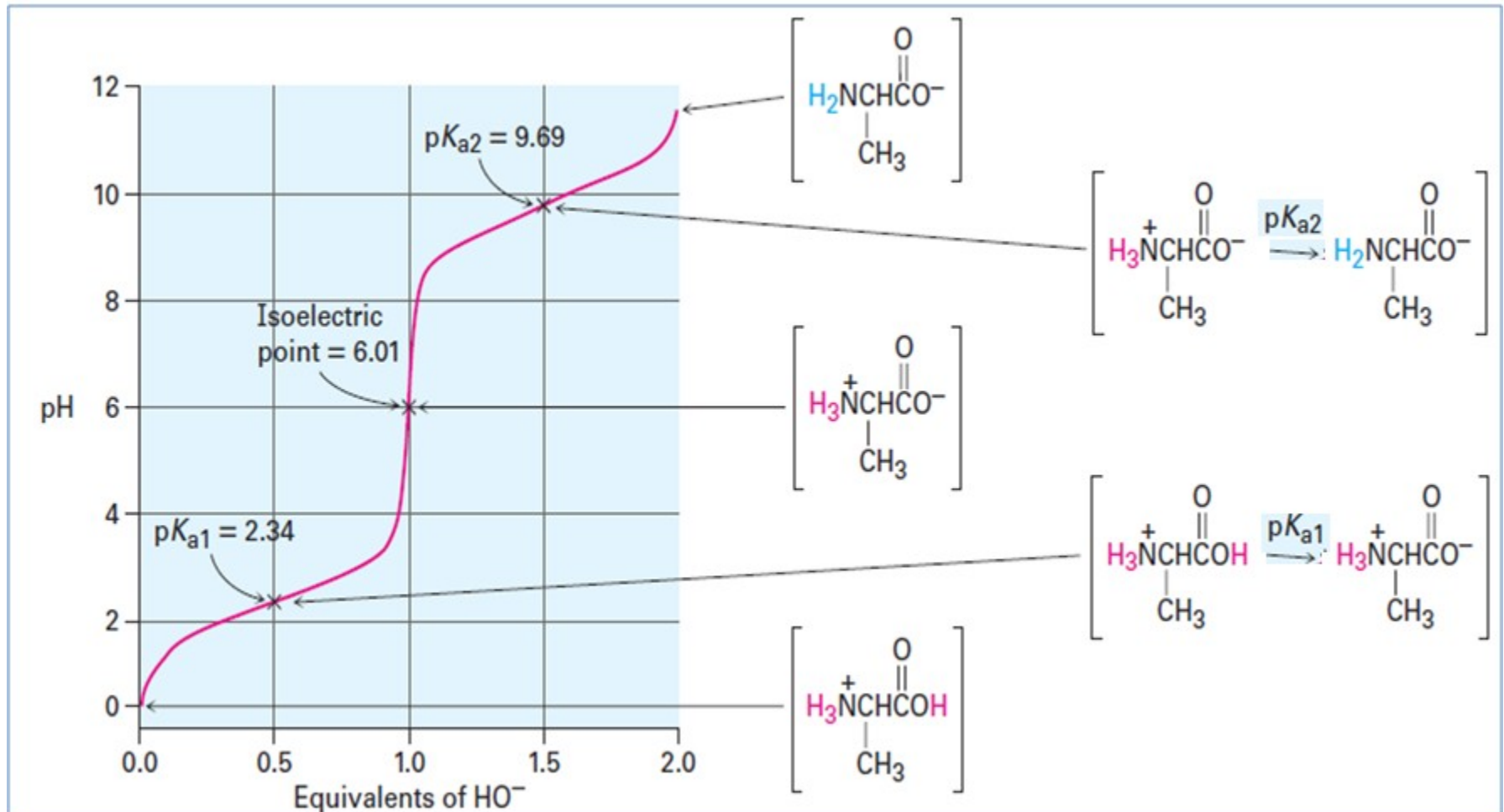
(Henderson-Hasselbach Equation)

$$\text{pH} = \text{pK}_a - \log [A^-]/[HA]$$

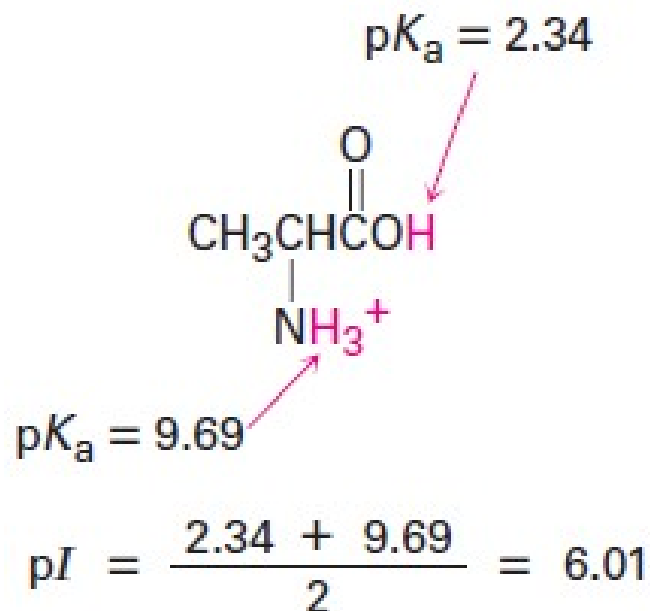
This permits a titration curve to be calculated or pK_a to be determined from a titration curve

- **Example: alanine**

Titration curve for Amino Acids



- **Calculated as if R is:**
 - **Non-ionizable**
 - Midway between its two pK_a values

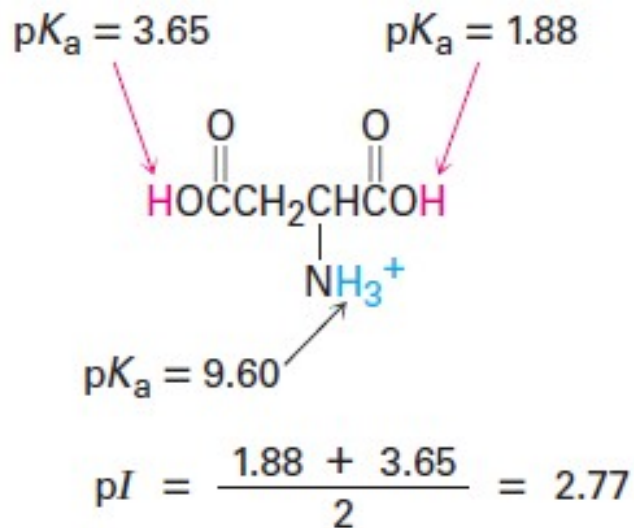


Neutral amino acid

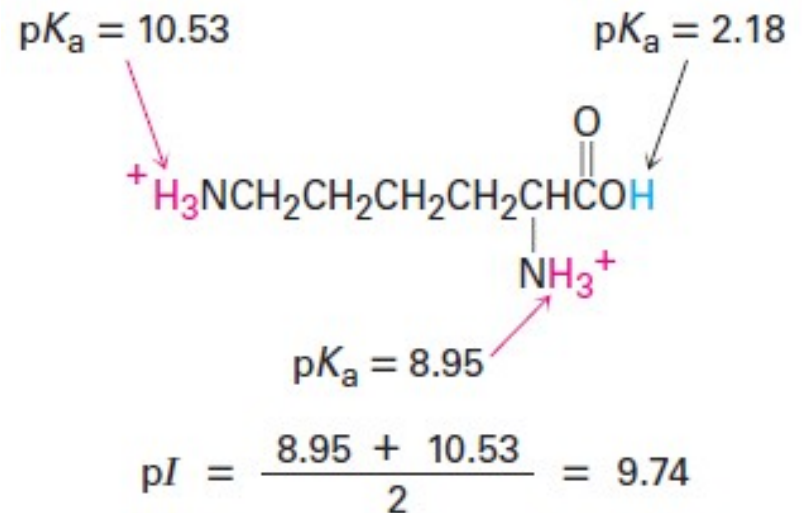
Alanine

– Ionizable

- Average of the values of the *similarly* ionizing groups



Acidic amino acid
Aspartic acid



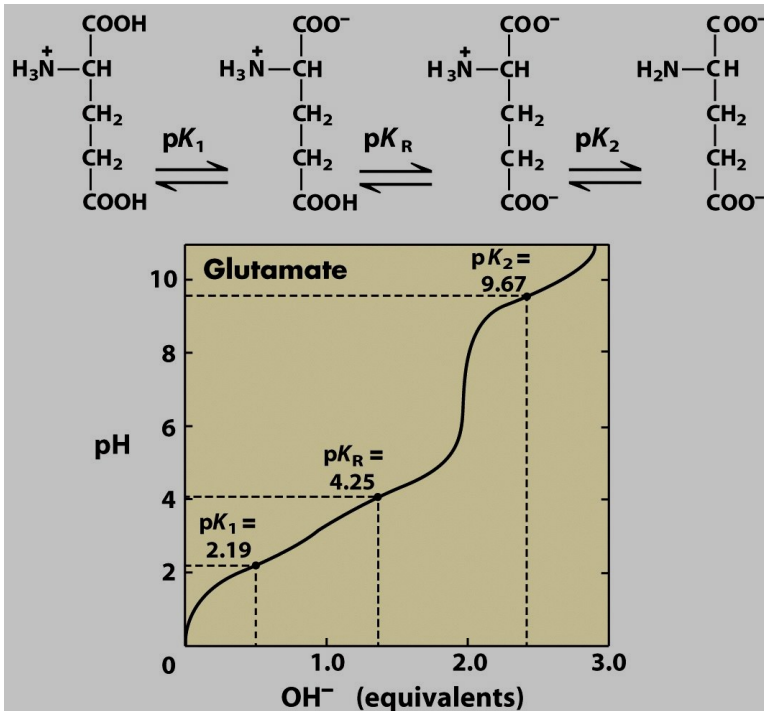
Basic amino acid
Lysine

Amino acids with
R groups that do not ionize

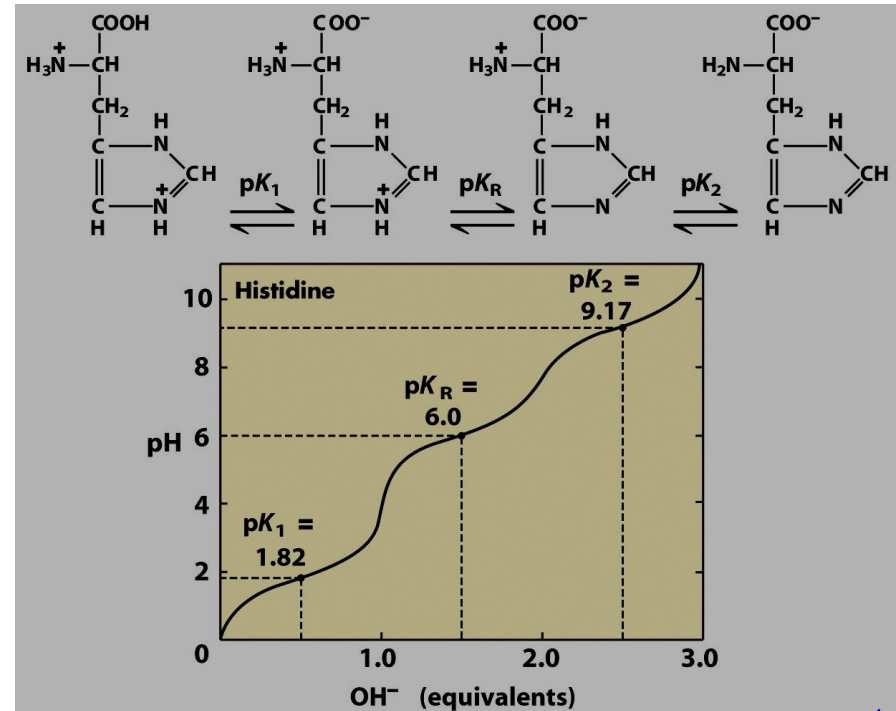
pK_a of the $-\text{COOH}$ group: 1.8 – 2.4
 pK_a of the $-\text{NH}_3^+$ group: 8.8 – 11.0

Amino acids with ionizable R groups

e.g. **Glutamate**



Histidine



three stages (three ionization steps $\rightarrow$ three pK_a values)

TABLE 3-1 Properties and Conventions Associated with the Common Amino Acids Found in Proteins

Amino acid	Abbreviation/ symbol	M_r	pK_a values			pI	Hydropathy index*	Occurrence in proteins (%) [†]
			pK_1 (—COOH)	pK_2 (—NH ₃ ⁺)	pK_R (R group)			
Nonpolar, aliphatic R groups								
Glycine	Gly G	75	2.34	9.60		5.97	−0.4	7.2
Alanine	Ala A	89	2.34	9.69		6.01	1.8	7.8
Proline	Pro P	115	1.99	10.96		6.48	1.6	5.2
Valine	Val V	117	2.32	9.62		5.97	4.2	6.6
Leucine	Leu L	131	2.36	9.60		5.98	3.8	9.1
Isoleucine	Ile I	131	2.36	9.68		6.02	4.5	5.3
Methionine	Met M	149	2.28	9.21		5.74	1.9	2.3
Aromatic R groups								
Phenylalanine	Phe F	165	1.83	9.13		5.48	2.8	3.9
Tyrosine	Tyr Y	181	2.20	9.11	10.07	5.66	−1.3	3.2
Tryptophan	Trp W	204	2.38	9.39		5.89	−0.9	1.4

*A scale combining hydrophobicity and hydrophilicity of R groups; it can be used to measure the tendency of an amino acid to seek an aqueous environment (− values) or a hydrophobic environment (+ values). See Chapter 11. From Kyte, J. & Doolittle, R.F. (1982) A simple method for displaying the hydropathic character of a protein. *J. Mol. Biol.* **157**, 105–132.

[†]Average occurrence in more than 1,150 proteins. From Doolittle, R.F. (1989) Redundancies in protein sequences. In *Prediction of Protein Structure and the Principles of Protein Conformation* (Fasman, G.D., ed.), pp. 599–623, Plenum Press, New York.

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			pK_1 (—COOH)	pK_2 (—NH ₃ ⁺)	pK_R (R group)			
Polar, uncharged								
R groups								
Serine	Ser S	105	2.21	9.15		5.68	−0.8	6.8
Threonine	Thr T	119	2.11	9.62		5.87	−0.7	5.9
Cysteine	Cys C	121	1.96	10.28	8.18	5.07	2.5	1.9
Asparagine	Asn N	132	2.02	8.80		5.41	−3.5	4.3
Glutamine	Gln Q	146	2.17	9.13		5.65	−3.5	4.2
Positively charged								
R groups								
Lysine	Lys K	146	2.18	8.95	10.53	9.74	−3.9	5.9
Histidine	His H	155	1.82	9.17	6.00	7.59	−3.2	2.3
Arginine	Arg R	174	2.17	9.04	12.48	10.76	−4.5	5.1
Negatively charged								
R groups								
Aspartate	Asp D	133	1.88	9.60	3.65	2.77	−3.5	5.3
Glutamate	Glu E	147	2.19	9.67	4.25	3.22	−3.5	6.3

*A scale combining hydrophobicity and hydrophilicity of R groups; it can be used to measure the tendency of an amino acid to seek an aqueous environment (− values) or a hydrophobic environment (+ values). See Chapter 11. From Kyte, J. & Doolittle, R.F. (1982) A simple method for displaying the hydropathic character of a protein. *J. Mol. Biol.* **157**, 105–132.

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Electrophoresis

Method for separating substances on the basis of their tendency to migrate to a positively or negatively charged electrode at a particular pH.

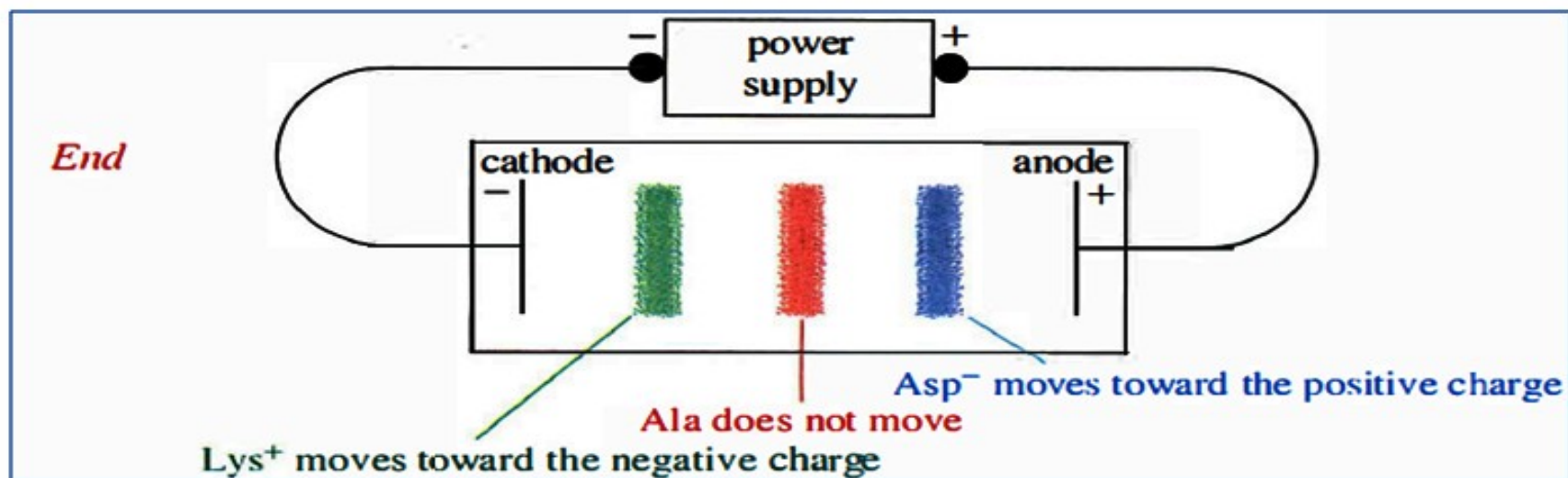
Separation of aa_s by electrophoresis (separation of charged compounds in an electric field) is possible because all aa_s have different isoelectric points

□ pI (isoelectric point)

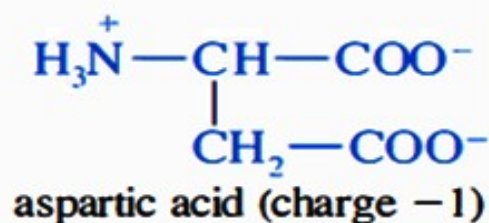
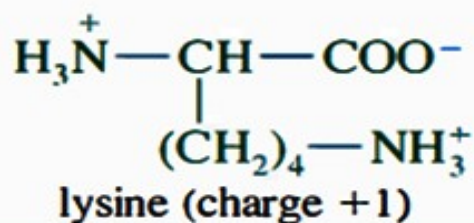
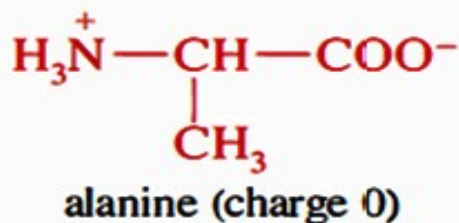
pI > pH (acidic solution) $\Rightarrow$ +ve $\Rightarrow$ cathode

pI < pH (basic solution) $\Rightarrow$ -ve $\Rightarrow$ anode

- Consider a mixture of alanine ($pI=6.01$), lysine ($pI=9.74$), and aspartic acid ($pI=2.77$) in a buffer solution at pH 6



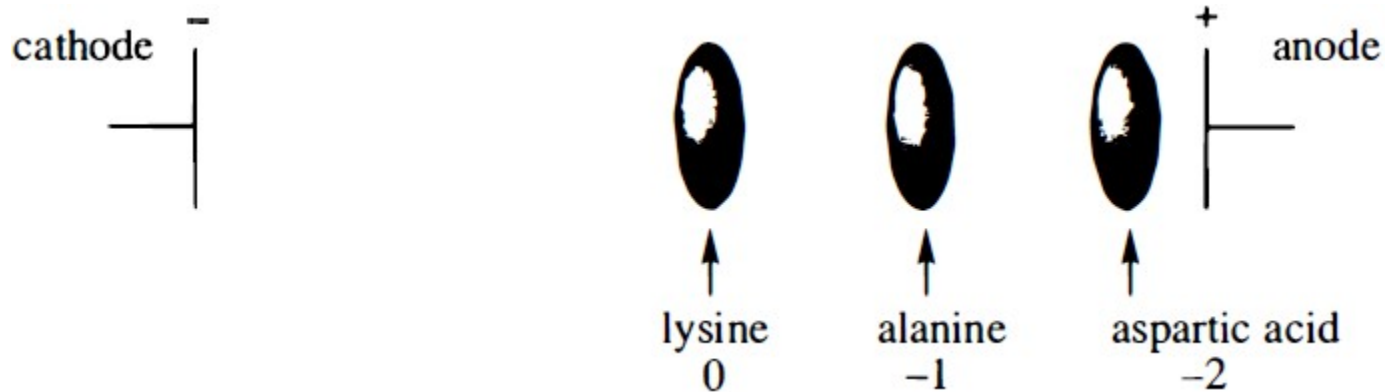
Structure at pH 6



- **Exercise**

- Draw the electrophoretic separation of Ala, Lys, and Asp at pH 9.7.

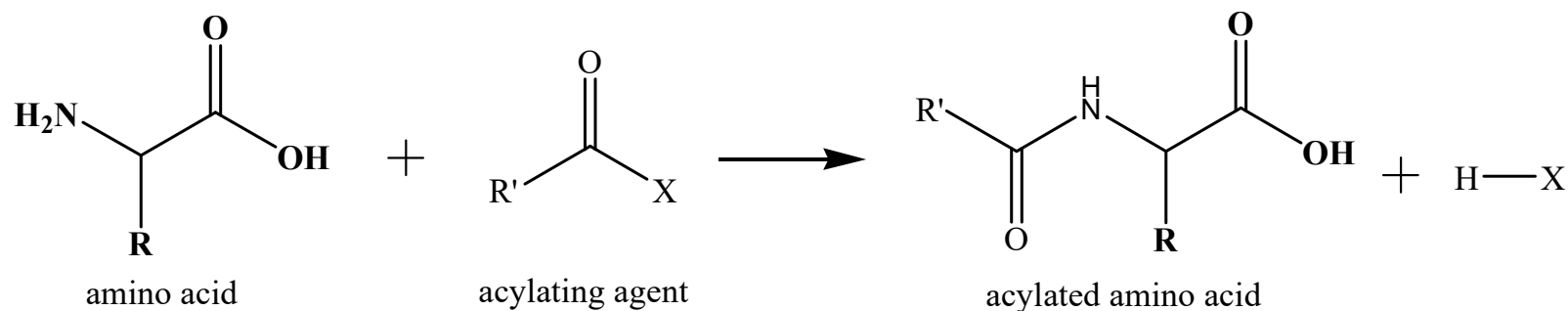
- **Answer:**



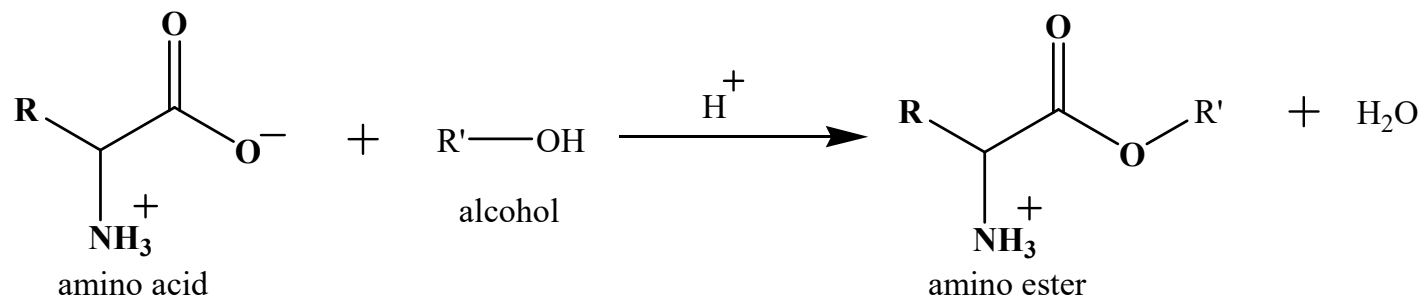
Reactions of amino acids

- Amino acids: both amino and carboxylic groups**

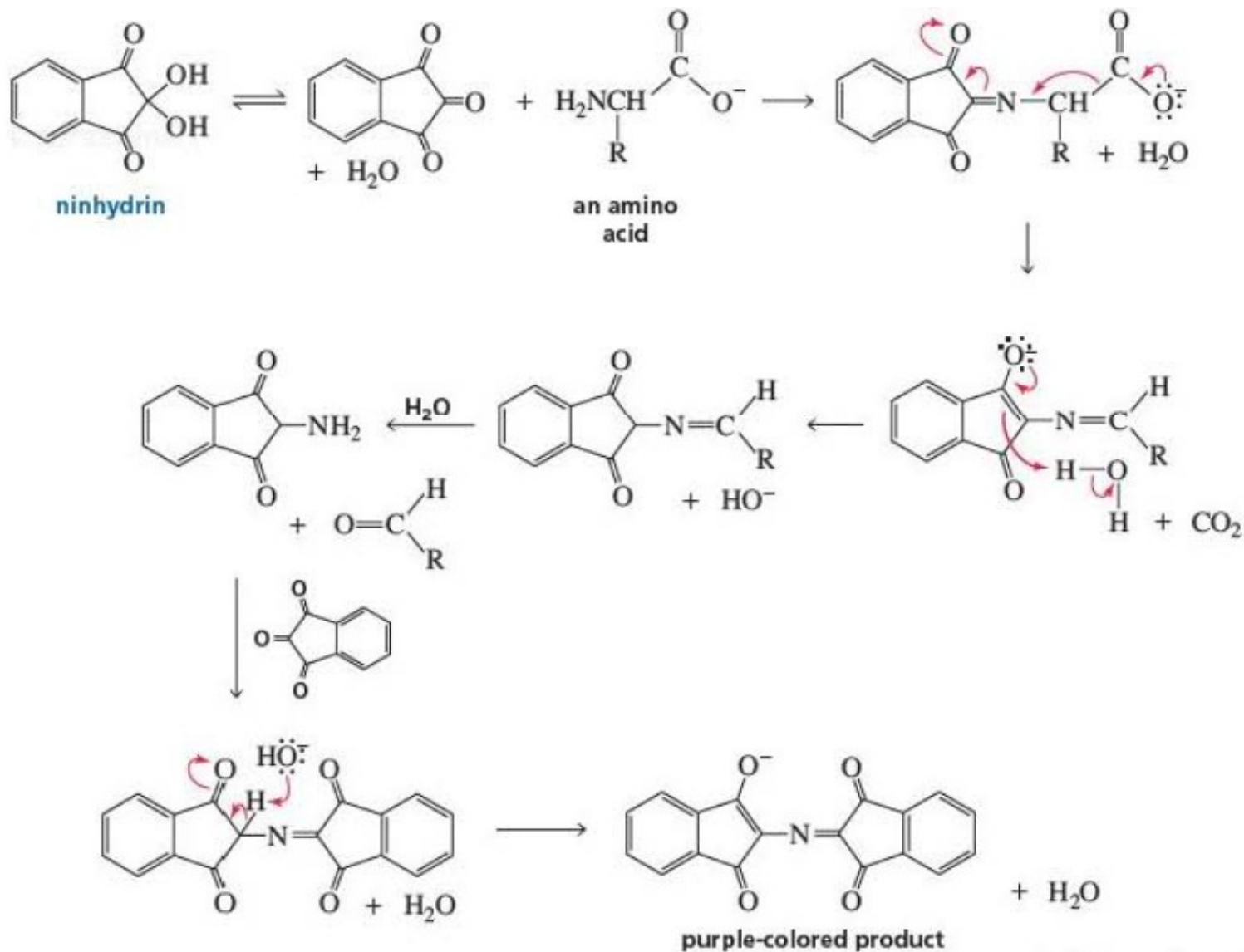
1. Acylation:



2. Esterification (Ester formation)



3. Reaction with ninhydrine



Synthesis of amino acids

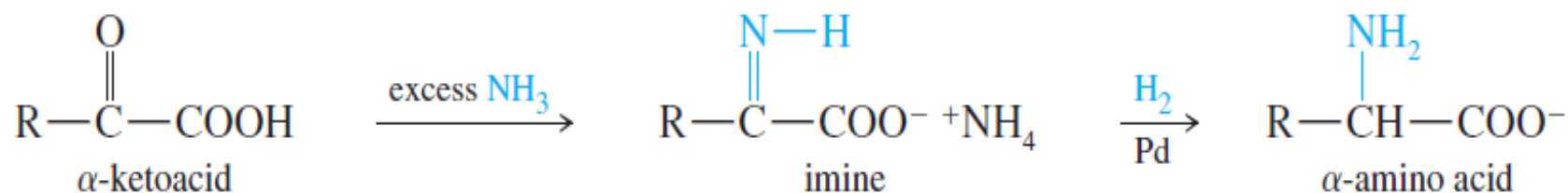
- **Amino acid sources**
 - Nature
 - Synthesize
 - Less expensive
 - Unusual amino acid or an unnatural enantiomer
- **Variety of methods**
 - 1.1. Reductive Amination
 - 1.2. Amination of an α -halo Acid
 - 1.3. The Gabriel–Malonic Ester Synthesis

Reductive amination

- **Amino acids**

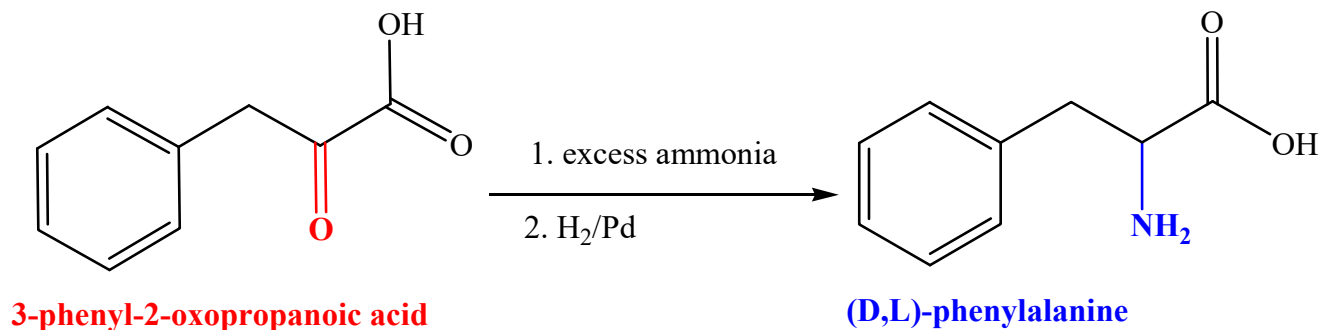
- From *alpha-ketoacid*

- Using excess NH_3 and H_2/Pd catalyst



- **Exercise:**

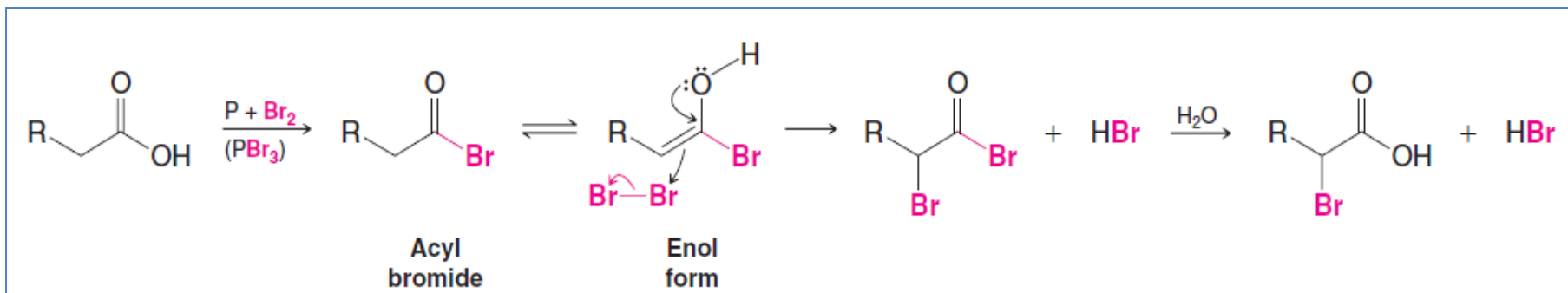
- Show the synthesis of (D,L)-phenylalanine using reductive amination



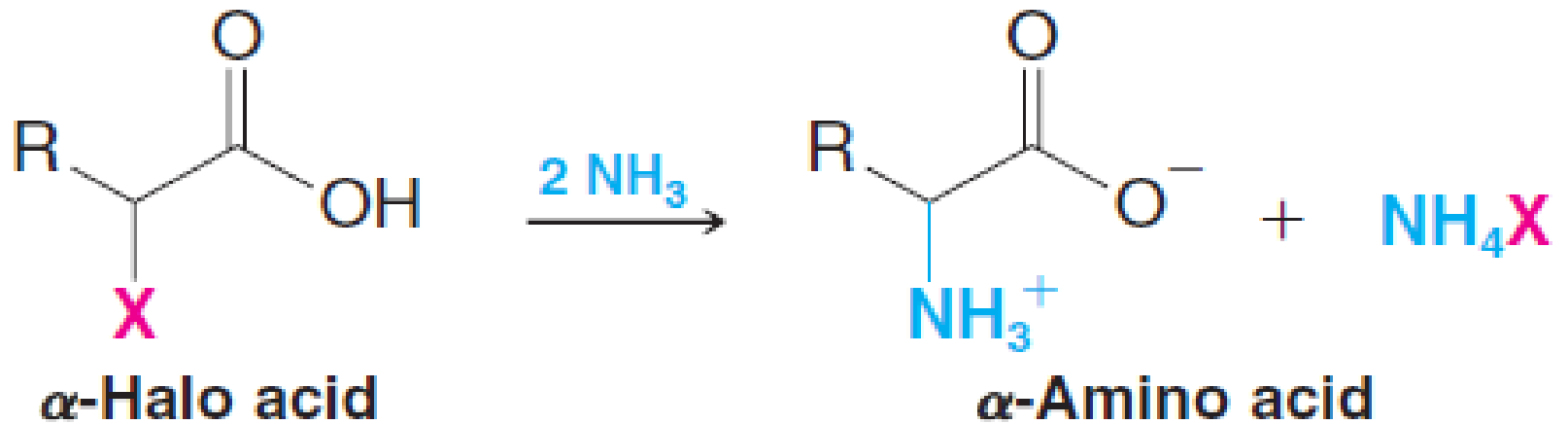
Amination of an α -halo acid

- **Amination of an α -halo acid:**
 - Two step Rxn
 - 1. Hell–Volhard–Zelinski reaction (formation of α -halo acid)
 - 2. followed by reaction with ammonia (Amination)

Hell–Volhard–Zelinski reaction α -Bromination of carboxylic acids

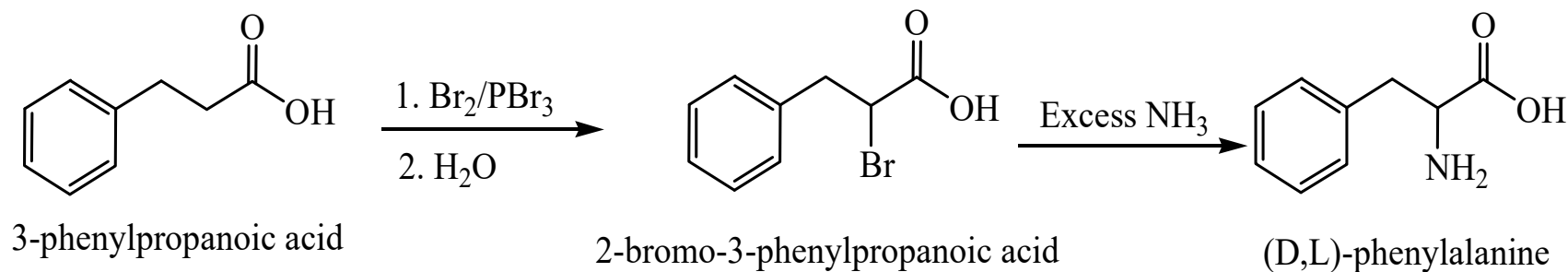


- Amination of an α -halo acid
 - α -amino acids



- **Exercise:**

- Show the synthesis of (D,L)-phenylalanine using amination of α -halo acid



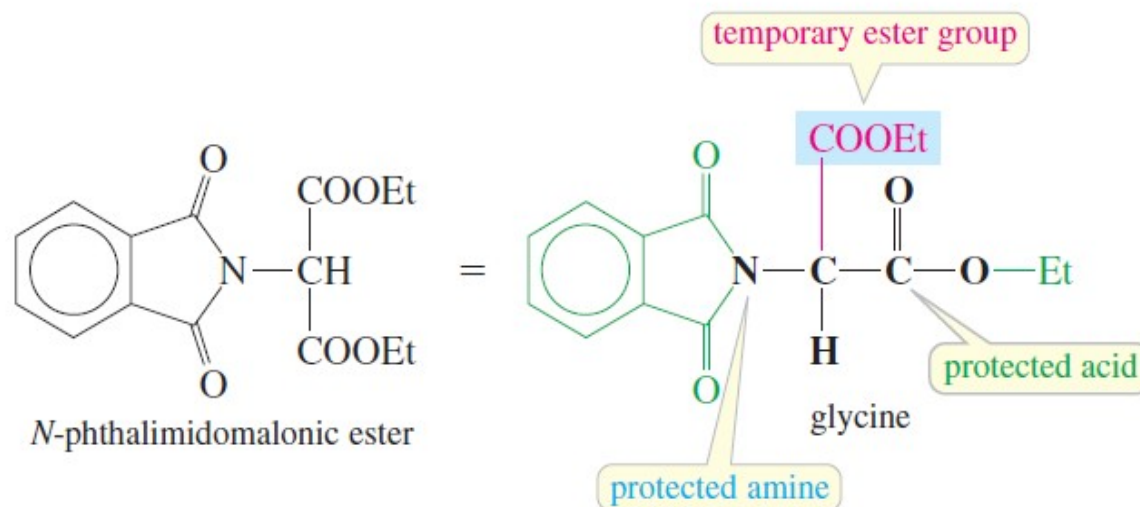
The Gabriel–Malonic ester synthesis

- **The Gabriel–Malonic Ester Synthesis**

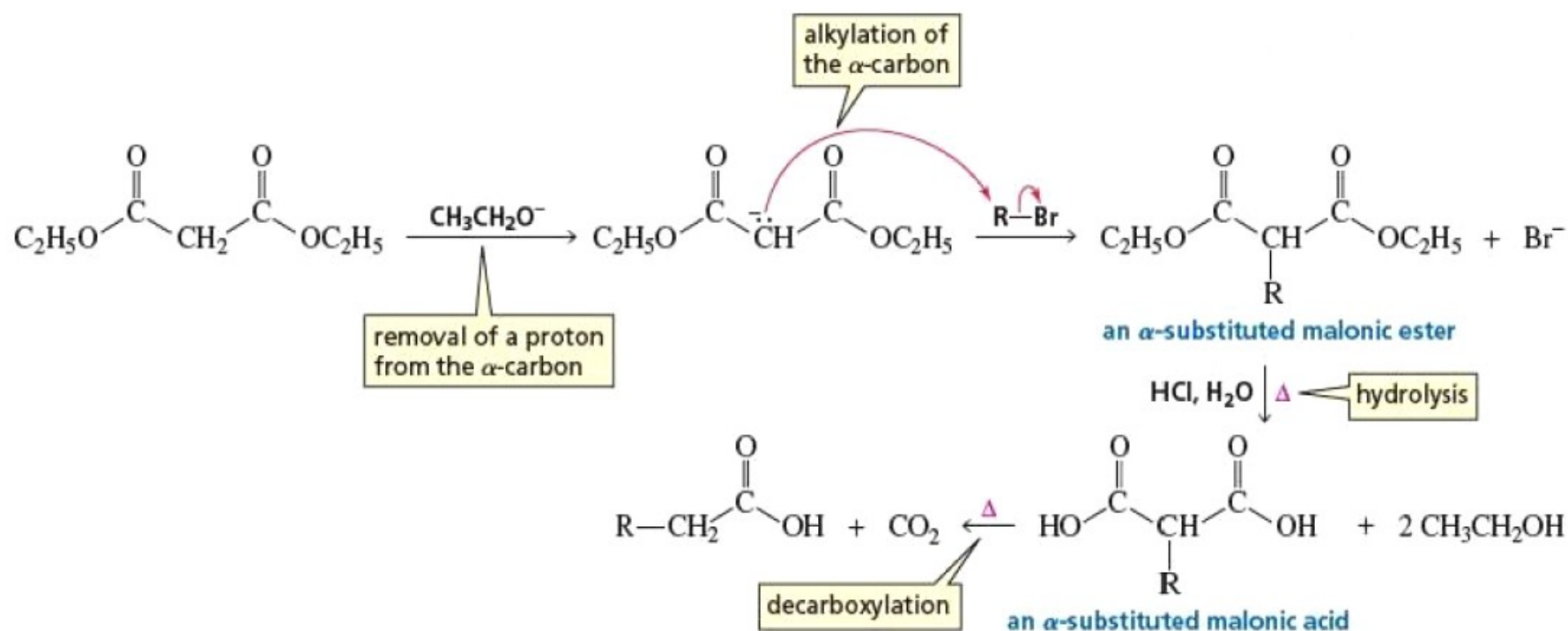
- N-phthalimidomalonic ester synthesis

- ☐ Involves two step reactions

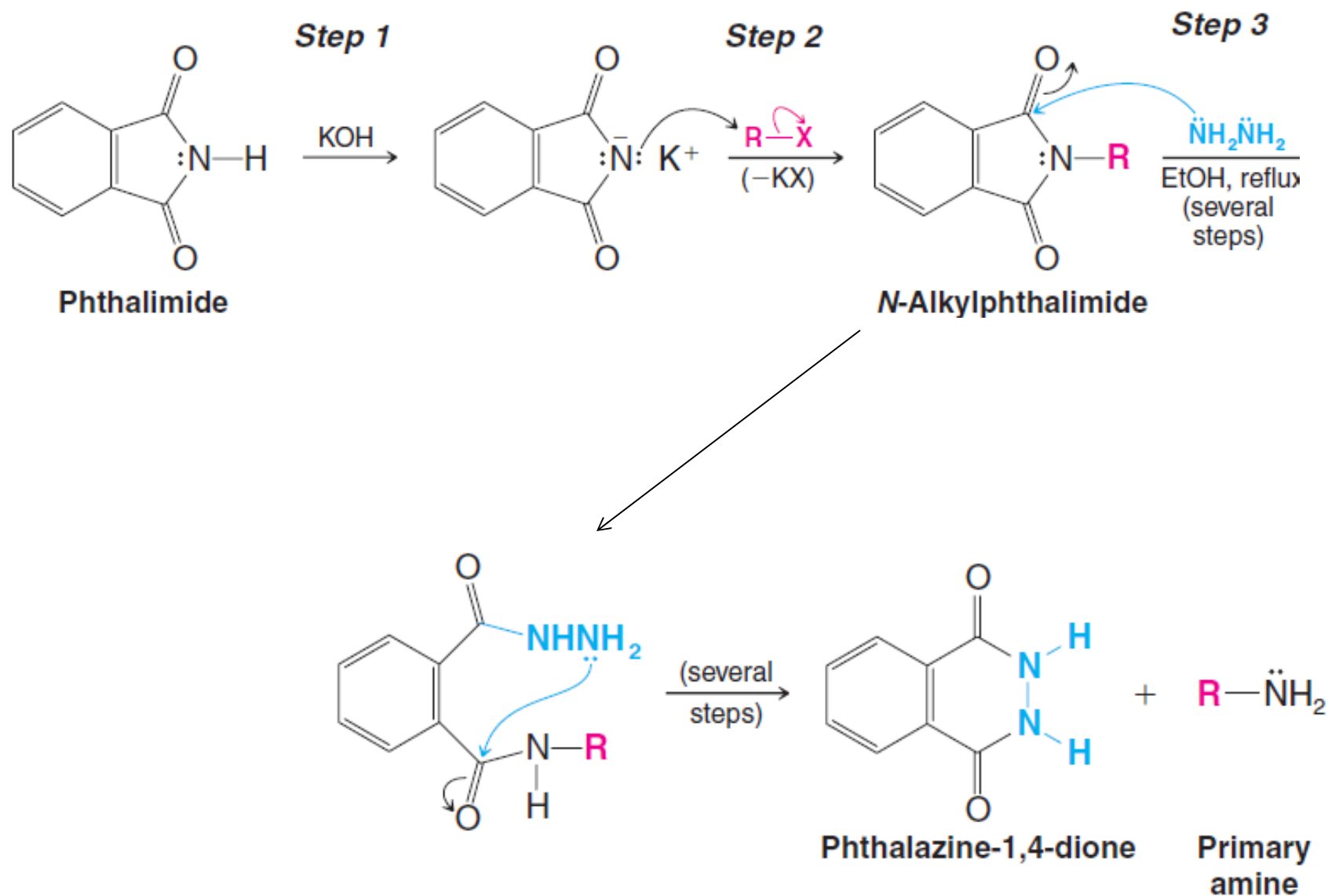
- Malonic ester synthesis and
 - the Gabriel synthesis of amines



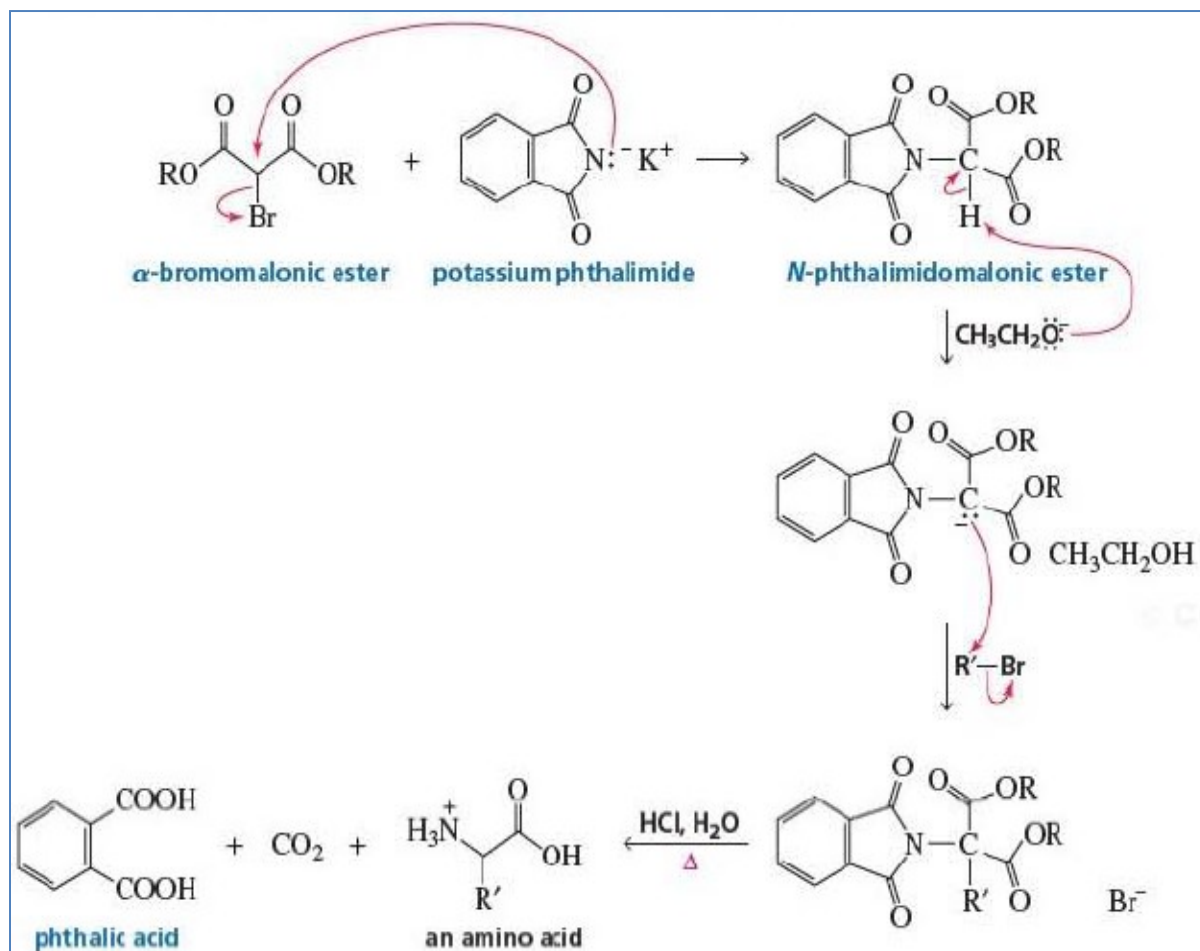
- Malonic ester synthesis of carboxylic acids



- **Gabriel synthesis of amines**

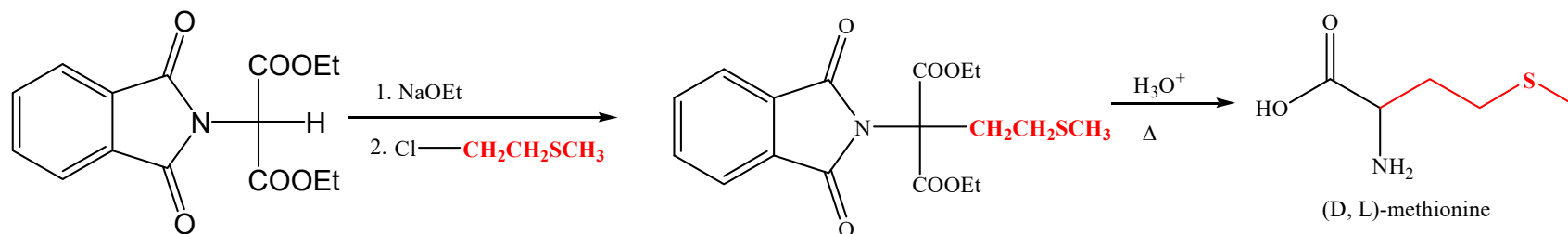


- **The Gabriel–Malonic Ester Synthesis**
 - Synthesis of α -amino acids



- **Exercise:**

- Outline the synthesis of racemic methionine using Gabriel–Malonic Ester Synthesis



Resolution of amino acids

- **Synthetic methods Vs Natural**

Racemic mixture

Pure 100%

- ***Needs separation***

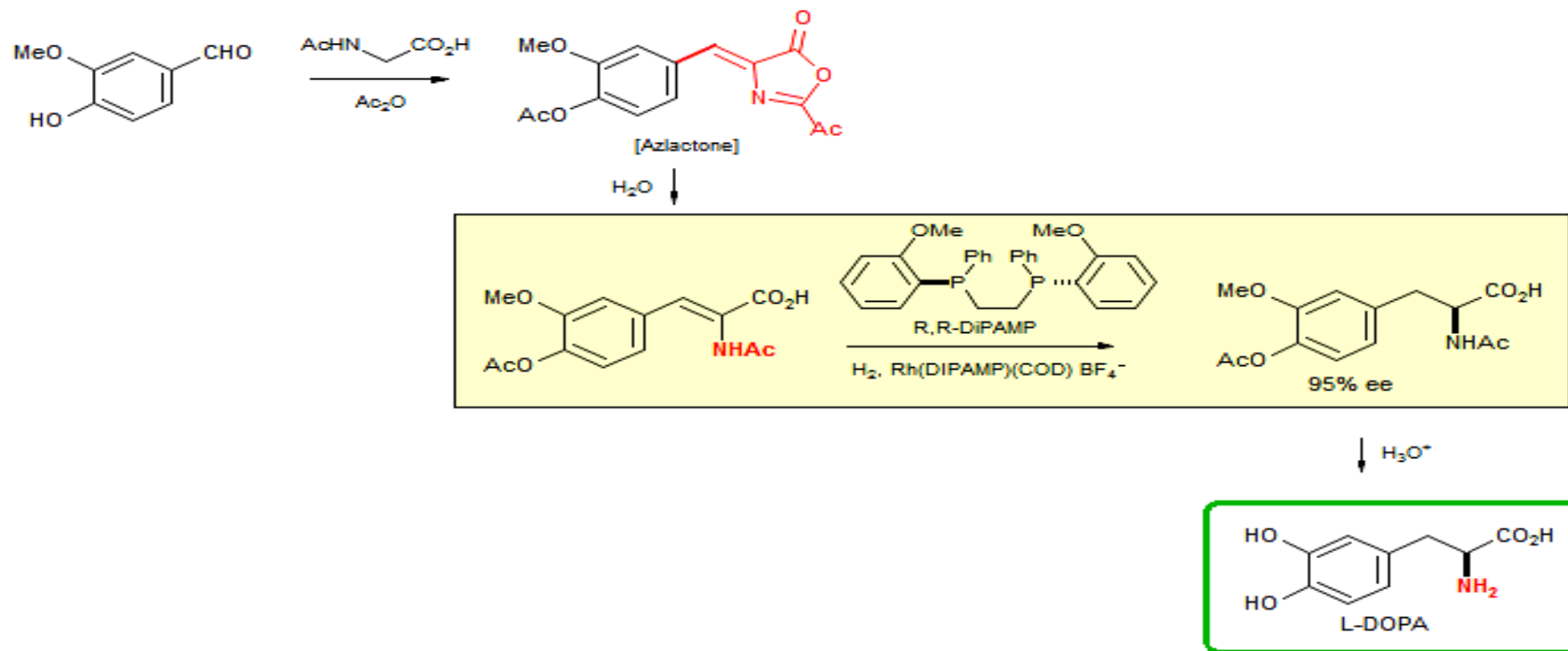
- **Methods**

1. Chiral catalyst or reagent (enantioselective synthesis)
2. Diastereomer salt formation
3. Enzyme catalyzed enantiomer separation

Enantioselective synthesis

- **Chiral catalyst or reagent (enantioselective synthesis)**

- Enantioselective reaction
 - Selective production of one enantiomer
- E.g. Synthesis of L-DOPA



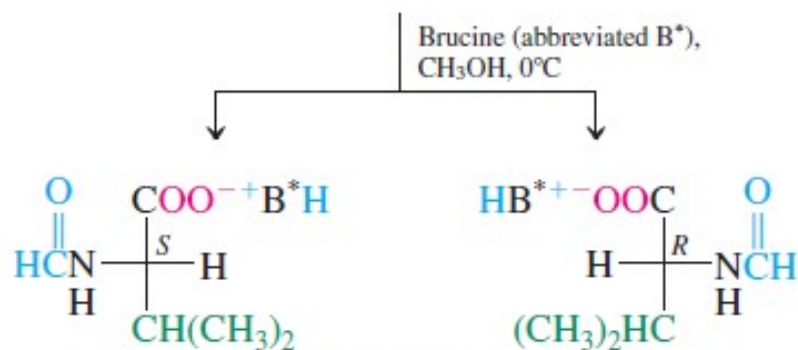
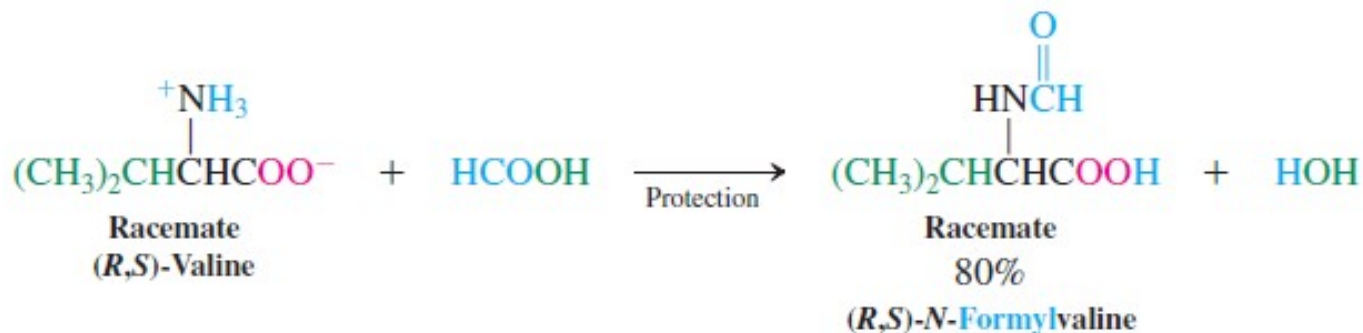
Resolution of racemic amino acids

(A). Resolution by diastereomer salt formation

Diastereomers

- ❑ Have different physicochemical properties

Resolution of Racemic Valine

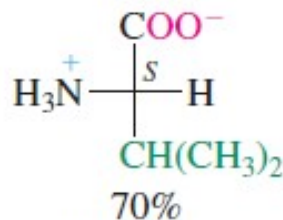


Diastereomers
Separate by fractional crystallization

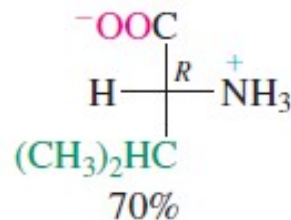
NaOH, H₂O, 0°C

Remove brucine and
hydrolyze amide

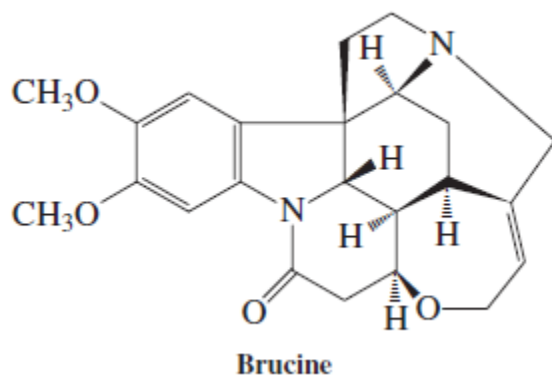
NaOH, H₂O, 0°C



Enantiomer
(S)-Valine

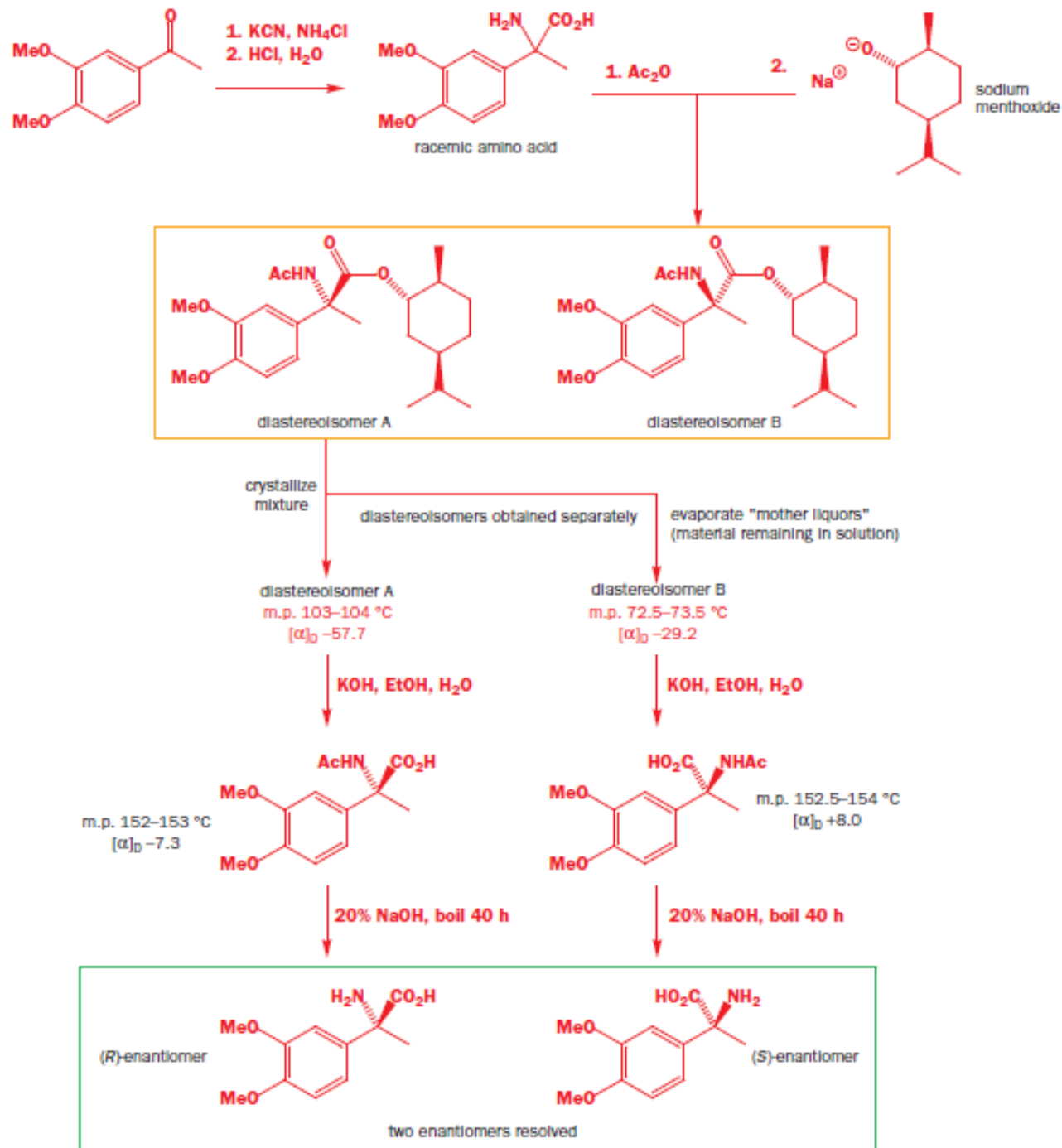


Enantiomer
(R)-Valine



Diastereomer salt
formation

Diastereomer formation

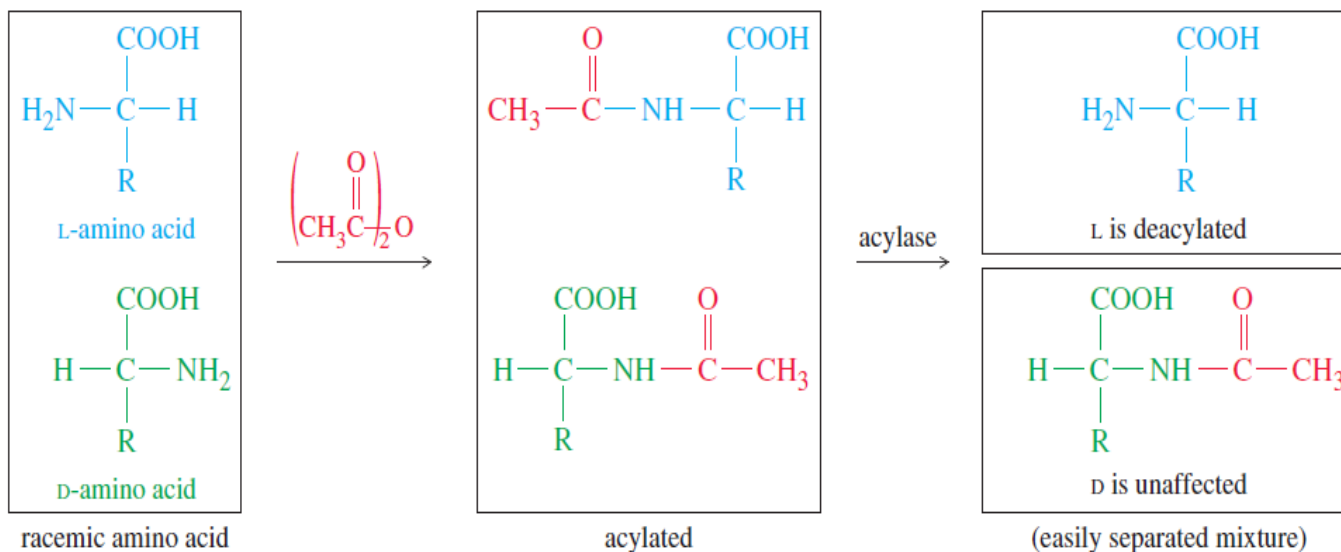


(B) Resolution *by* enzymes

- **Enzyme catalyzed enantiomer separation**

- Enzymes are *chiral molecules* with *specific catalytic activities*

- E.g. hog kidney acylase or carboxypeptidase



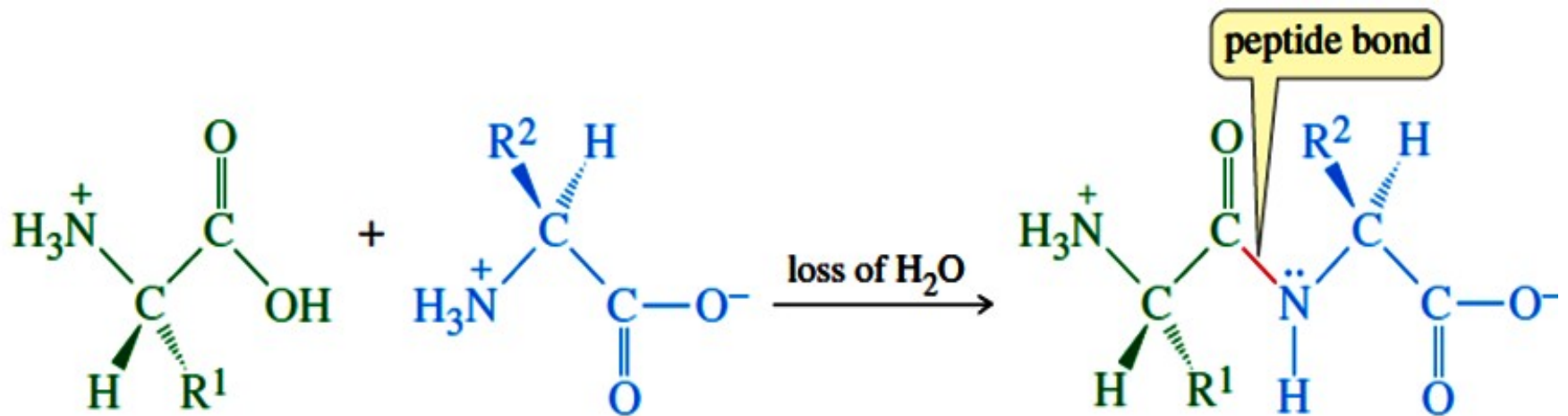
Peptides, polypeptides and proteins

- **Peptides, polypeptides and proteins**
 - Polymers of amino acids
 - Differ
 - *Number and*
 - *sequence of the constituent amino acids*
 - Structural and functional integrity of cells
 - Polypeptides usually have molecular weight less than 5000.
 - Protein molecular weight 6000-40,000,000
 - Distinction from polypeptides is essentially based on functionality besides complexity

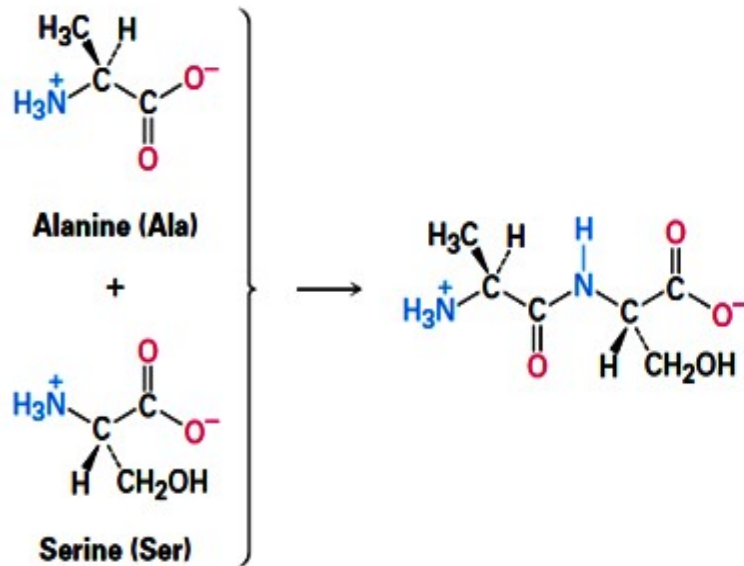
- **Peptides**

- ***Condensation product*** the amino group of one aa (> 2) molecule with the carboxyl group of another

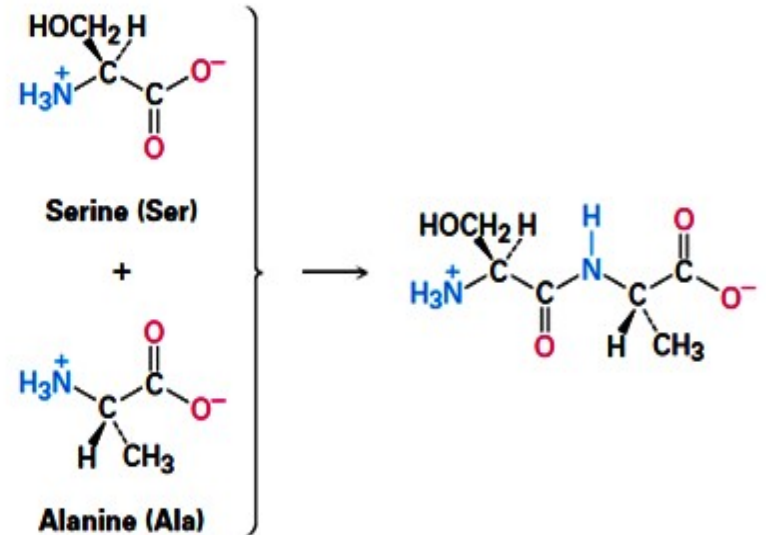
- Residue



Example:

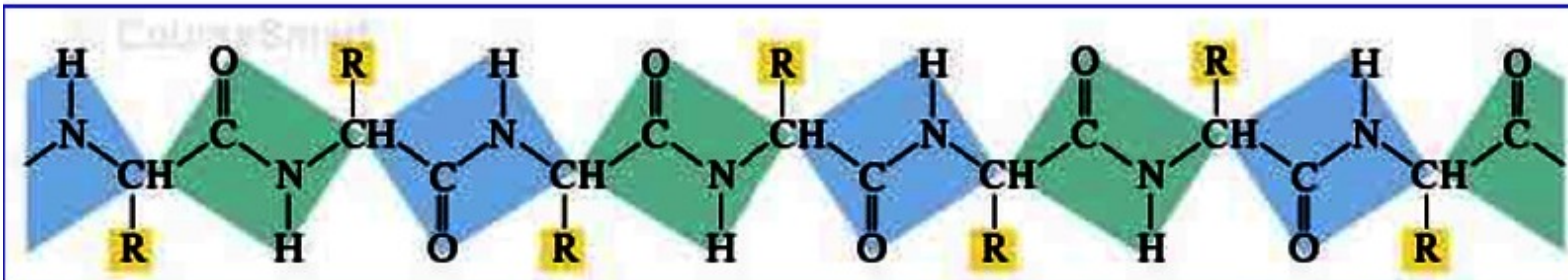
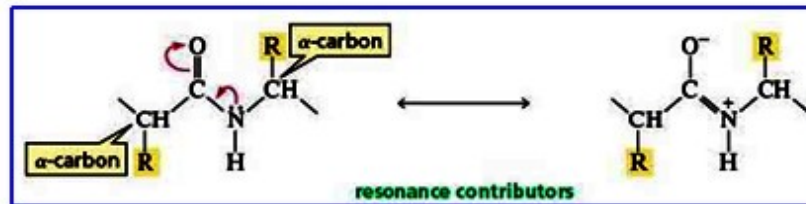
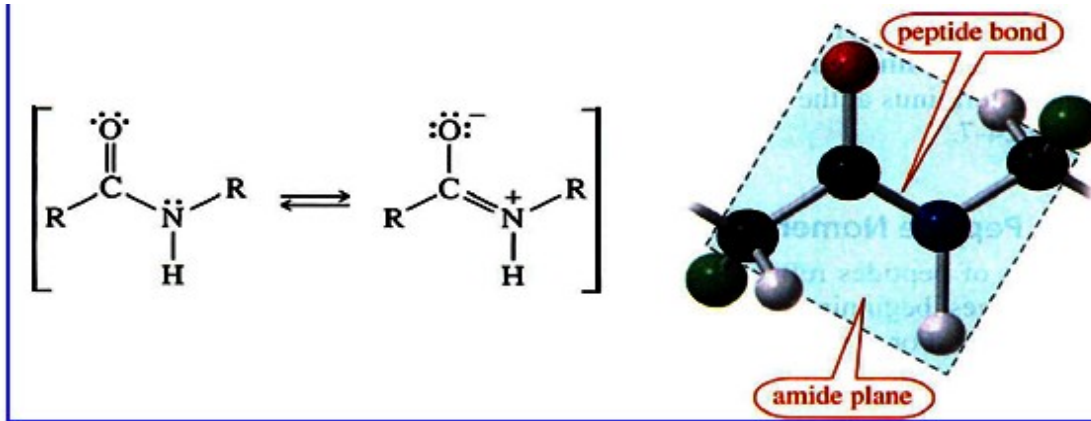


Alanyls erine



Serylal anine

- Peptide bonds or amide bonds
 - Partial double bond character. What?



- **Peptide nomenclature: by convention**

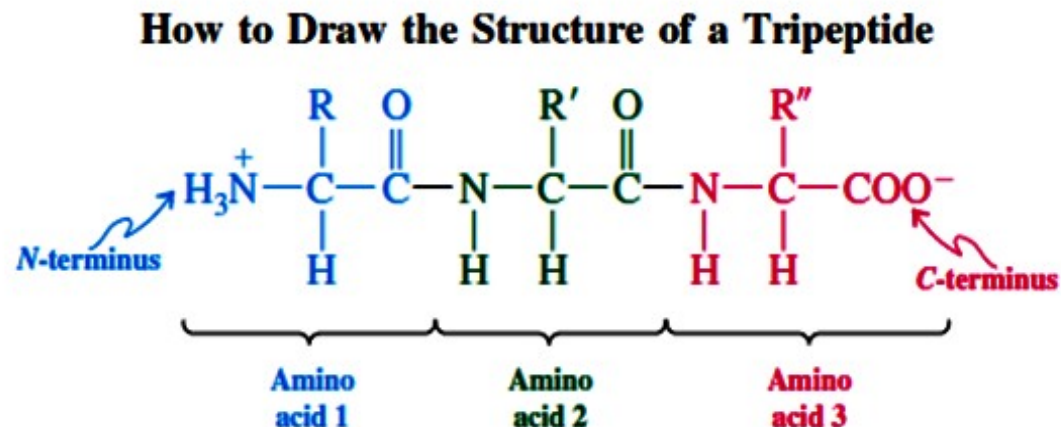
- Free —NH_3^+ (amino group)

- ***N-terminus = left side***

- Free —CO_2^- (carboxylate group)

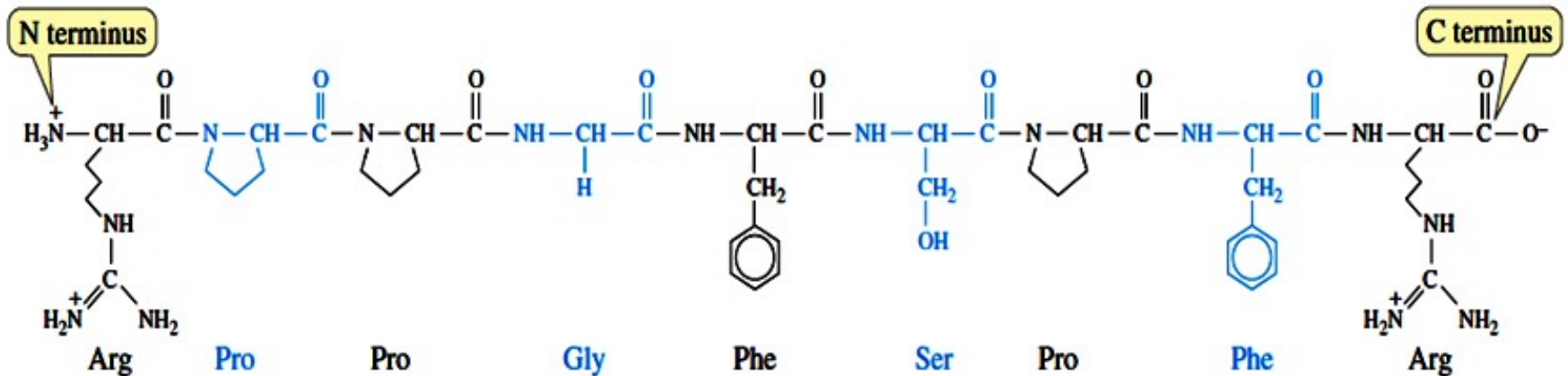
- ***C-terminus = right side***

- Three letter naming the most common



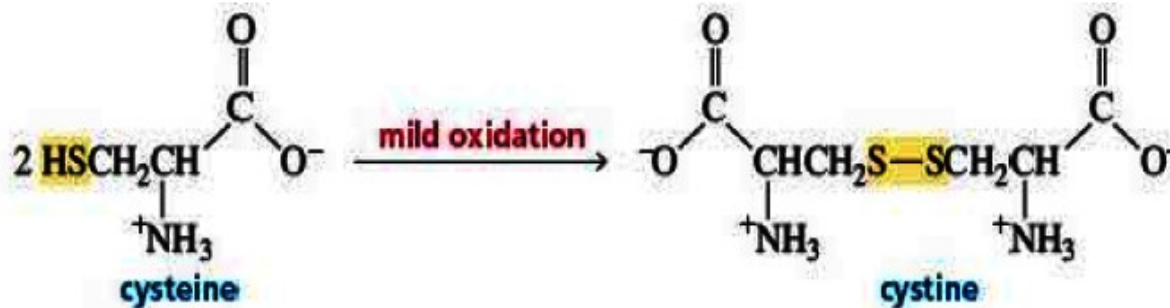
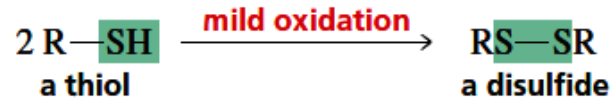
Example

- Bradykinin: peptide that causes the blood vessel to enlarge
 - Use of ***-yl***
 - Arginylprolylprolylglycylphenylalanylserylprolylphenylalanylarginine
 - ***Three letter***
 - Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg
 - ***Single (one)-letter symbols***
 - RPPGFSPFR

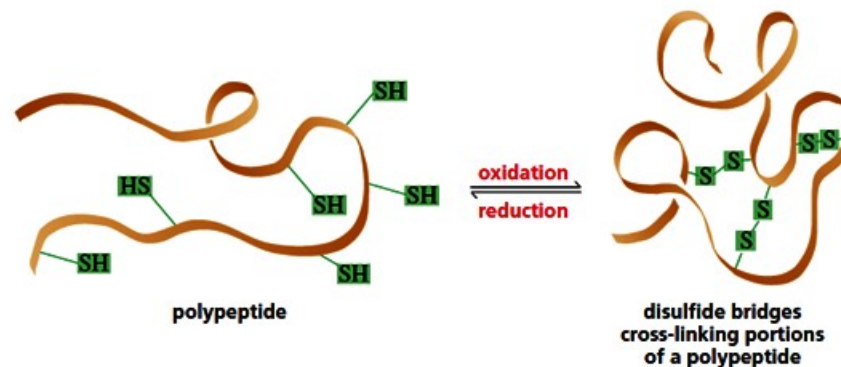


- **Disulfides**

- S—S bond
- Two cysteine residues



- In peptides: **disulfide bridges**

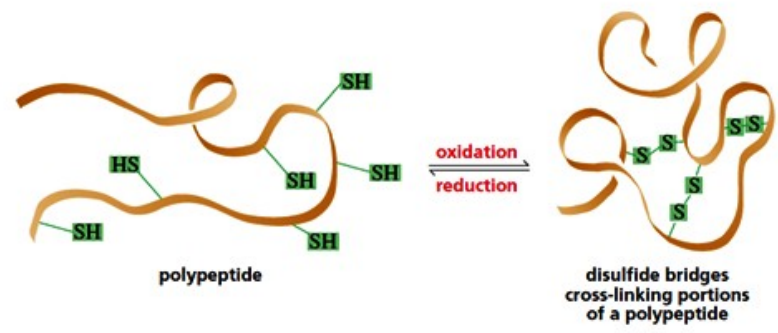
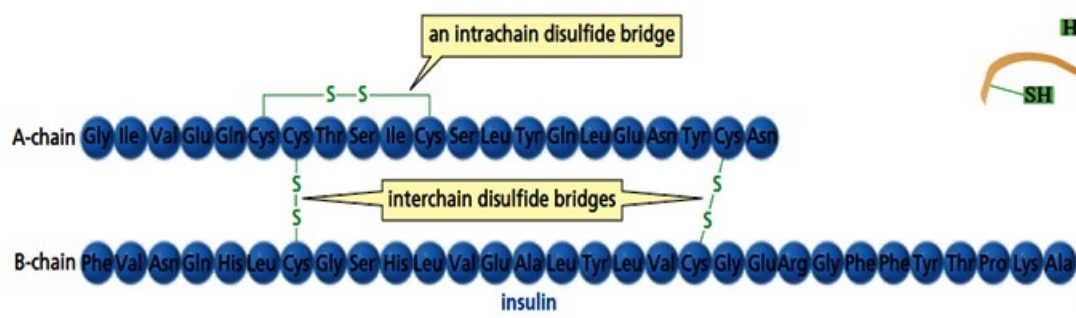
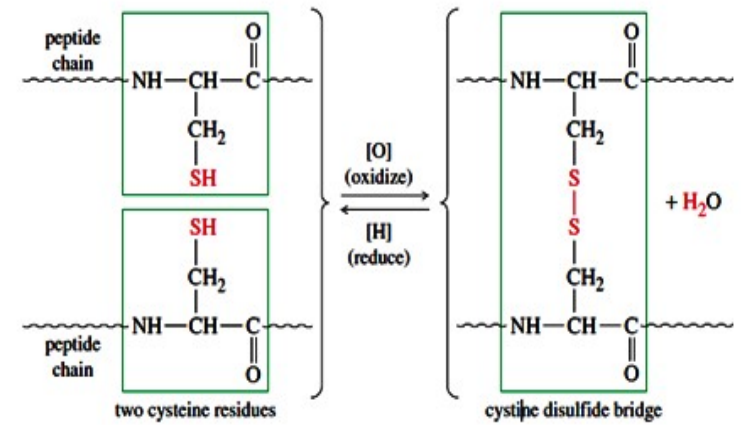


Peptide structure determination: Amino acid analysis and peptide sequencing

Peptide structure determination:
Cleavage of disulfide linkages and amino acid analysis

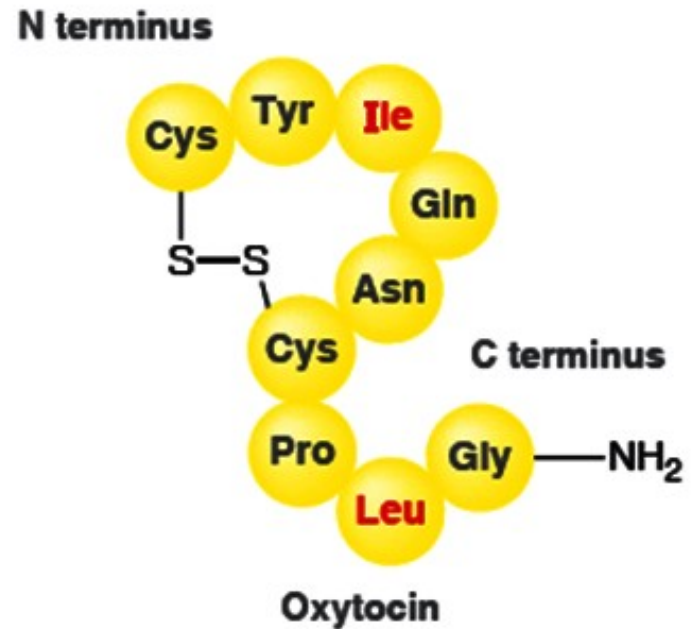
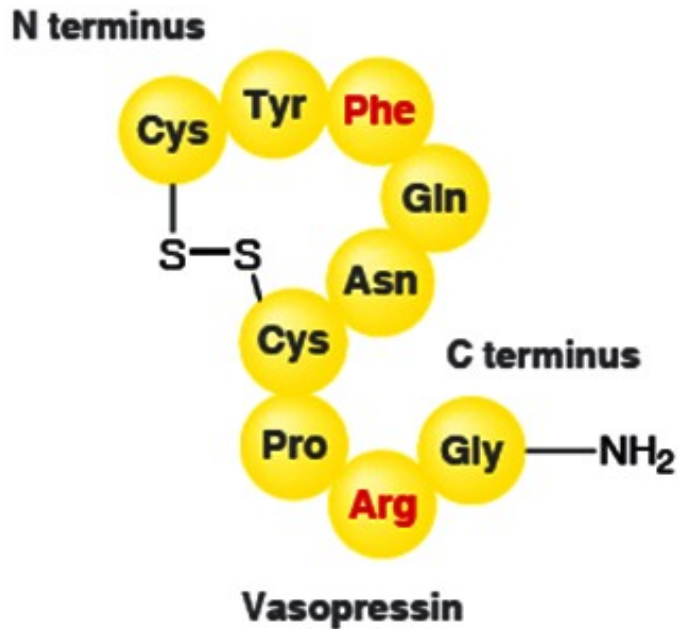
- Primary structure determination
 - Sequence of amino acids plus*
 - Disulfide bridges*

- Covalent bond between any cysteine residues*
 - Oxidation product of thiol groups of cysteine*
 - Cystine*



- **Examples**

- Vasopressin and oxytocin



Peptide primary structure determination

—Three questions

- *What kind of amino acids are present?*
- *How much of each is present?*
- *In what sequence do the amino acids occur in the peptide chain?*

- **Steps**

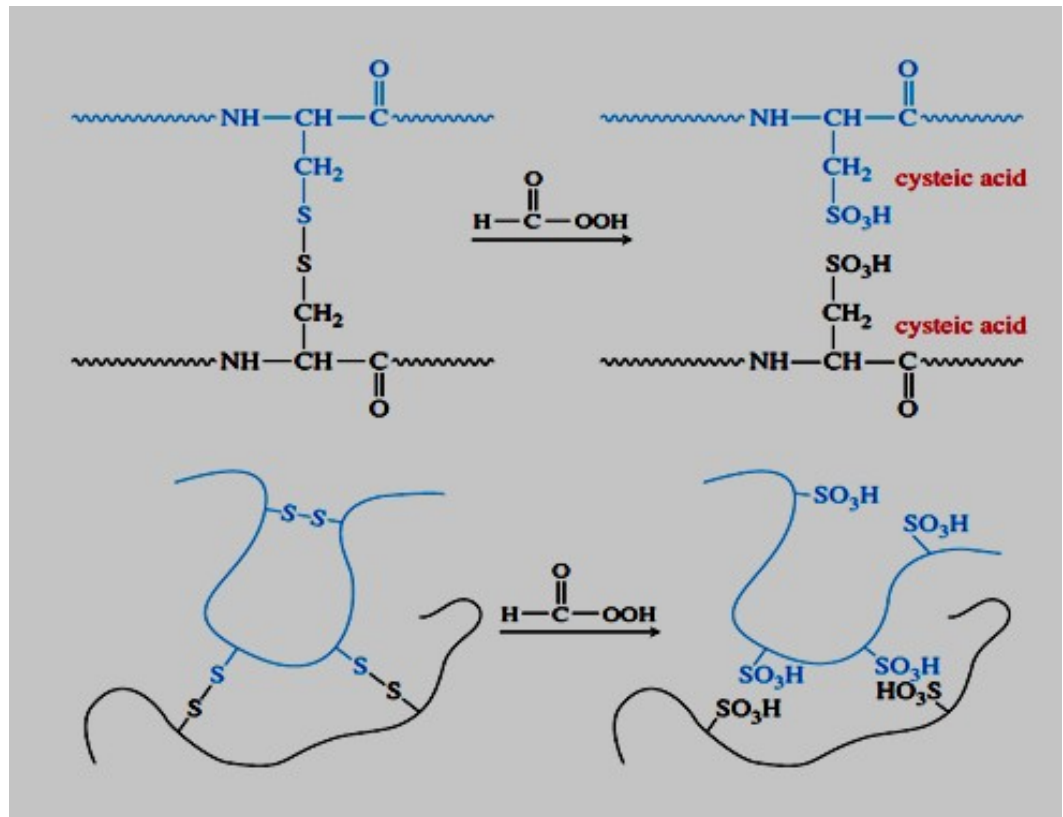
- ***Cleavage of disulfied bridge***

- ***Amino acid analysis***

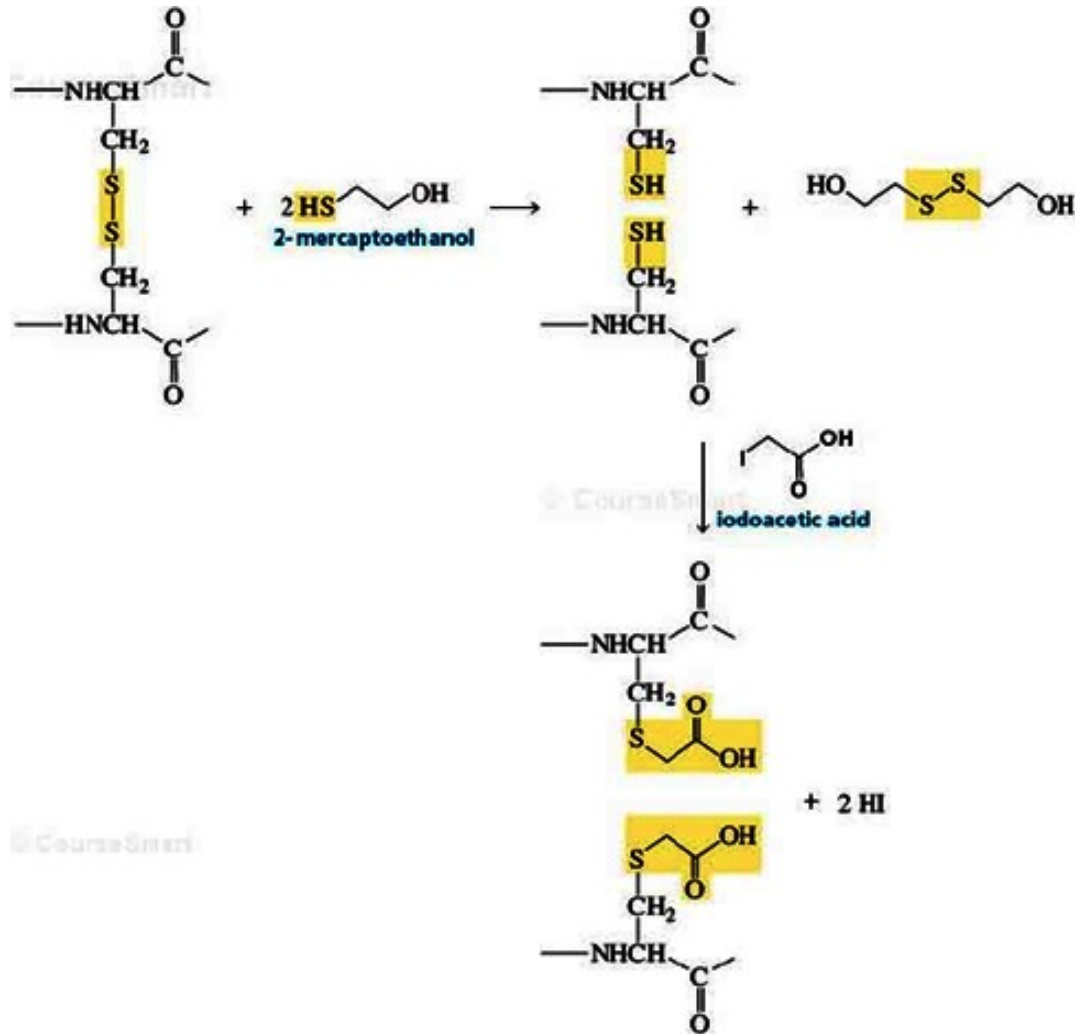
- ***Sequencing***

Cleavage of disulfide linkages

- **Cleavage of disulfied bridge**
 - Simple reduction to –SH.
 - Two methods
 - Peroxyformic acid



– 2-Mercaptoethanol



Amino acid analysis: Determination of the amino acid composition

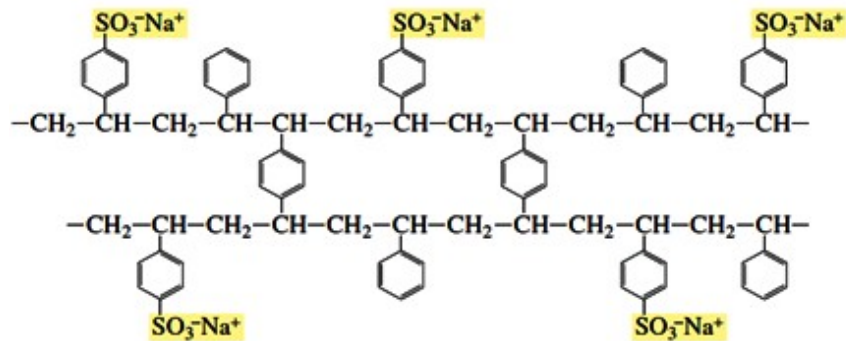
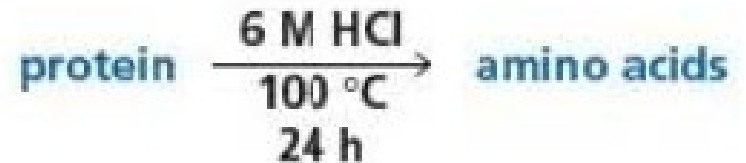
- **Amino acid analysis**

- **Complete hydrolysis**

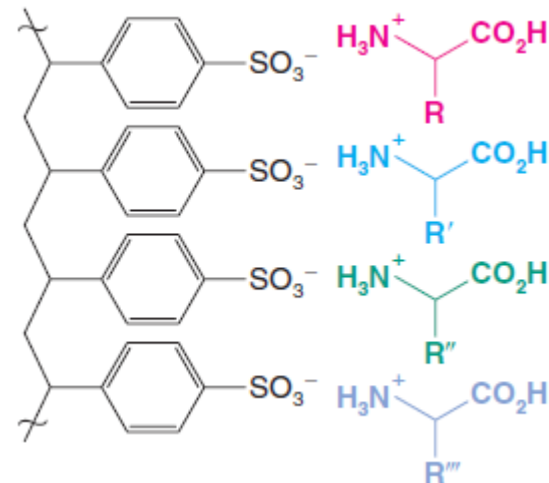
- By boiling it for 24 hours in 6 M HCl

- **Amino acid analyzer**

- Ion-exchange chromatography

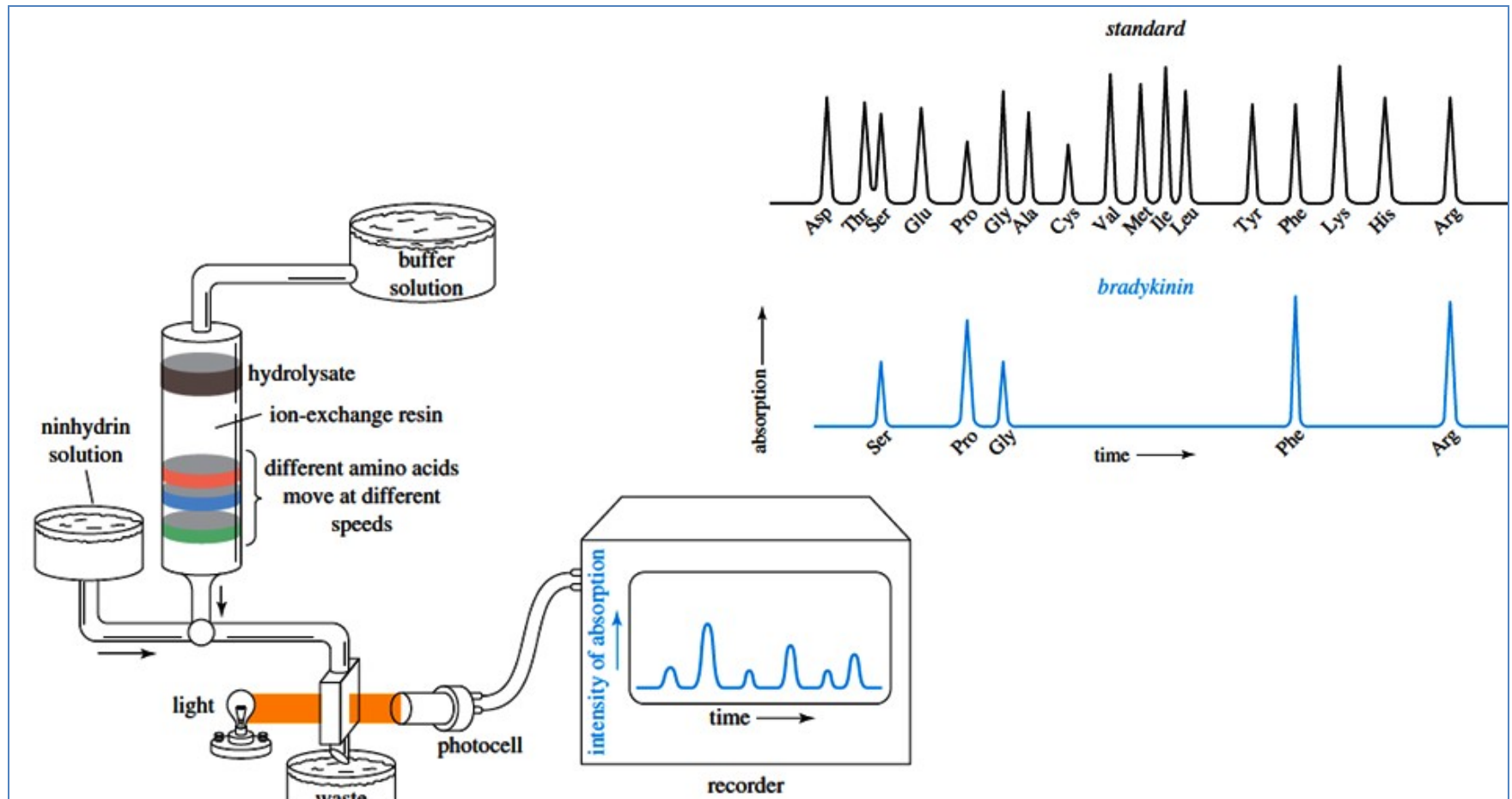


Ion exchange resin



– Amino acid analyzer

- Ninhydrin reaction with amino acids
- Comparison with standards



*Sequencing the peptide: Terminal
residue analysis and partial hydrolysis*

- **Sequencing the peptide:**
 - Terminal residue analysis
 - Partial hydrolysis

- **Terminal residue analysis**

*“Cleaving either end of the peptide (either the N terminus or the C terminus) **at a time** to determine its sequence”*

- **Methods**

- Sequencing from the N-Terminus
 - The Edman degradation and The Sanger method
- C-Terminal residue analysis
 - Enzymes

- **Sequencing from the N-Terminus**

- **Two methods**

- **The Edman degradation**
 - **Sanger N-terminal analysis**

- **The Edman degradation**

- *Cleave one amino acid at a time from the N terminus of the peptide chain*

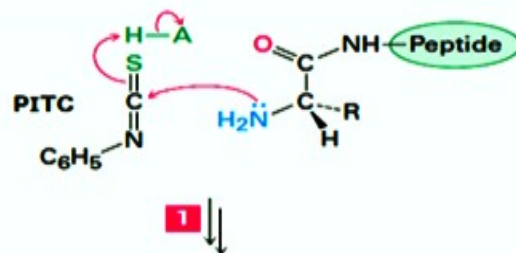
- ***Uses phenyl isothiocyanate (Edman's reagent) followed by mild acid hydrolysis***

- **Phenylthiohydantoin (PTH) derivative**

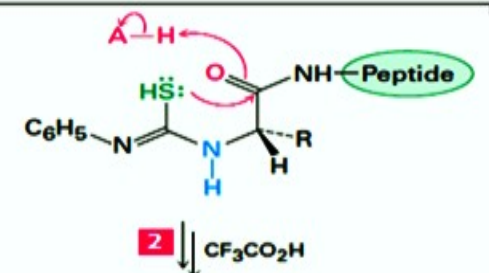
- *Identified by comparison of elution time of PTH derivatives of the 20 common amino acids*

- **Three steps**

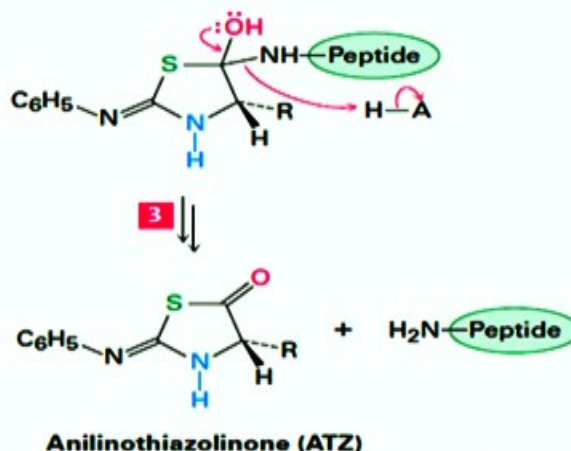
- 1** Nucleophilic addition of the peptide terminal amino group to phenyl isothiocyanate (PITC) gives an *N*-phenylthiourea derivative.



- 2** Acid-catalyzed cyclization of the phenylthiourea yields a tetrahedral intermediate ...



- 3** ... which expels the chain-shortened peptide and forms an anilinothiazolinone (ATZ) derivative.



- 4** The ATZ rearranges in the presence of aqueous acid to an isomeric *N*-phenylthiohydantoin (PTH) as the final product.



- Disadvantage:

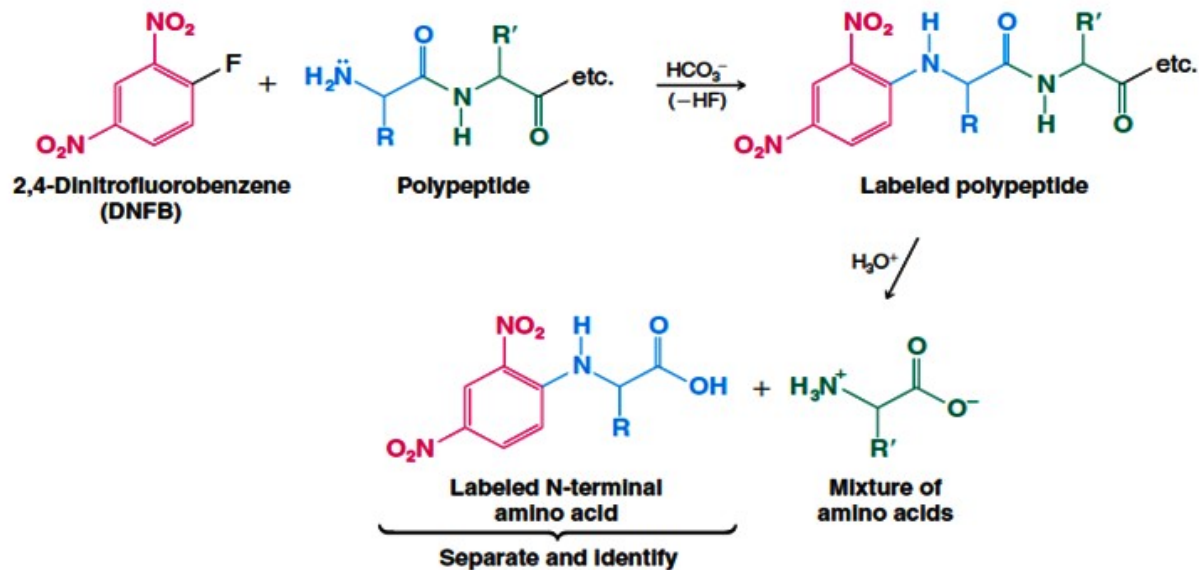
“Complete sequencing of large proteins by Edman degradation is impractical because of internal hydrolysis of the peptide, with loss of sample and accumulation of by-products.”

– Solution

- ***Large proteins via partial hydrolysis***

The Sanger method

- Sanger N-terminal analysis
 - 2,4-Dinitrofluorobenzene (DNFB) and 6M aqueous HCl
 - *Only for N-terminus determination. Why?*

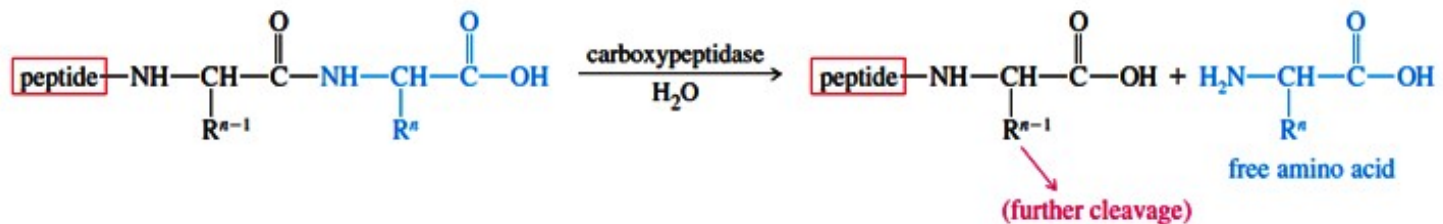


Terminal residue analysis

- **C-Terminal residue analysis**

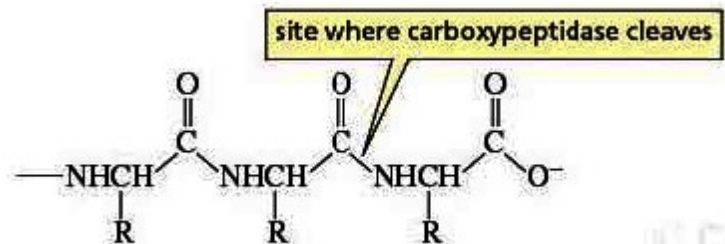
- Enzymes-**Peptidases**

- *Selectively hydrolyze specific peptide bonds*



- **E.g.**

- Carboxypeptidase B- C-side of Arg or Lys



Breaking the peptide into shorter chains:

Partial hydrolysis

Partial hydrolysis

- **Breaking the peptide into shorter chains:**
 - **Edman degradation: not used**
 - **> 30 amino acids**
 - **Solution**
 - **Fragment (Partial hydrolysis) analysis**
 - **Methods**
 - *Dilute acid with short reaction time*
 - *Enzymes*
 - *Chemical reagents*

- **Partial hydrolysis with dilute acid**
 - Generates a family of peptides cleaved in *random locations* and with *varying lengths*

- **Exercise**
 - Acid-catalyzed partial hydrolysis of oxytocin (after cleavage of the disulfide bridge) gives a mixture that includes the following peptides:
Ile-Gln-Asn-Cys Gln-Asn-Cys-Pro
Pro-Leu-Gly·NH₂ Cys-Tyr-Ile-Gln-Asn
Cys-Pro-Leu-Gly
Determine the sequence for oxytocin.

- **Solution**

Cys-Tyr-Ile-Gln-Asn

Ile-Gln-Asn-Cys

Gln-Asn-Cys-Pro

Cys-Pro-Leu-Gly

Pro-Leu-Gly·NH₂

Complete structure

Cys-Tyr-Ile-Gln-Asn-Cys-Pro-Leu-Gly·NH₂

• Enzymes (Peptidases)

– Endopeptidases

- ☐ Trypsin- hydrolyzes on the C-side of Arg and Lys
- ☐ Chymotrypsin- hydrolyzes on the C-side of amino acids that contain aromatic six-membered rings (Phe, Tyr, Trp)
- ☐ Elastase- hydrolyzes on the C-side of small amino acids (Gly and Ala)
- ☐ ***None of the exopeptidases or endopeptidases will catalyze the hydrolysis of an amide bond if proline is at the hydrolysis site***

Ala-Lys-Pro

trypsin will not cleave

Leu-Phe-Pro

chymotrypsin will not cleave

Pro-Phe-Val

chymotrypsin will cleave

- **Example**

- Trypsin: Ala-Phe-Lys-Ile-Met-Tyr-Gly-Arg-Ser-Trp-Leu-His

Ala-Phe-**Lys**~Ile-Met-Tyr-Gly-**Arg**~Ser-Trp-Leu-His



Ala-Phe-**Lys**

Ile-Met-Tyr-Gly-**Arg**

Ser-Trp-Leu-His

- **Chymotrypsin**

– Ala-Phe-Lys-Pro-Met-Tyr-Gly-Arg-Ser-Trp-Leu-His

Ala-Phe-Lys-Pro-Met-Tyr-Gly-Arg-Ser-Trp-Leu-His



Ala-Phe

Lys-Pro-Met-Tyr

Gly-Arg-Ser-Trp

Leu-His

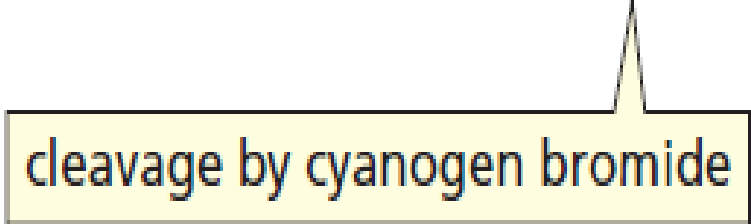
- **Chemical reagents**

- Cyanogen bromide

- C-side of a methionine residue

- *Sensitive to presence of Pro?????*

Ala-Lys-Phe-Gly-Lys-Trp-Ser-Arg-Met-Val-Arg-Tyr-Leu-His



cleavage by cyanogen bromide

Peptide synthesis

- **Peptide synthesis**

- **Total synthesis. Problems?**

- *Expensive*

- ✓ *Most peptides or proteins from natural source*

- E.g. Insulin from pork pancreas

- ✓ *Or biotechnological production*

- E.g. *Escherichia coli*

–Laboratory synthesis

- For confirmation of protein structure
- Enough supply of protein
- Preparation of altered peptides to compare their biological activity with the natural peptides

–Methods

1. Solution-phase method
2. Solid-phase method

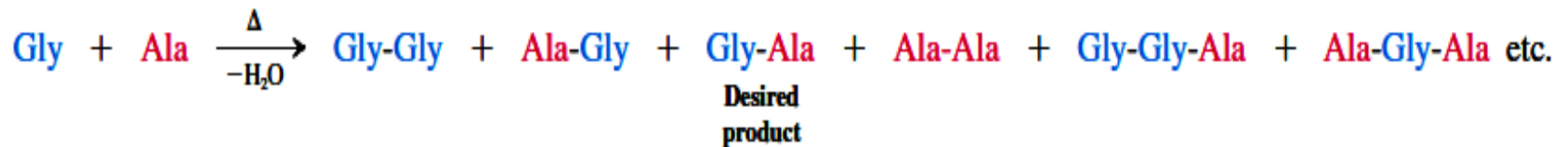
Solution-phase method

- **Peptide synthesis**

- Stepwise linking of amino acids

- Not an easy process. **Why?**

- ✓ Multifunctional nature of the amino acids



- **What shall we do?**

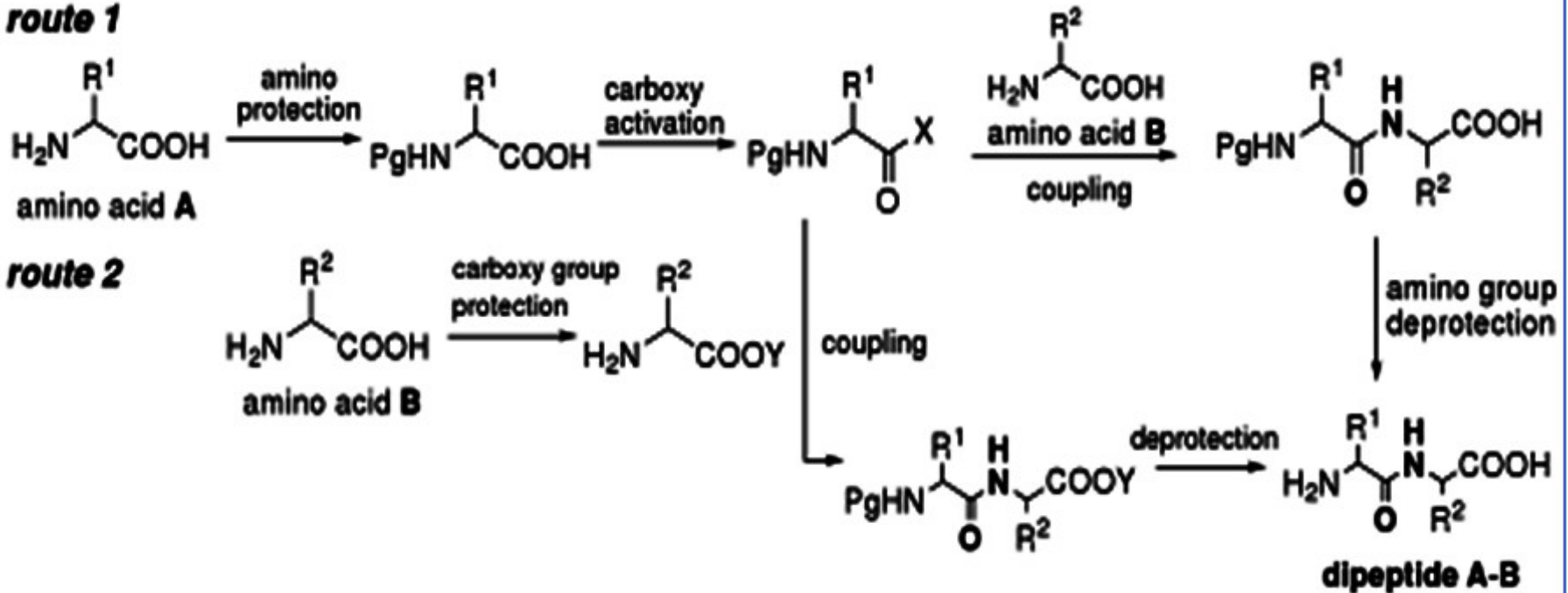
- **Solution:**

- ✓ Mask the reactivity of the groups on amino acids.

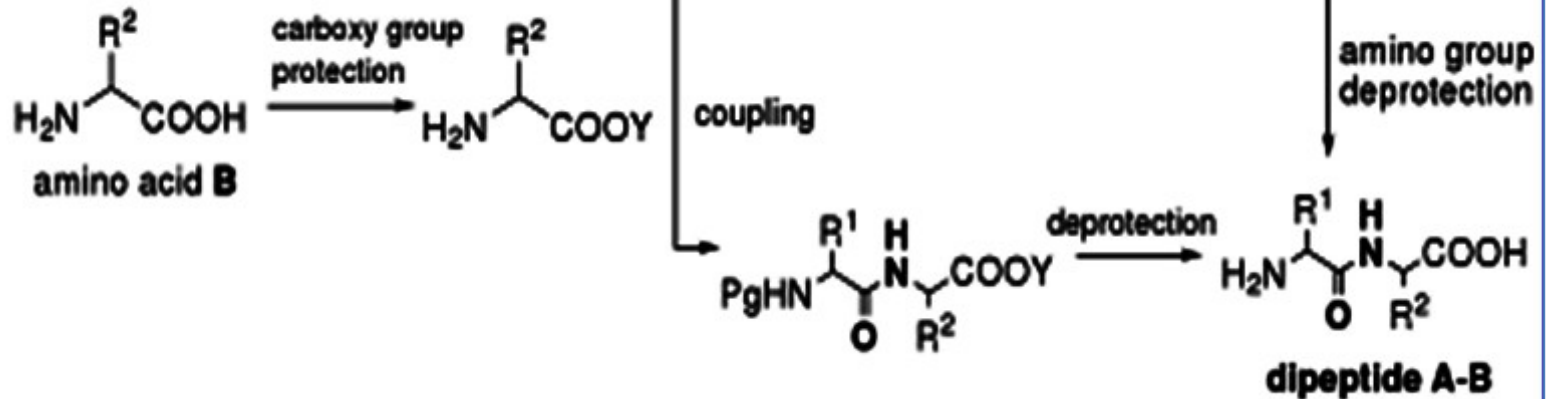
- **How?**

- ✓ **Protecting group**

route 1



route 2



Step-1: Amino Protection

Step-2: Activation of carboxyl group of N protected amino acid

Step-3: Coupling

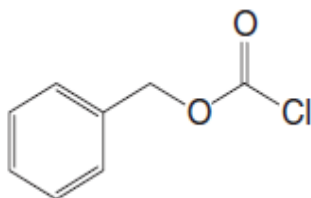
Step-4: Deprotection

Amino group protection and deprotection

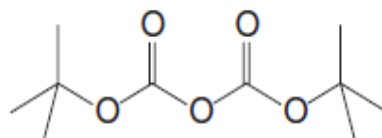
- **α -Amino group**

- Reduce its *nucleophilicity*
- Variety of protecting grouping available
 - Selection:

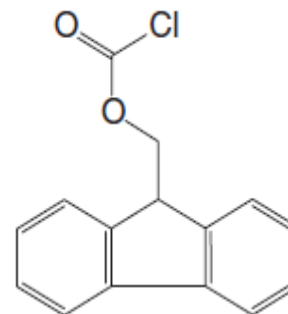
“Easy to remove without destruction of any peptide linkages that may have been built up”



Benzyl chloroformate



Di-*tert*-butyl dicarbonate
(Boc₂O)



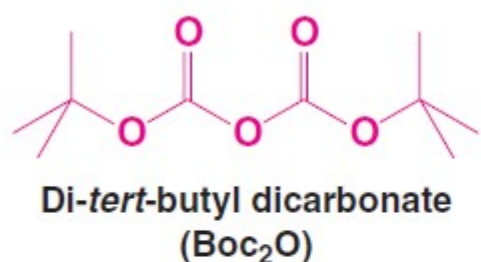
9-Fluorenylmethyl chloroformate

- ***t*-BOC or Boc in $\text{CF}_3\text{CO}_2\text{H}$ or acetic acid**

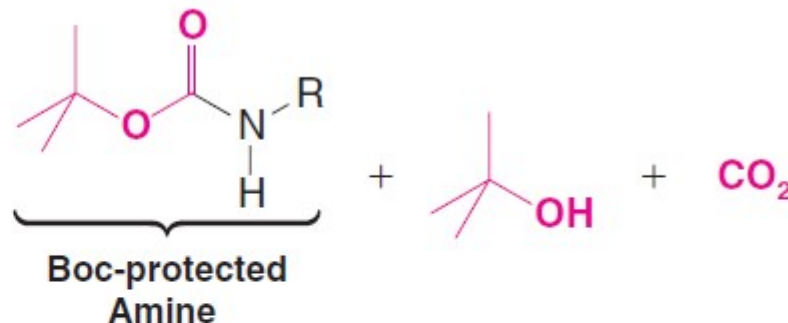
- Ideal

- Doesn't induce peptide bond breakage

tert-Butyloxycarbonyl (Boc) Group



Amino Group
Protection



Deprotection

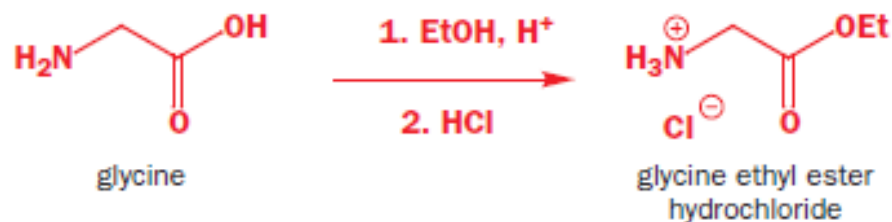
HCl in or $\text{CF}_3\text{CO}_2\text{H}$
acetic acid, 25°C



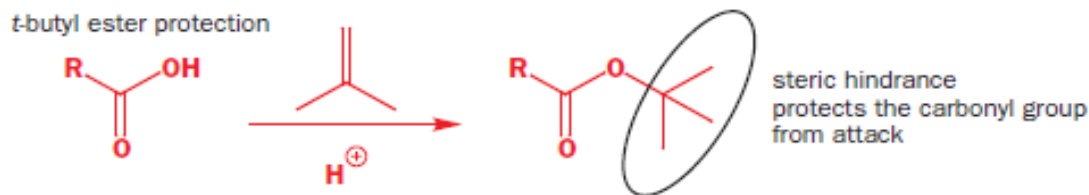
Carboxyl group protection and deprotection

- **Carboxyl groups**
 - Protected as *esters*

- E.g.

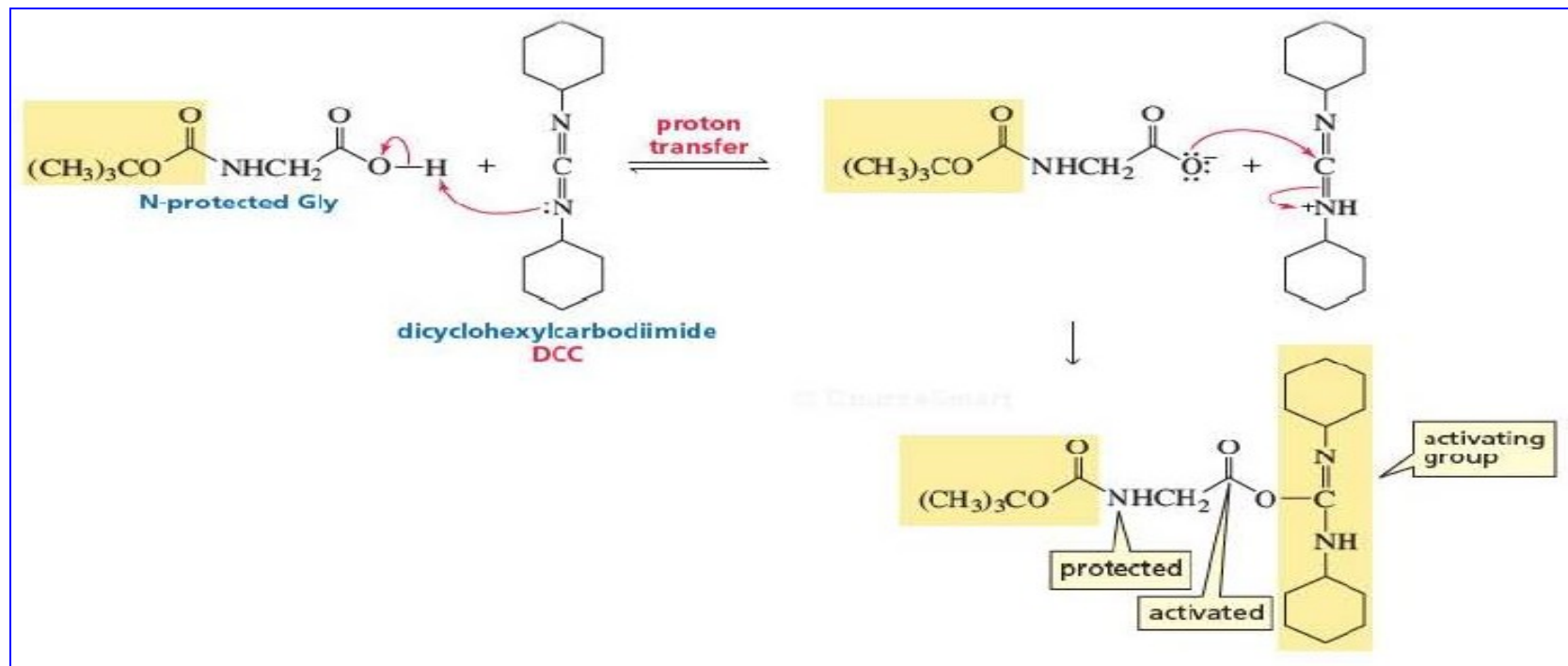


- Simple methyl and ethyl esters may pose problems
 - *React with such nucleophiles as amines*
 - Solution:
 - » Bulk groups



Activation of the carboxyl group

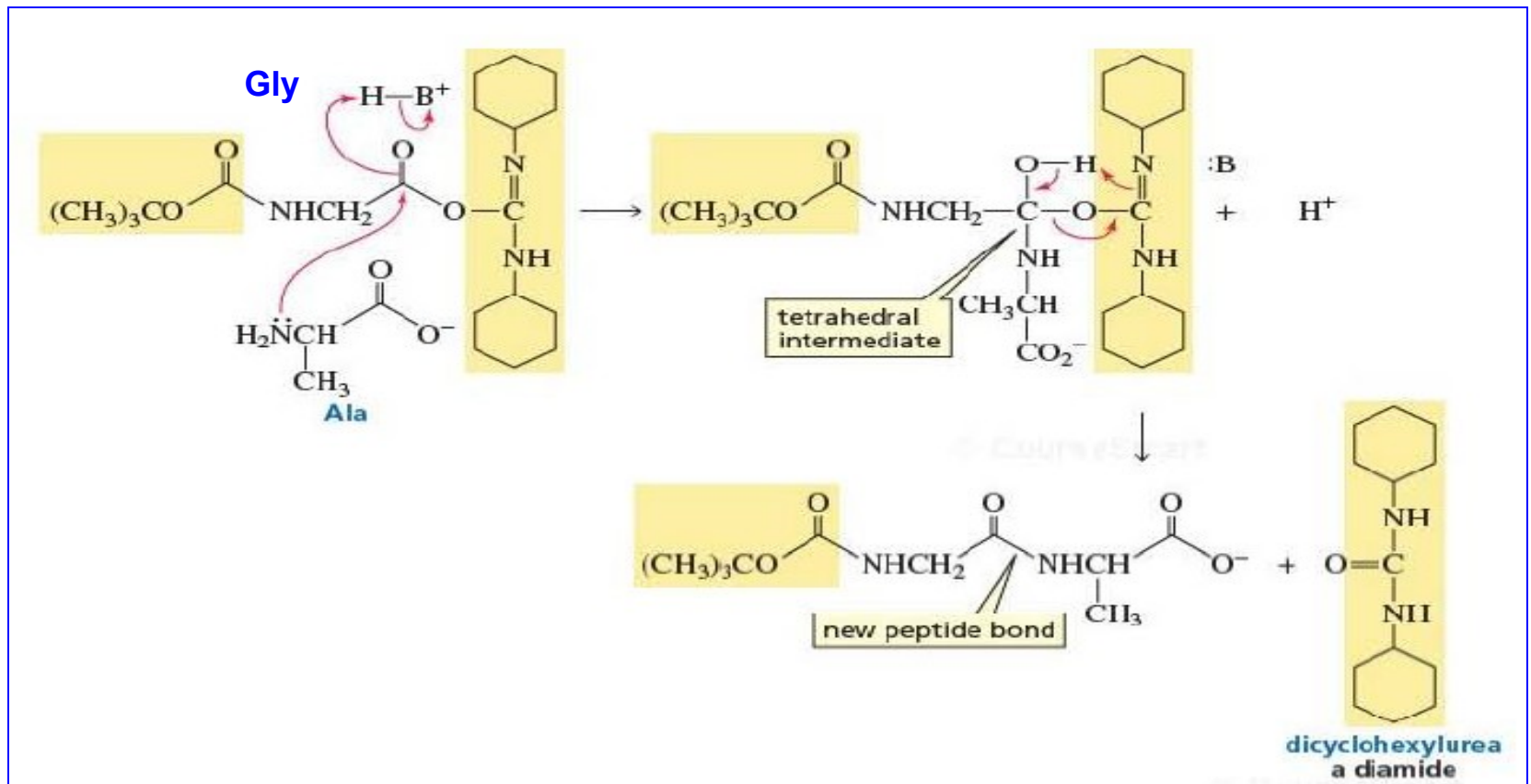
- **Activation of the carboxyl group. Why?**
 - Converting to acyl chlorides.
 - Any drawbacks?
 - **Dicyclohexylcarbodiimide (DCC)**
 - Creates good leaving group



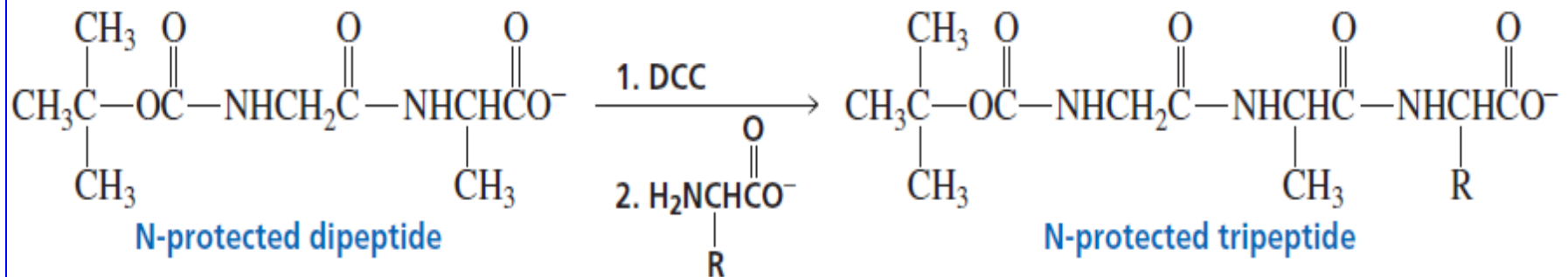
Formation of peptide

- **Formation of peptide**

- *After the amino acid has its N-terminal group protected and its C-terminal group activated*



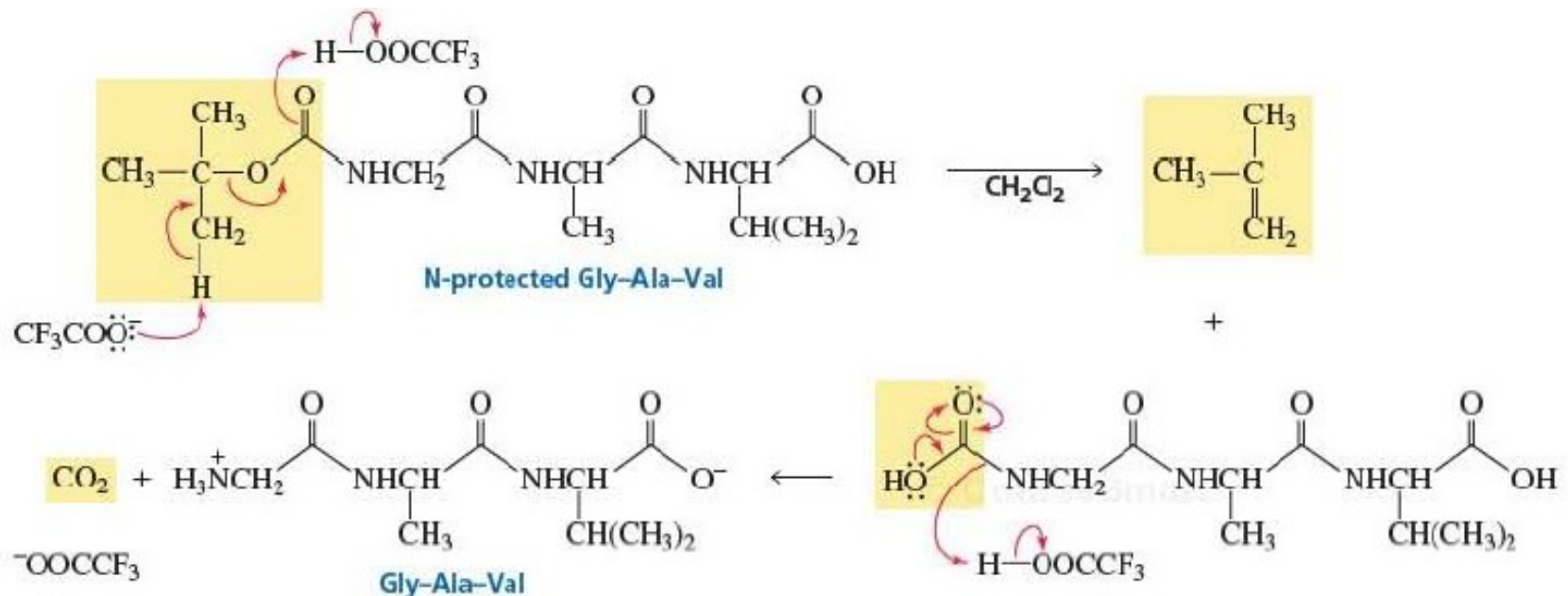
- Chain elongation- repetition of:
 - *Activating the carboxyl group*
 - *Adding a new amino acid*



*Deprotection of the amino
group*

- **t-BOC is an ideal protecting group. Why?**

- Removed by washing with trifluoroacetic acid and methylene chloride
 - Do not break any other covalent bonds.



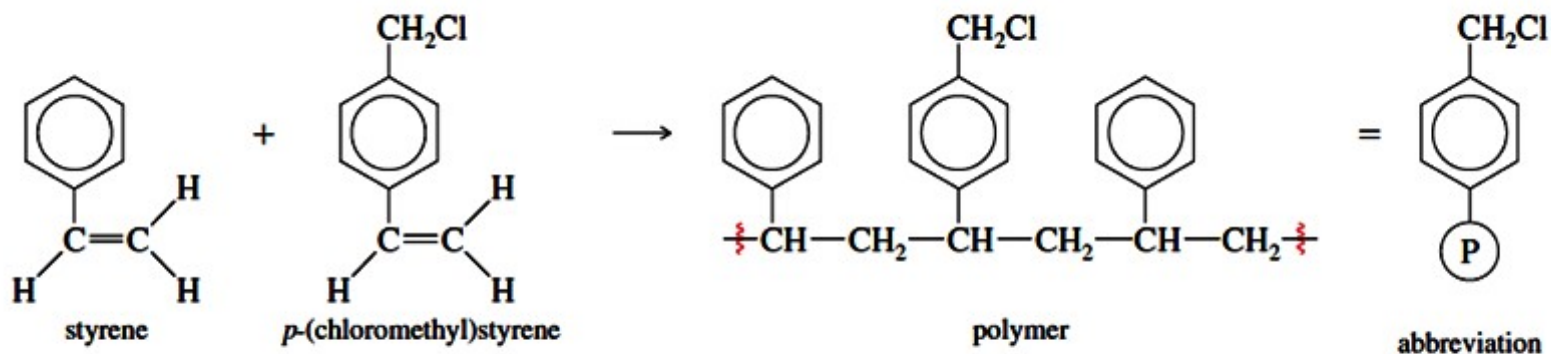
Solid-phase peptide synthesis (Merrifield method)

- **Solid-Phase Peptide Synthesis**

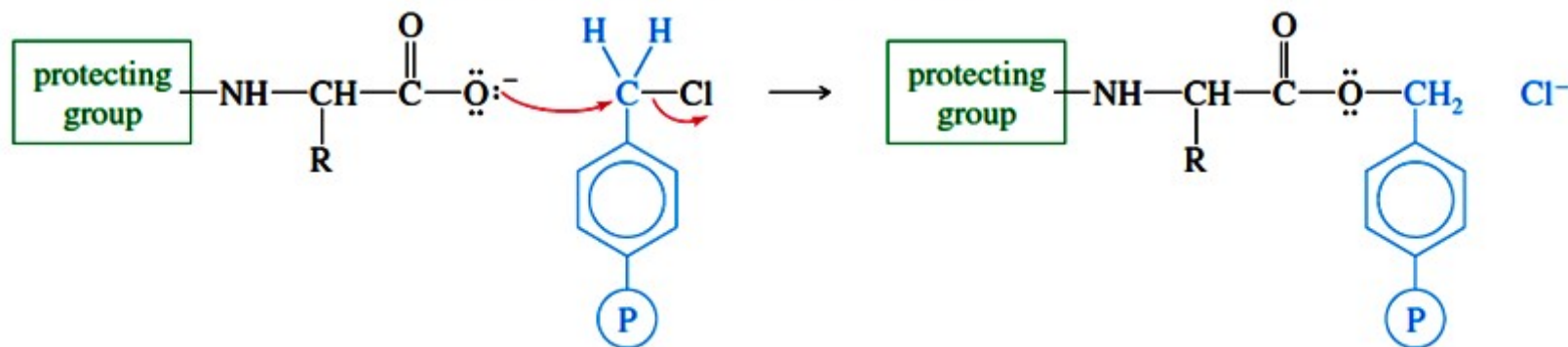
- Also called *Merrifield method*
- C-terminal amino acid is ***covalently attached to a solid support*** contained in a column
- Each ***N-terminal blocked amino acid*** is added one at a time
- ***Starts with the C terminus and going toward the N terminus***

- Solid support = *Merrifield resin*

Formation of the Merrifield resin

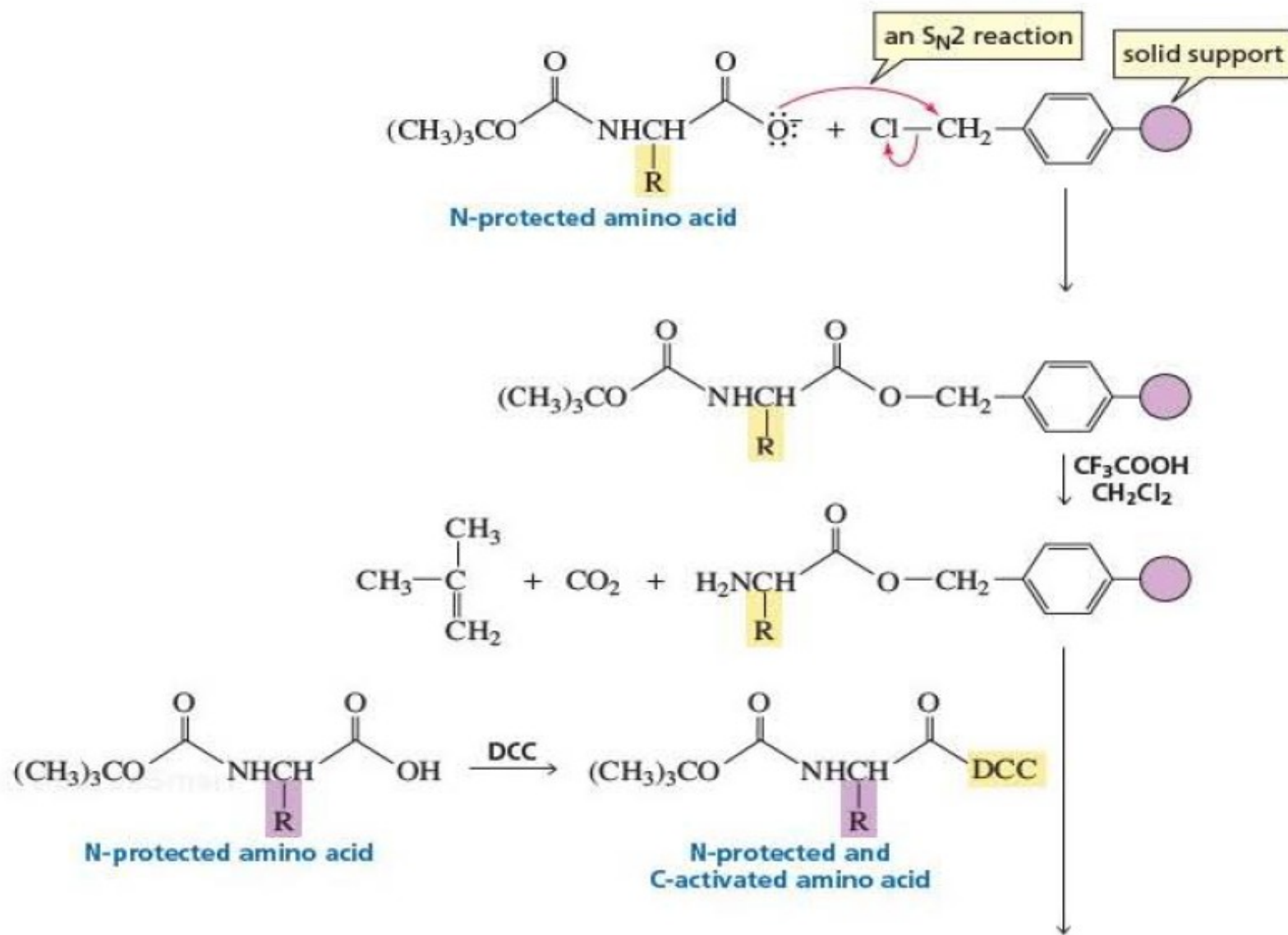


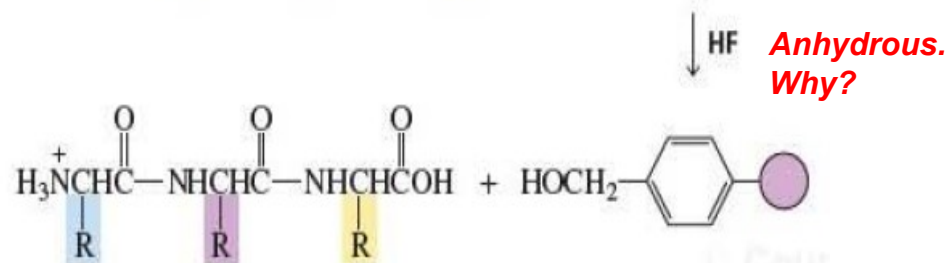
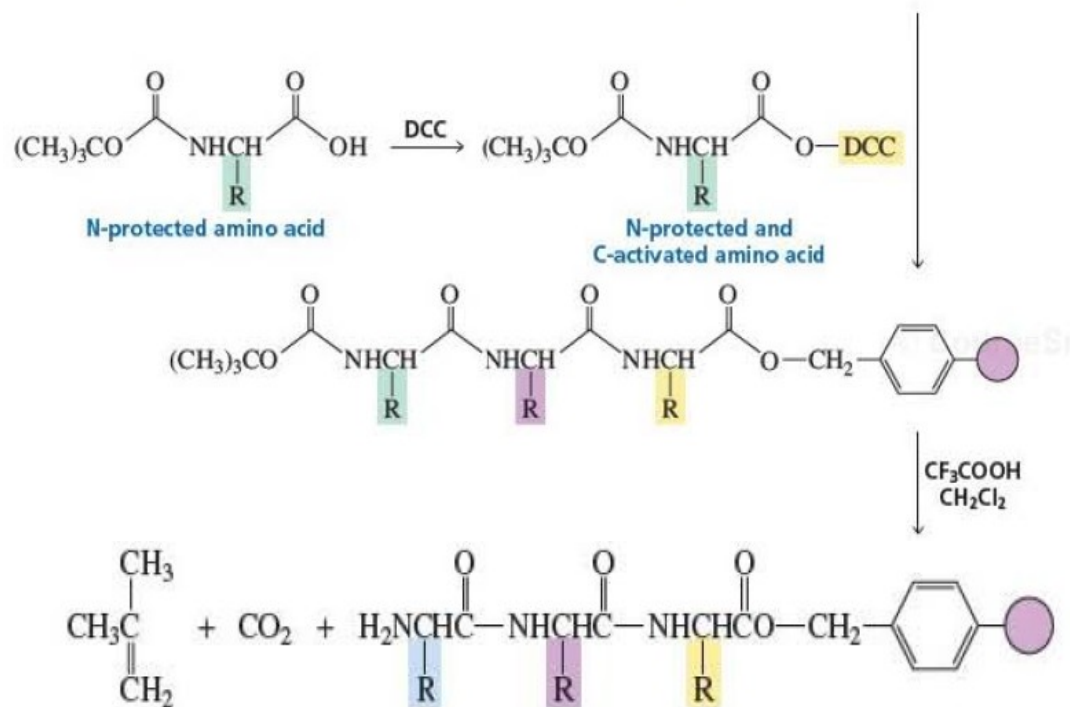
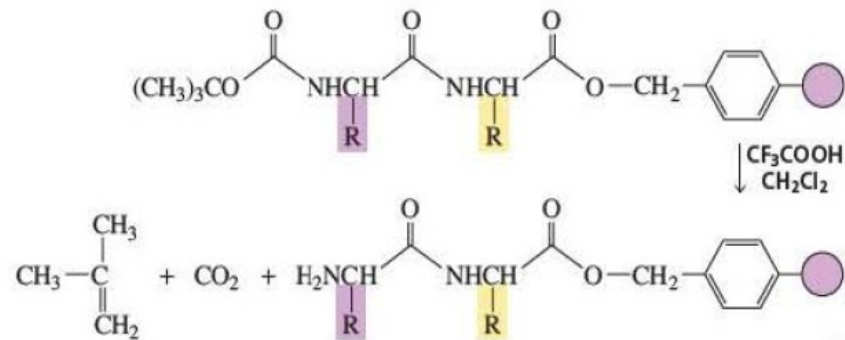
Attachment of the C-terminal amino acid



Merrifield automated solid-phase synthesis of a tripeptide

© Coursesmart





- Advantage of the Merrifield method
 - *high yields of purified product.*
 - *fast*
 - *automated*
 - E.g. bradykinin
 - 85% in 27 hrs

Specificity of peptide or protein cleavage

Reagent	Specificity
---------	-------------

Chemical

reagents

Edman's reagent	removes the N-terminal amino acid
-----------------	-----------------------------------

Cyanogen bromide	hydrolyzes on the C-side of Met
------------------	---------------------------------

Exopeptidases*

Carboxypeptidase A	removes the C-terminal amino acid (not Arg or Lys)
--------------------	--

Carboxypeptidase B	removes the C-terminal amino acid (only Arg or Lys)
--------------------	---

Endopeptidases*

Trypsin	hydrolyzes on the C-side of Arg and Lys
---------	---

Chymotrypsin	hydrolyzes on the C-side of amino acids that contain
--------------	--

aromatic six-membered rings (Phe, Tyr, Trp)

Elastase	hydrolyzes on the C-side of small amino acids (Gly and Ala)
----------	---

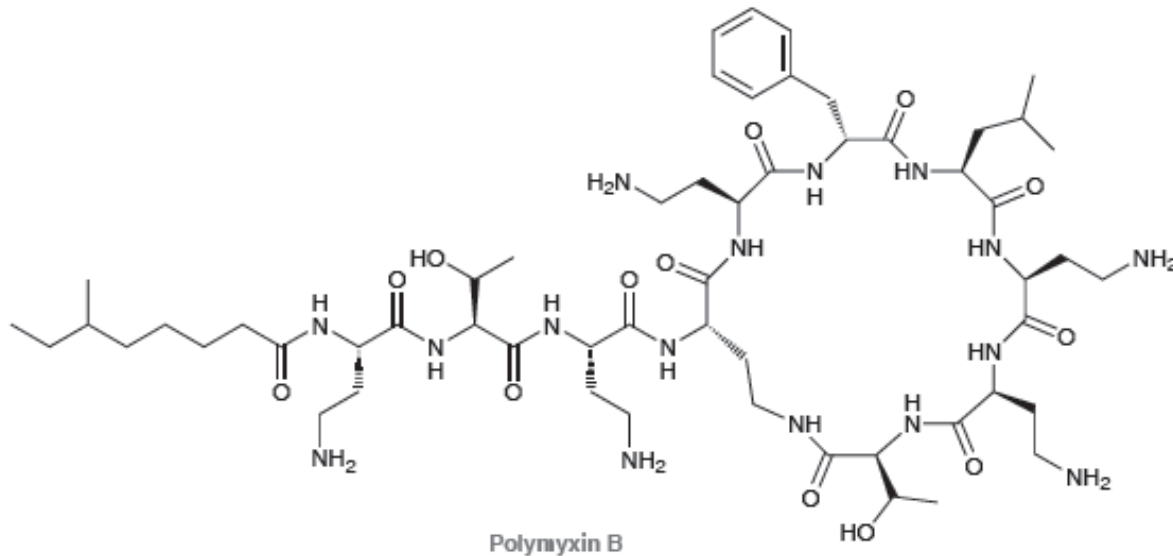
Clostripain	hydrolyzes on the C-side of Arg
-------------	---------------------------------

Thermolysin	amino end of Leu, Ile, Val
-------------	----------------------------

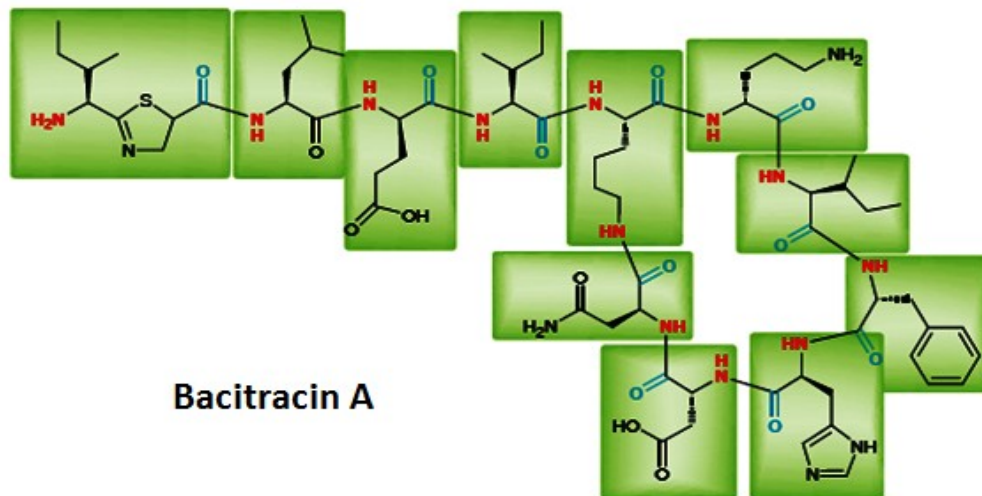
*Cleavage will not occur if Pro is on either side of the bond to be hydrolyzed.

Examples of peptide drugs

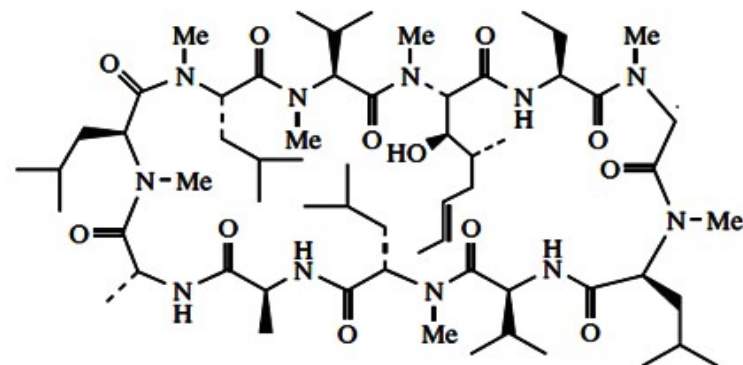
- Examples



Examples of peptide drugs



Bacitracin A



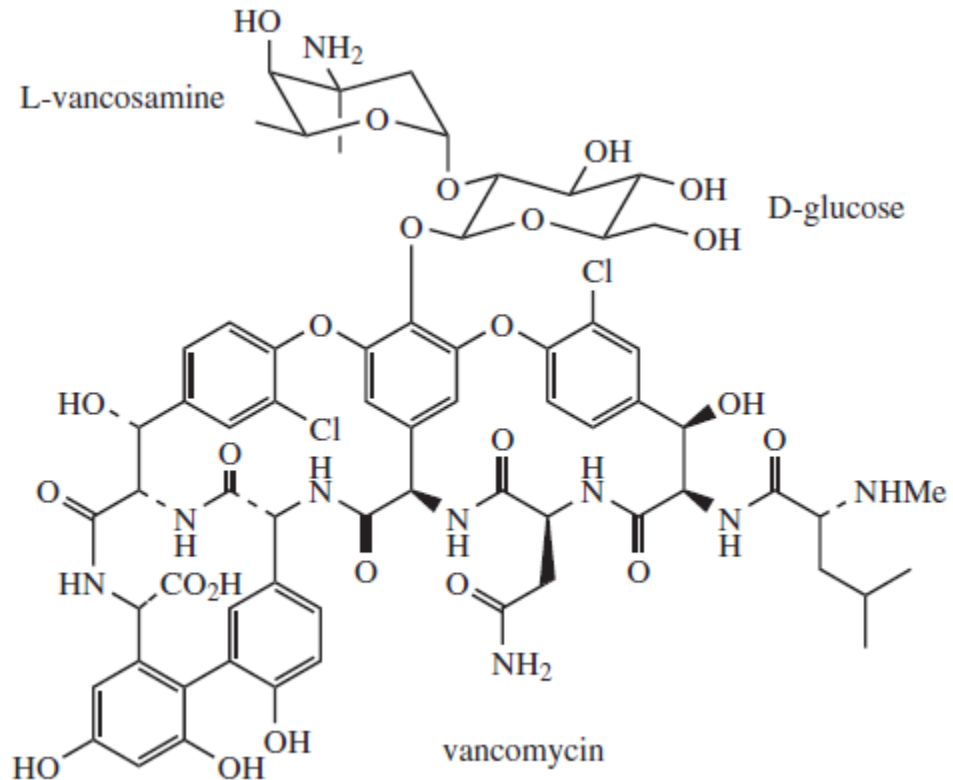
Ciclosporin (cyclosporin A)

Diagram illustrating the reversible oxidation of glutathione:

Left side: Two glutathione molecules (labeled **glutathione**) are shown. Each molecule consists of a glutamate residue (with a carboxylate group, COO^-), a cysteine residue (with a thiol group, HS-), and a glycine residue (with a carboxylate group, COO^-). The thiol group is highlighted in yellow.

Right side: The oxidized glutathione (labeled **oxidized glutathione**) is shown. It consists of two glutathione molecules linked by a disulfide bond (S-S). The disulfide bond is highlighted in yellow.

The reaction is reversible, indicated by the double-headed arrow. The forward reaction (oxidation) is labeled **oxidizing agent**, and the reverse reaction (reduction) is labeled **reducing agent**.



Extraction and purification of proteins

Structure determination of peptides/proteins

Step I. Purifying polypeptides

Why?

- Proteins (polymers) are easily contaminate
- proteins are found in a very small percentage

Note: Usually a protein is purified in a multi-step procedure and each step will take advantage of certain characteristics of the protein.

Strategies for purifying polypeptide

- **Various properties are manipulated:**
 - Size
 - Shape
 - Charge
 - Isoelectric point
 - Charge distribution
 - Hydrophobicity
 - Solubility
 - Ligand binding, etc

Method I: Modulating solubility

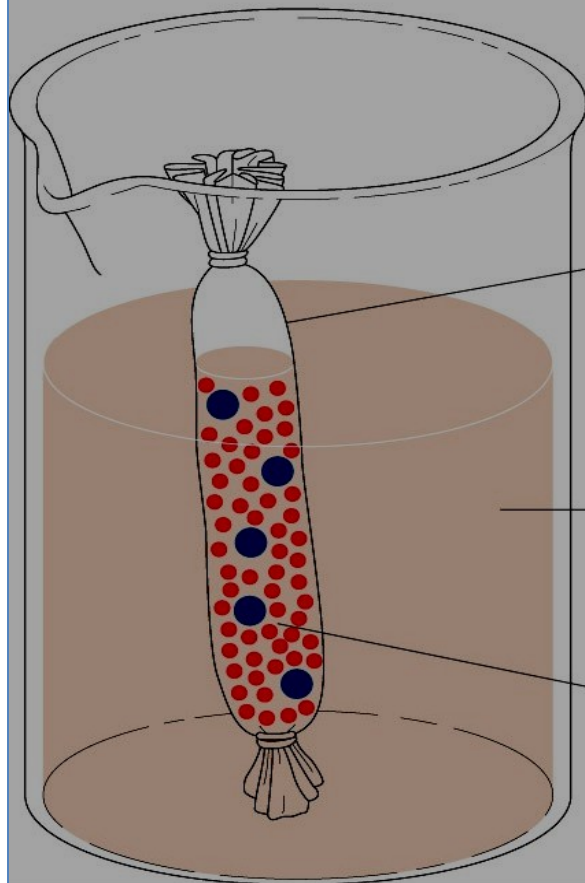
- **Precipitation at the pI**
 - **Salting in**
 - Many proteins are poorly soluble in pure water, but are much more soluble in salt-containing solutions.
 - Thus, lowering ionic strength can be used to precipitate certain proteins
 - **Salting out**
 - At very high concentrations (>1 M) of certain salts, proteins solubility is reduced due to a competition with the protein for interaction with water molecules.
 - **Ammonium sulfate precipitation**
 - common first step in protein purification
 - proteins precipitate at different concentrations of salt
 - dissolve the precipitate in a buffer and dialyze it to remove the salt.

Method II. Dialysis

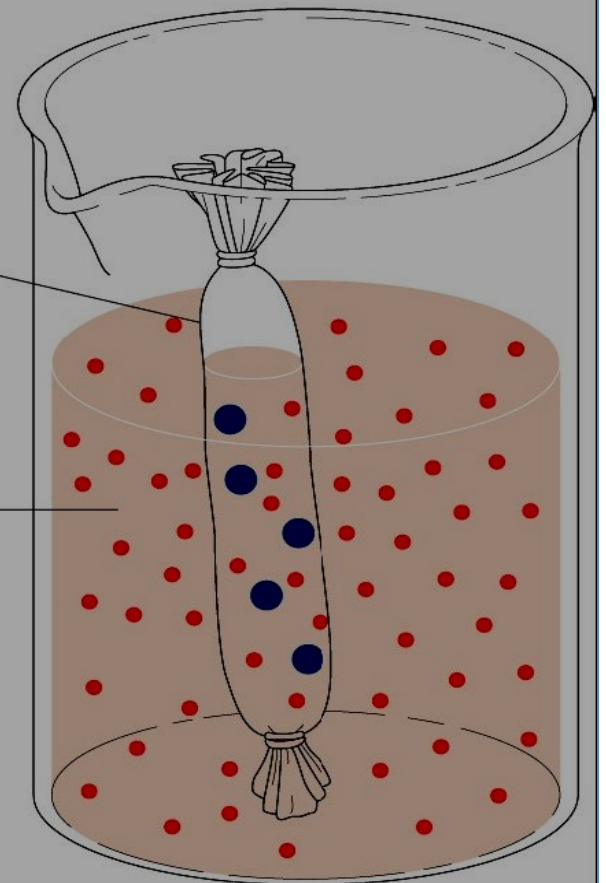
- The protein is put in a bag of cellulose membranes having small pores of controlled size.
- Proteins bigger than the pores are retained, while smaller molecules may diffuse out.
- As the volume of the buffer surrounding the bag is many times (100-1000x) the volume within the bag, the smaller molecules can be effectively removed after several changes of the outer buffer.

Dialysis

(a) At start of dialysis



(b) At equilibrium



Dialysis
membrane

Solvent

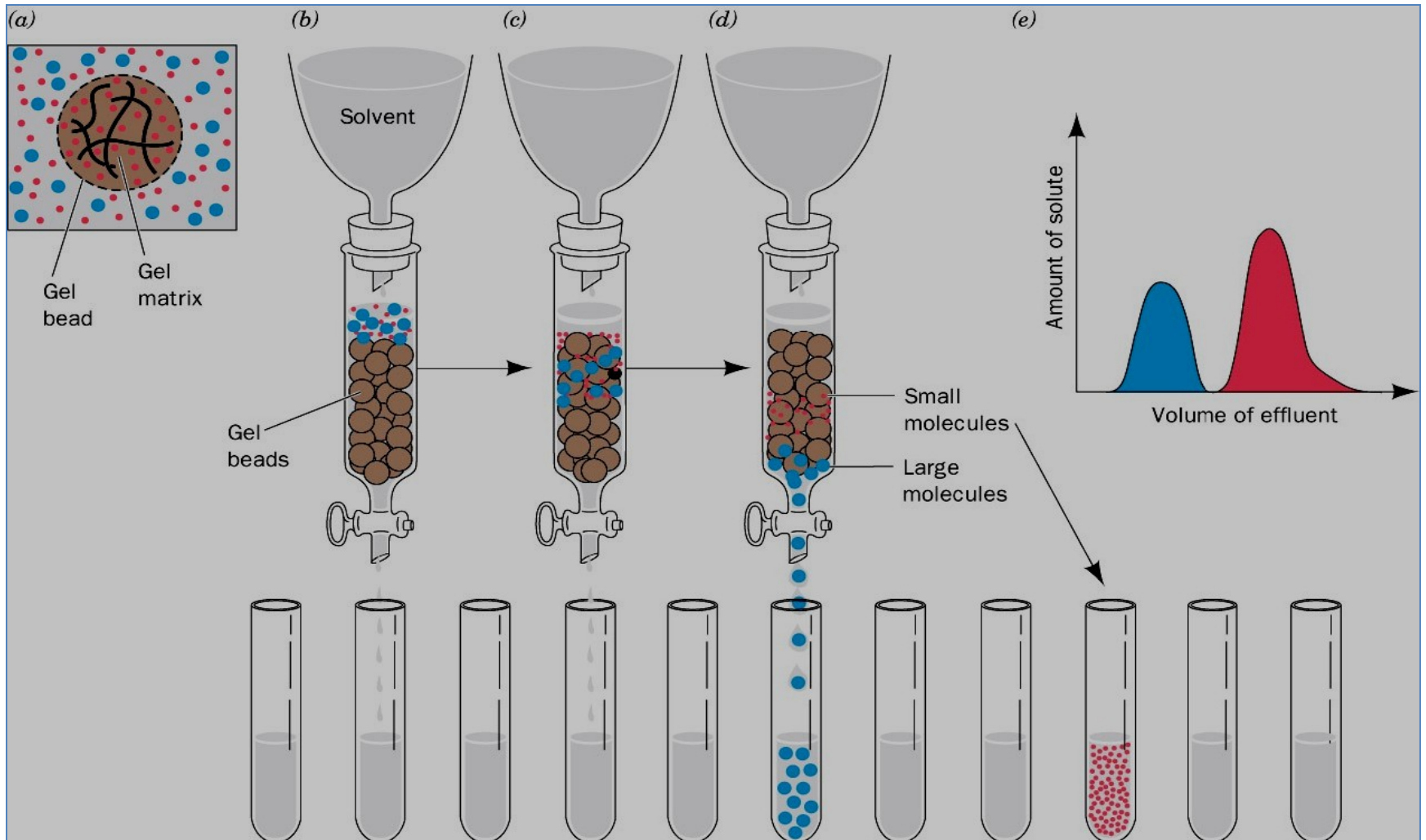
Concentrated
solution

Method III. Chromatography

a. Gel filtration (Size exclusion)

- The protein is applied to the top of a column consisting of porous beads made of a hydrated material, such as agarose, dextran, or poly- acrylamide.
- The pores of the beads are of a controlled size and this regulates which proteins can enter the beads.
- The larger proteins will be excluded from the beads and will flow through the column faster than the smaller molecules, which experience a much larger volume.
- In practice, there exists a range of pore sizes, and proteins are separated by their sizes (and shapes): the largest are eluted first, and the smallest elute last.

Gel filtration chromatography



Commonly Used Gel Filtration Materials

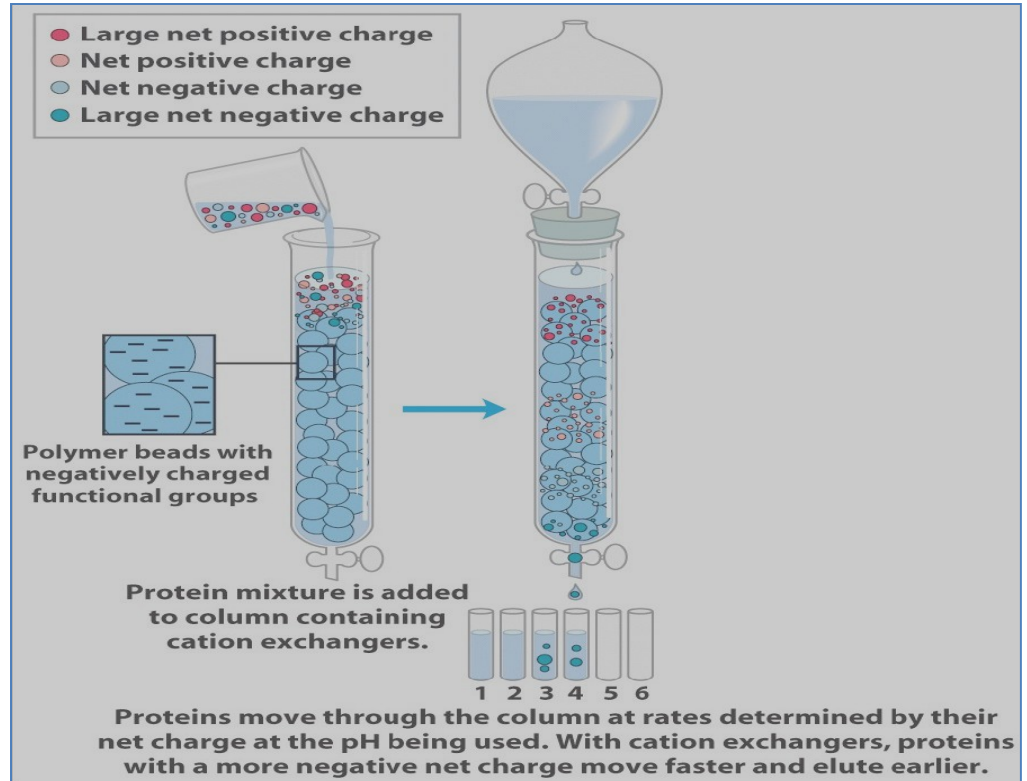
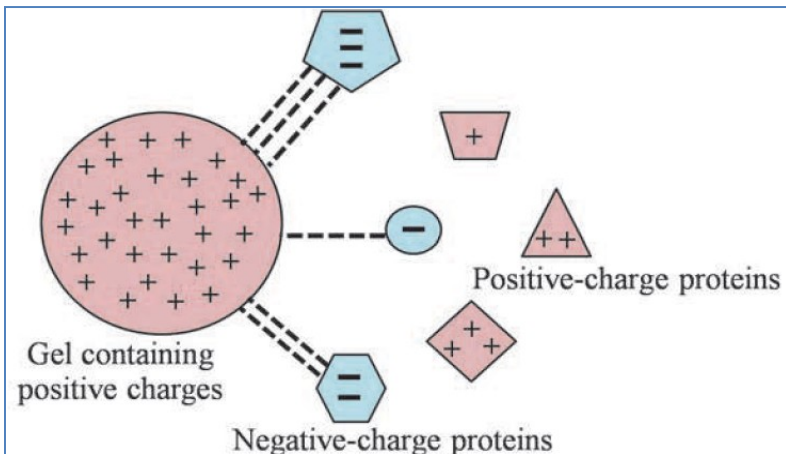
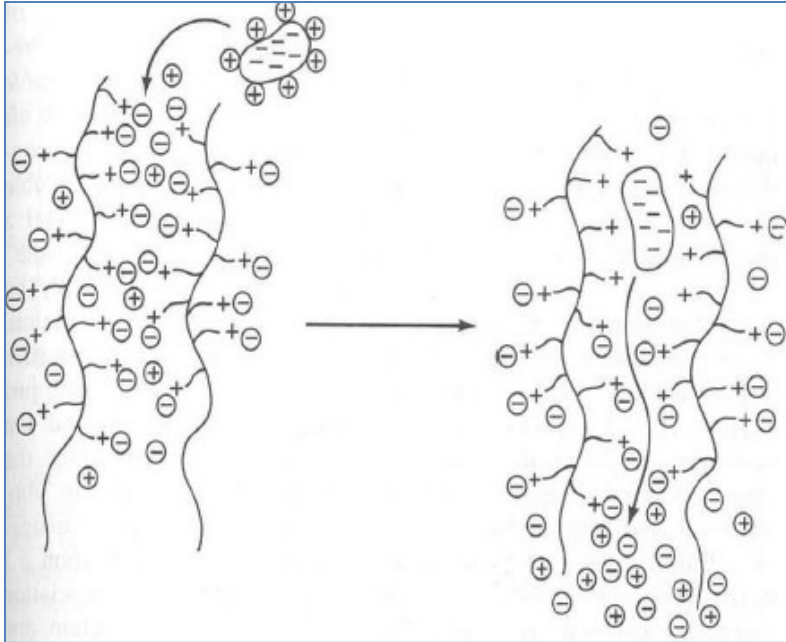
Name ^a	Type	Fractionation Range (kD)
Sephadex G-10	Dextran	0.05–0.7
Sephadex G-25	Dextran	1–5
Sephadex G-50	Dextran	1–30
Sephadex G-100	Dextran	4–150
Sephadex G-200	Dextran	5–600
Bio-Gel P-2	Polyacrylamide	0.1–1.8
Bio-Gel P-6	Polyacrylamide	1–6
Bio-Gel P-10	Polyacrylamide	1.5–20
Bio-Gel P-30	Polyacrylamide	2.4–40
Bio-Gel P-100	Polyacrylamide	5–100
Bio-Gel P-300	Polyacrylamide	60–400
Sepharose 6B	Agarose	10–4,000
Sepharose 4B	Agarose	60–20,000
Sepharose 2B	Agarose	70–40,000
Bio-Gel A-5	Agarose	10–5000
Bio-Gel A-50	Agarose	100–50,000
Bio-Gel A-150	Agarose	1000–150,000

^aSephadex and Sepharose gels are products of Amersham Pharmacia Biotech; Bio-Gel gels are manufactured by BioRad Laboratories.

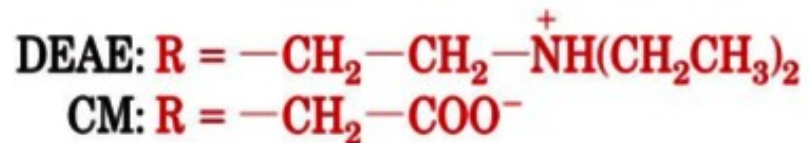
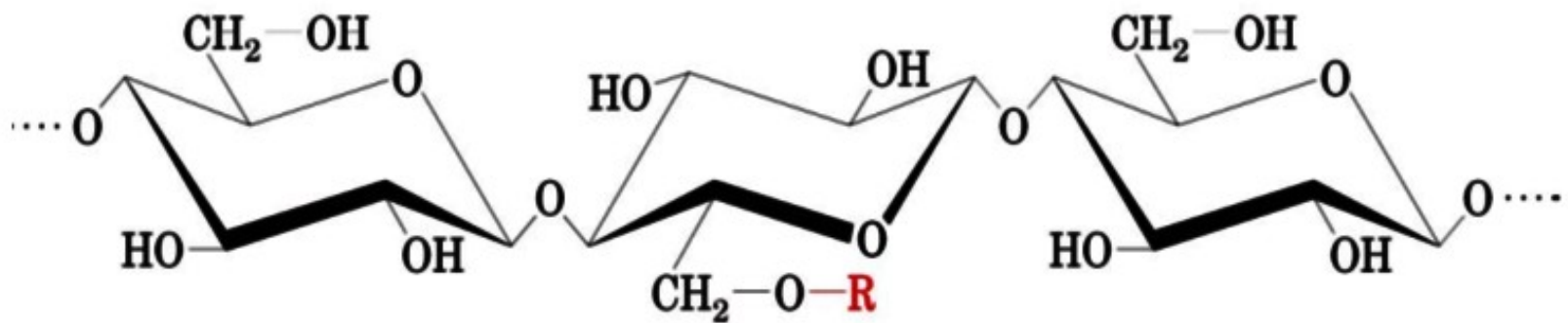
b. Ion exchange chromatography

- In most cases, a column is prepared with the ion exchanger, which is then equilibrated with the same buffer used to dissolve the protein.
- The exchanger and pH are chosen such that the protein will bind relatively tightly to the ion exchanger.
- The protein solution is loaded to the top and the column is washed with the buffer, removing all proteins with the opposite charge or low charge.
- The protein can be eluted from the column either by changing the pH or by increasing the salt concentration, which shields the charges and thus decreases their attraction.
- Elution is most often carried out by applying a salt gradient to the column.

Ion exchange chromatography



Molecular formulas of cellulose based ion exchangers.



Some common ion exchangers

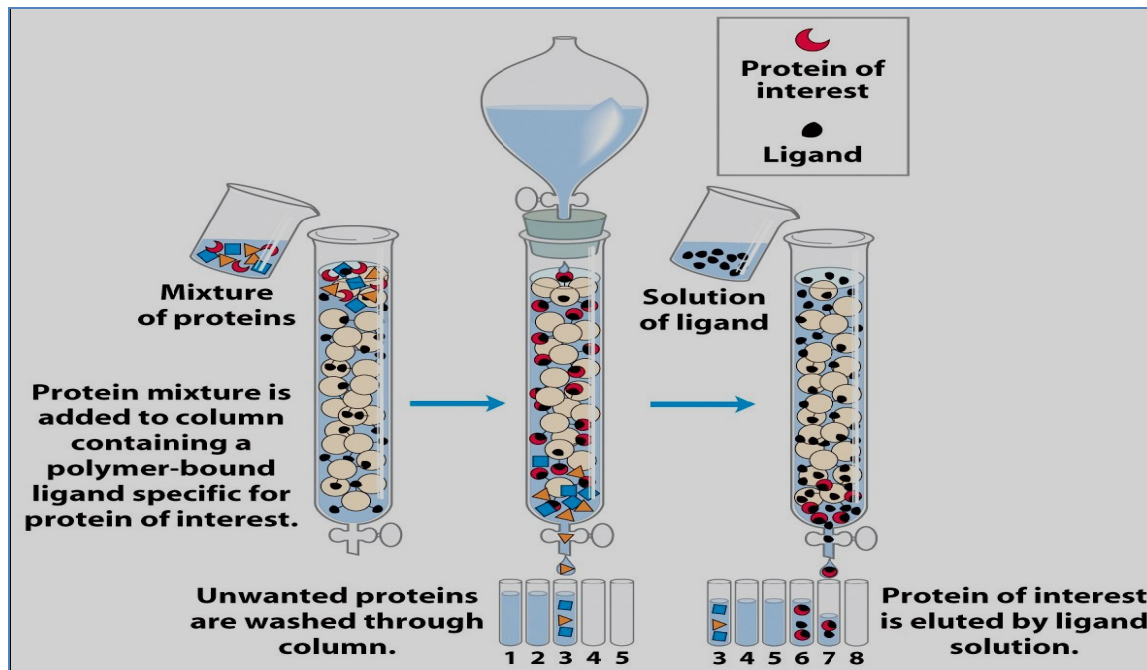
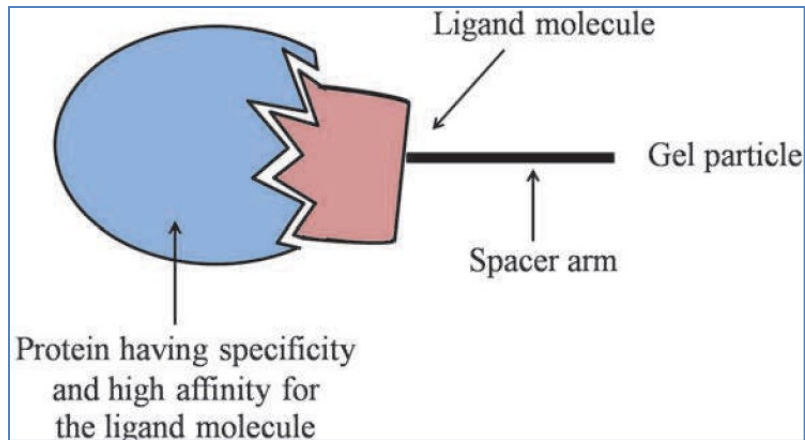
Name ^a	Type	Ionizable group	Remarks
DEAE-cellulose	Weakly basic	Diethylaminoethyl —CH ₂ CH ₂ N(C ₂ H ₅) ₂	Used to separate acidic and neutral proteins
CM-cellulose	Weakly acidic	Carboxymethyl —CH ₂ COOH	Used to separate basic and neutral proteins
P-cellulose	Strongly and weakly acidic	Phosphate —OPO ₃ H ₂	Dibasic; binds basic proteins strongly
Bio-Rex 70	Weakly acidic, polystyrene-based	Carboxylic acid —COOH	Used to separate basic proteins and amines
DEAE-Sephadex	Weakly basic cross-linked dextran gel	Diethylaminoethyl —CH ₂ CH ₂ N(C ₂ H ₅) ₂	Combined chromatography and gel filtration of acidic and neutral proteins
SP-Sepharose	Strongly acidic cross-linked agarose gel	Methyl sulfonate —CH ₂ SO ₃ H	Combined chromatography and gel filtration of basic proteins
CM Bio-Gel A	Weakly acidic cross-linked agarose gel	Carboxymethyl —CH ₂ COOH	Combined chromatography and gel filtration of basic and neutral proteins

^aSephadex and Sepharose gels are manufactured by Amersham Pharmacia Biotech, Piscataway, New Jersey; Bio-Rex resins and Bio-Gels are manufactured by BioRad Laboratories, Hercules, California.

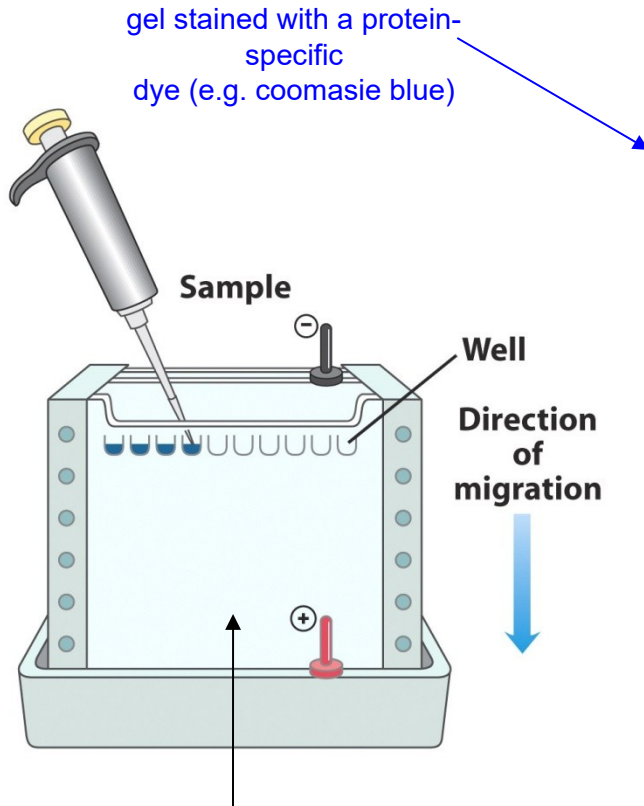
c. Affinity chromatography

- This technique takes advantage of the fact that many proteins specifically bind other molecules as part of their function.
- One can use this information to construct a column containing the ligand covalently attached to a matrix.
- Upon passing the protein solution through such a column, only the proteins that can bind the ligand will be retained on the column.
- Then the conditions can be adjusted to effect release from the ligand.
- Often this can be done simply by eluting with the soluble version of the ligand.

Affinity chromatography

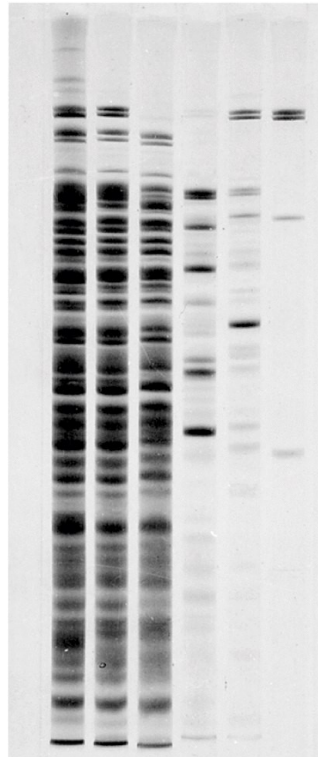


d. Electrophoresis



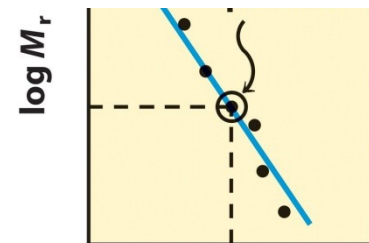
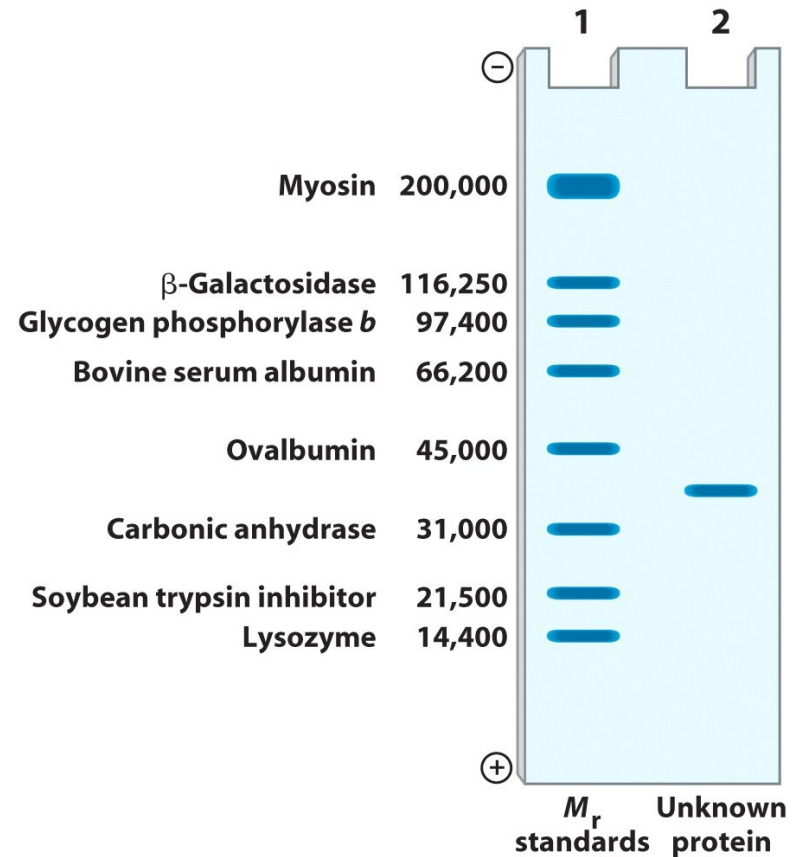
Cross-linked polymer
polyacrylamide

acts as a molecular sieve, slowing the migration of proteins approximately in proportion to their



Purification of RNA polymerase from *E. coli*

SDS-polyacrylamide



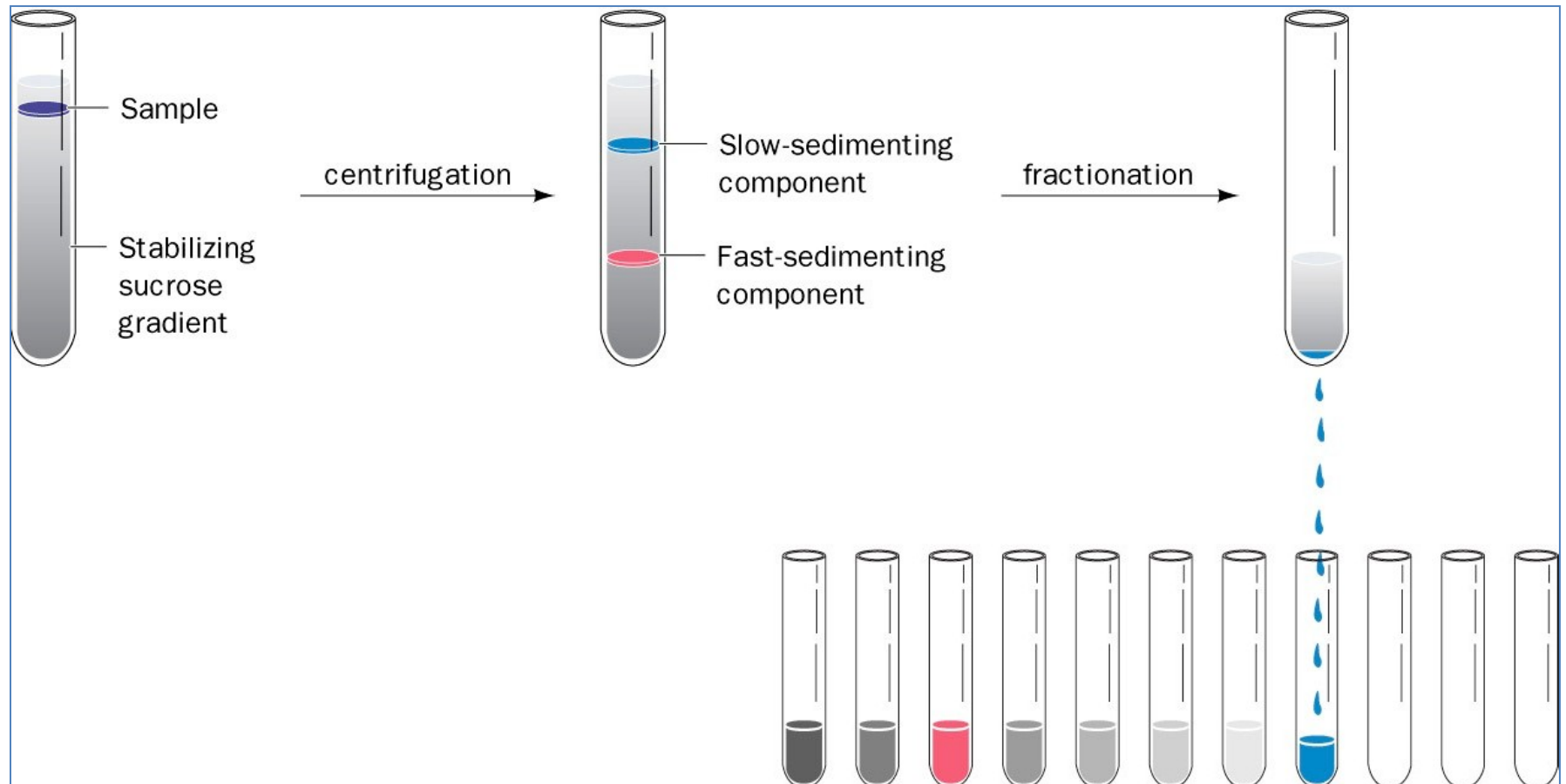
Relative migration

SDS



Na⁺

Method IV. Zonal Ultracentrifugation



Method V. Isoelectric focusing

- A pH gradient is set up by electrophoresing polyampholytes (300-600 Da oligomers bearing amino and carboxylate groups in varying ratios) in a gel tube.
- The more basic ones (cationic) will accumulate near the cathode and the more acidic (anionic) will accumulate near the anode, thereby establishing a continuous pH gradient.
- When a protein is applied to the gel, it will migrate toward the anode if it is negatively charged or toward the cathode if positively charged, until it reaches the region corresponding to its pI, where it will stop.
- Proteins are often denatured in 6M urea, which does not change the protein's charge (unlike SDS).

Isoelectric focusing

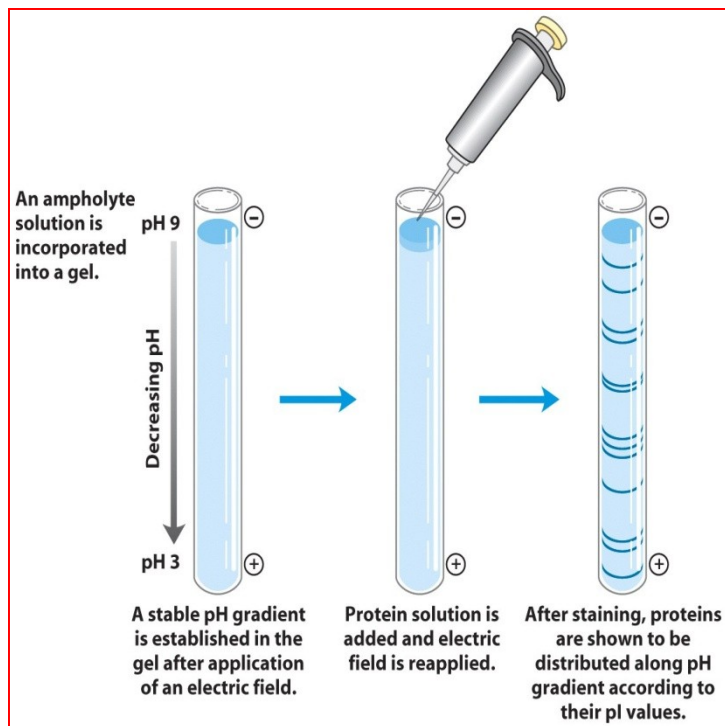
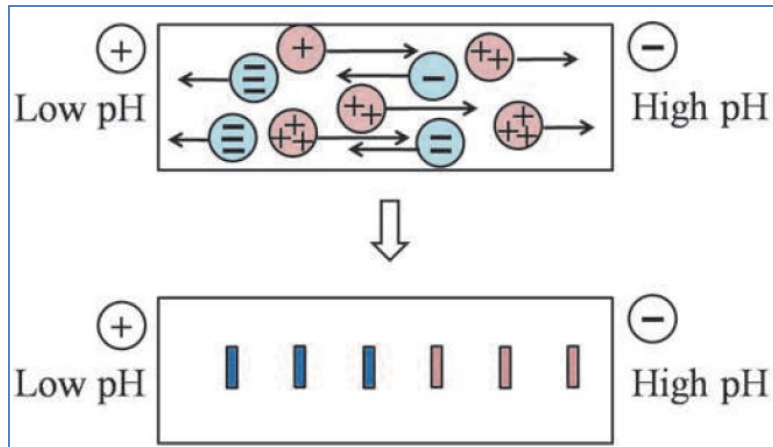
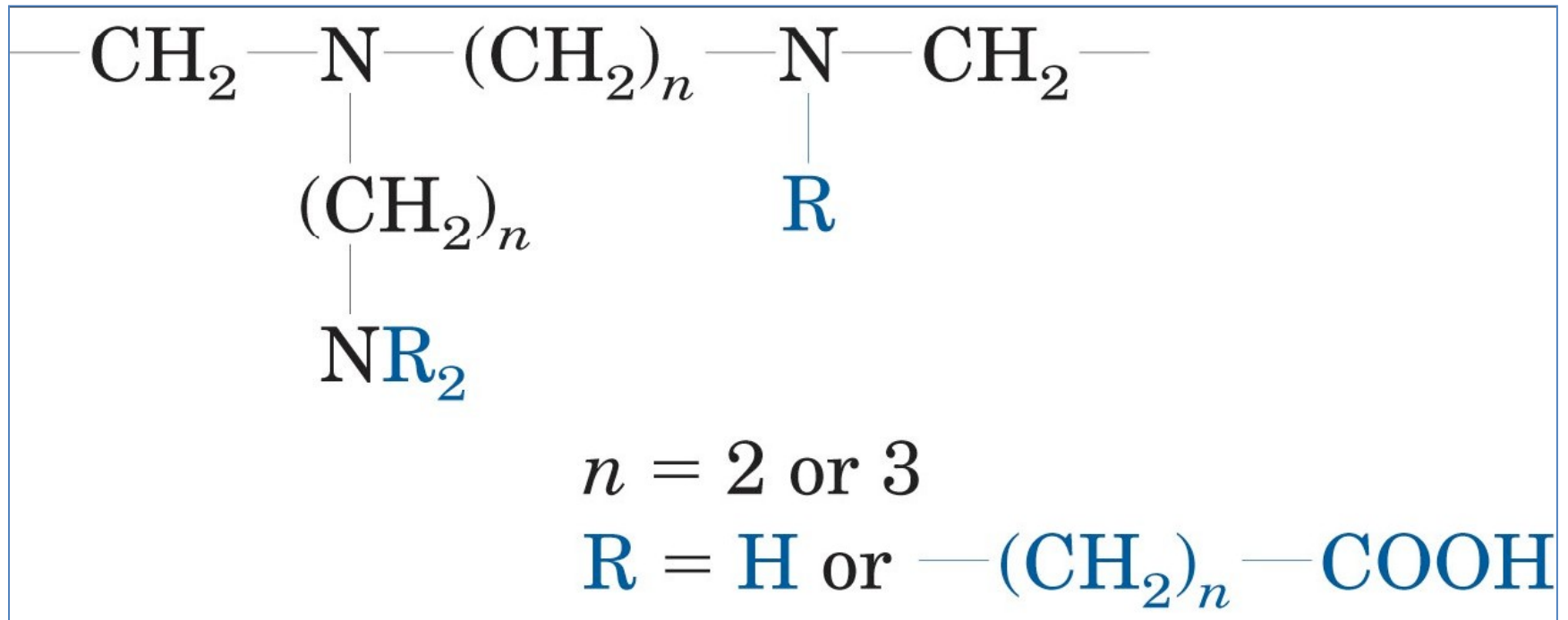


TABLE 3-6 The Isoelectric Points of Some Proteins

<i>Protein</i>	<i>pI</i>
Pepsin	<1.0
Egg albumin	4.6
Serum albumin	4.9
Urease	5.0
β -Lactoglobulin	5.2
Hemoglobin	6.8
Myoglobin	7.0
Chymotrypsinogen	9.5
Cytochrome c	10.7
Lysozyme	11.0

- A protein with a pI of 5.2 is introduced near the cathode end, which has pH of 9.5 (in this gel).
- It is then negatively charged and will migrate toward the anode.
- As it migrates in this direction, the pH will steadily decrease, and the amount of negative charge carried by the protein will also decrease, as more and more ionizable groups become protonated.
- Thus it will slow down, until it hits the region where $pH = 5.2$, where it will experience no further force.
 - If it diffuses in either direction, it will pick up charge and the electric field will force it back.
- This results in the protein being concentrated (focused) in a very narrow region of the gel.

General formula of the ampholytes
used in isoelectric focusing.



Protein Purification: Practical Aspects

- All cells contain proteases – enzymes that catalyze the hydrolysis of peptide bonds.
- Upon breaking cells, these are released into the extract, where they can degrade the protein you want to purify.
- In order to inhibit proteolysis, protein purification is usually carried out in the cold (on ice or in a cold room) in the presence of protease inhibitors (small molecules that inhibit specific proteases).

☐ Requirements

- ✓ Cooling system
- ✓ 0-4° C
- ✓ Buffer solution
- ✓ Sometimes protease inhibitors

Protein structure

Proteins

- **Proteins**, from the Greek *proteios*, meaning first, are a class of organic compounds which are present in and vital to every living cell.
 - In the form of skin, hair, callus, cartilage, muscles, tendons and ligaments, proteins hold together, protect, and provide structure to the body of a multicelled organism.
 - In the form of enzymes, hormones, antibodies, and globulins, they catalyze, regulate, and protect the body chemistry.
 - In the form of hemoglobin, myoglobin and various lipoproteins, they effect the transport of oxygen and other substances within an organism.

- The disease *kwashiorkor* is an extreme form of protein deficiency.
- Protein based antibiotics and vaccines help to fight disease,
- We warm and protect our bodies with clothing and shoes that are often protein in nature (e.g. wool, silk and leather).
- The deadly properties of protein toxins and venoms is less widely appreciated.
 - Botulinum toxin A, from *Clostridium botulinum*, is regarded as the most powerful poison known. Based on toxicology studies, a teaspoon of this toxin would be sufficient to kill a fifth of the world's population.
 - The toxins produced by tetanus and diphtheria microorganisms are nearly as poisonous.
 - A list of highly toxic proteins or peptides would also include the venoms of many snakes, and ricin, the toxic protein found in castor beans.

- **Proteins**

- Proteins are polymers of amino acids

- *Molecular weights*

- *10,000 to more than one million*

- **Functions**

- Catalysis

- Transport

- Contraction

- Protection

- Structure

- Metabolic regulation

- **Two Types**

- ***Based on 3D-shape:***

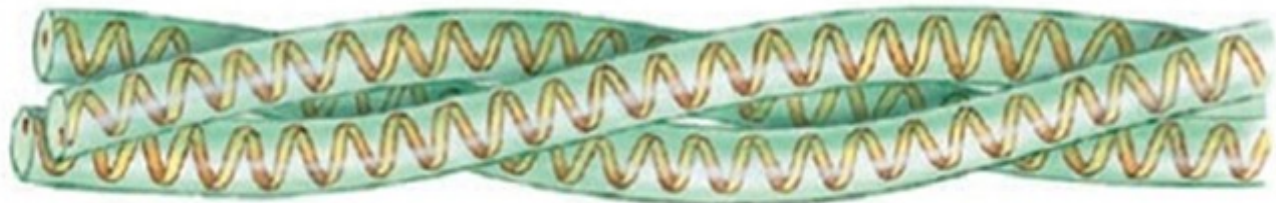
- (1). Fibrous proteins**

- Insoluble in water

- Long and thread-like

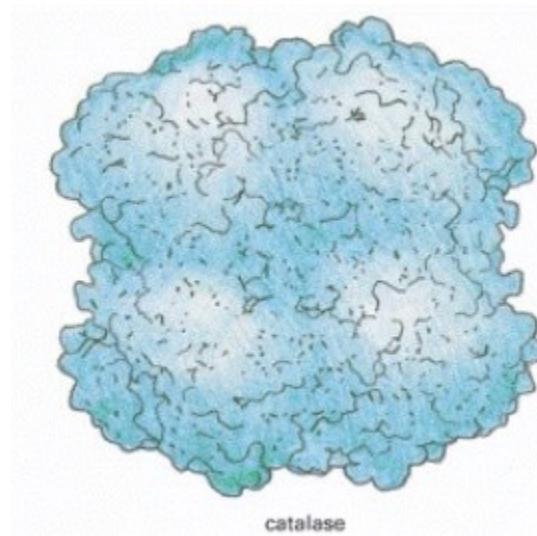
- » Tend to lie side by side to form fibers

- Strong intermolecular forces



(2). Globular proteins

- Soluble in water or aqueous solutions of acids, bases, or salts
- Spheroidal shapes
 - » Areas of contact between molecules are small
 - Intermolecular forces are comparatively weak



- **Four levels of protein structure**

1. Primary structure
2. Secondary structure
3. Tertiary structure
4. Quaternary structure

Protein Structure

The **primary** structure of a protein is simply the amino acid sequence.

The **secondary** structure of a protein describes how segments of the peptide backbone orient into a regular pattern (local structure).

Examples are α -helix and pleated sheet

The **tertiary** structure describes how the entire protein molecule coils into an overall three-dimensional shape.

The **quaternary** structure describes how different protein molecules come together to yield large aggregate structures

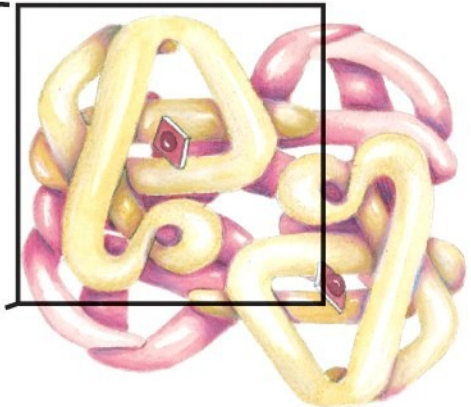
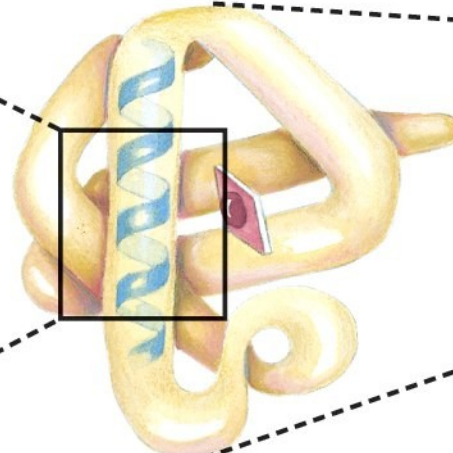
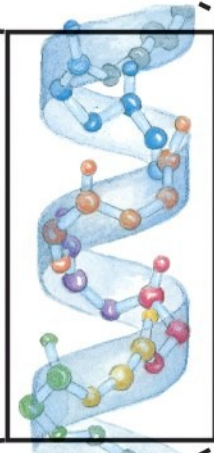
There are several levels of protein structure

Primary structure

Secondary structure

Tertiary structure

Quaternary structure



Amino acid residues

α Helix

Polypeptide chain

Assembled subunits

Particularly stable arrangements of amino acid residues giving rise to

recurring structural patterns

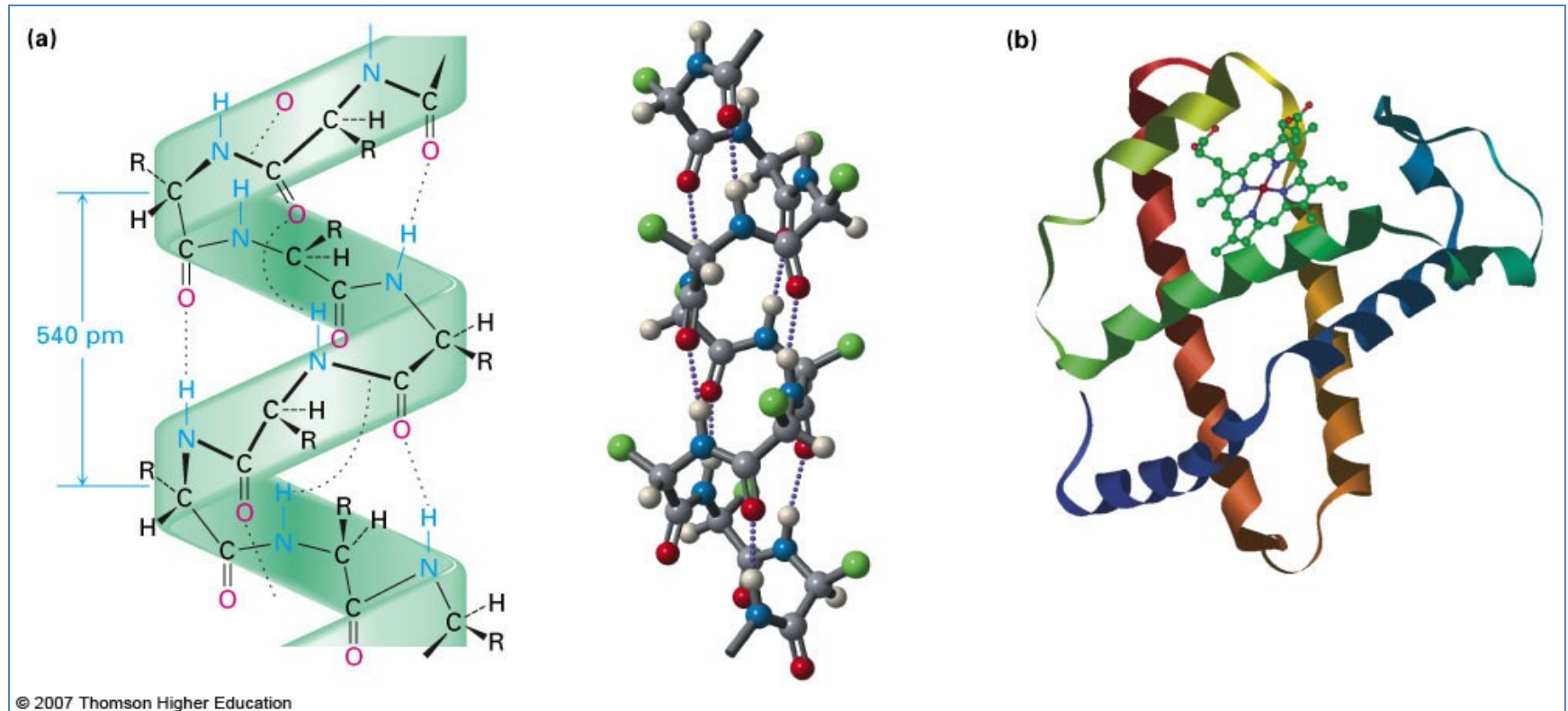
All aspects of the 3-D folding of a polypeptide

*Multisubunit proteins
Arrangement is space of polypeptide subunits*

Includes disulfide bonds

α -Helix

α -helix stabilized by H-bonds between amide N—H groups and C=O groups four residues away α -helical segments in their chains

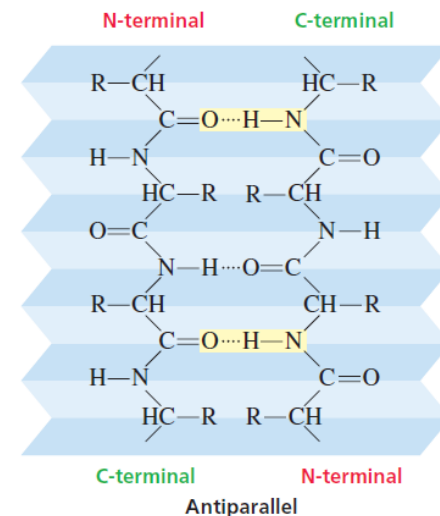
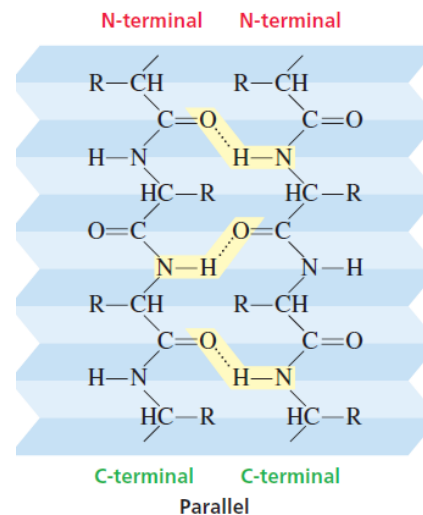
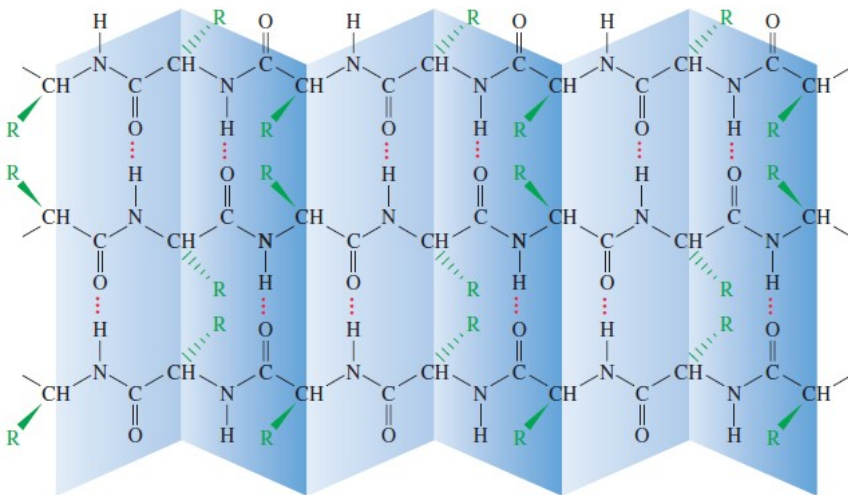
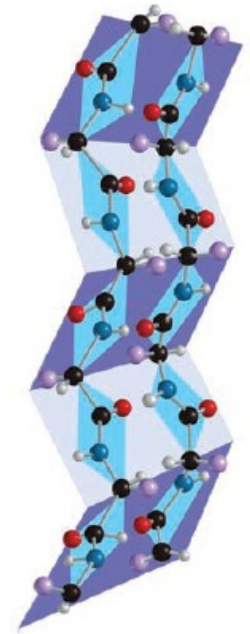


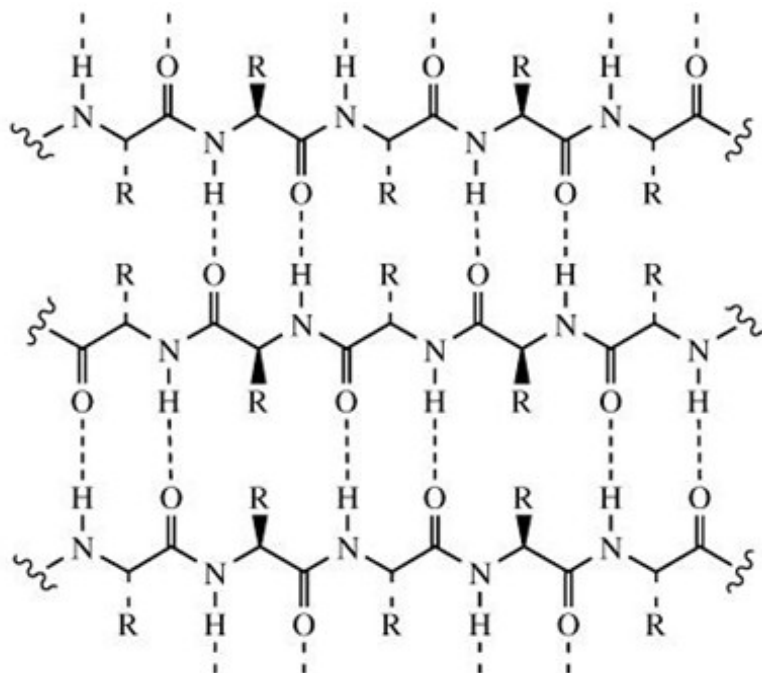
- **β -Pleated sheet**

- *Peptide chain is fully extended rather than coiled*

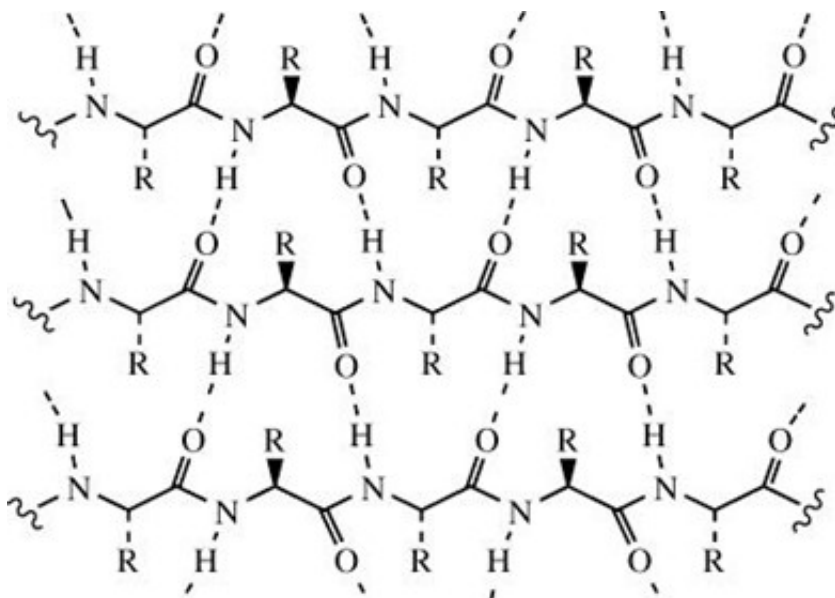
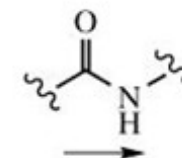
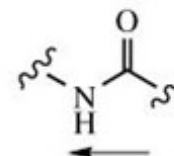
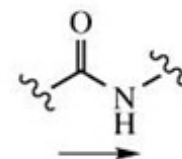
- » *hydrogen bonds occur between residues in adjacent chains*

- Two-dimensional sheet that is pleated (creased)

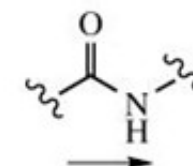
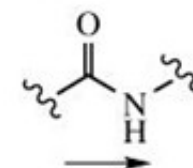
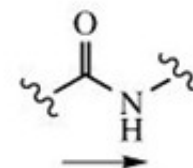




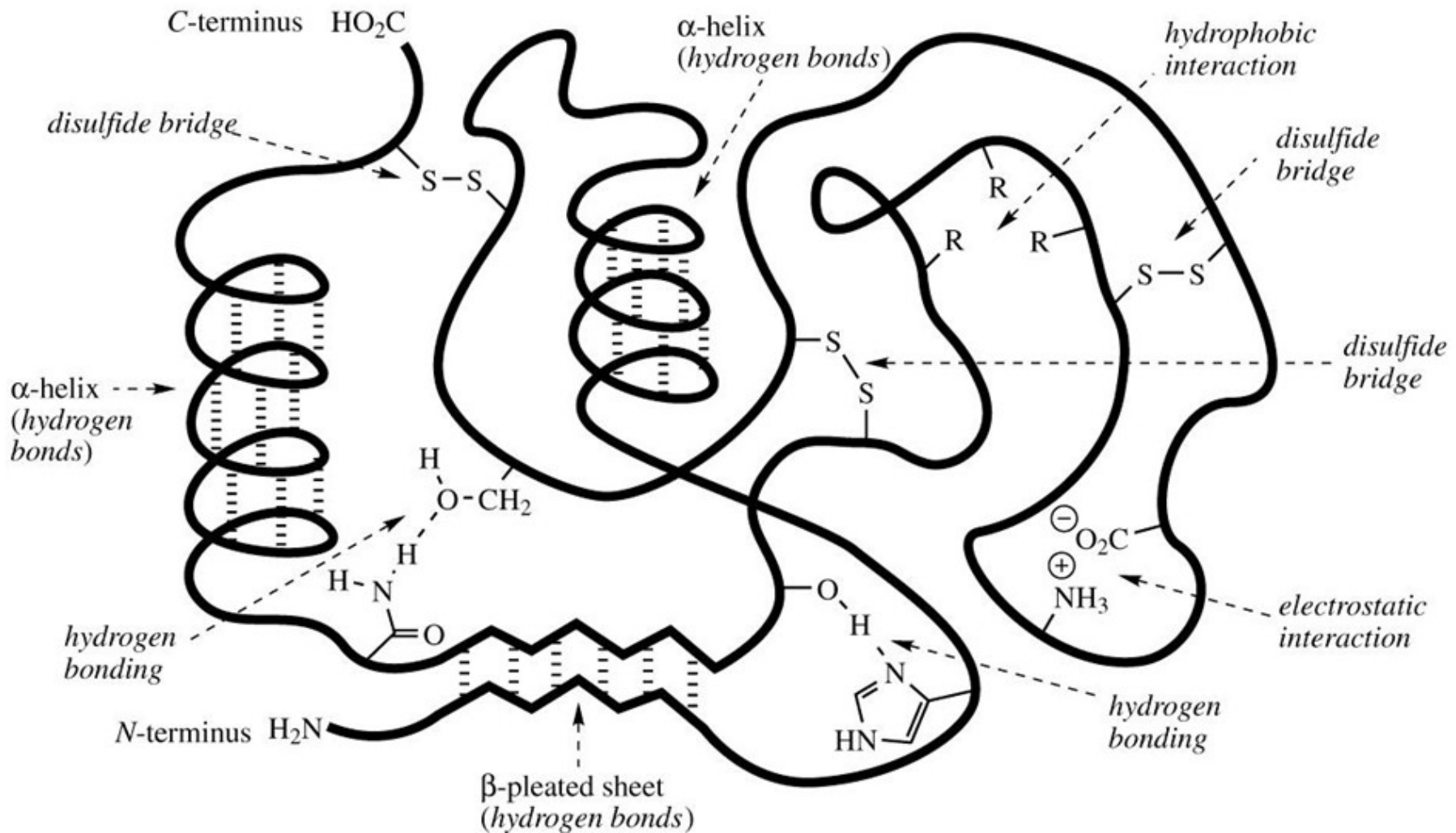
anti-parallel β -sheet:
carbonyl to amino peptide
linkages in opposite directions



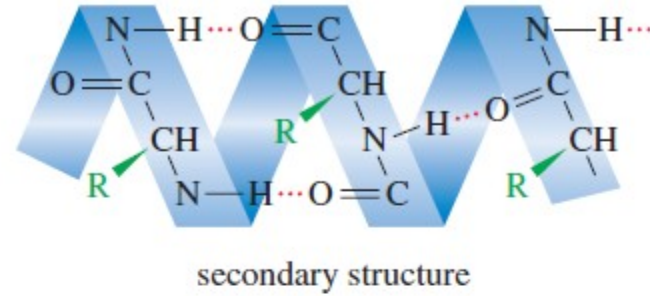
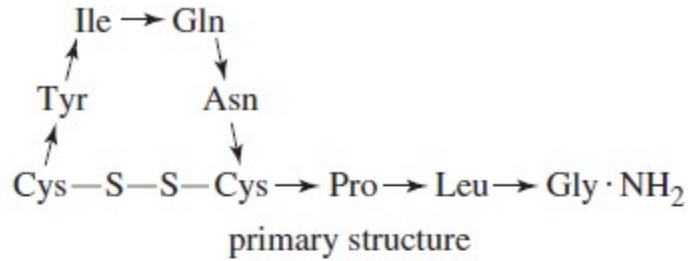
parallel β -sheet:
carbonyl to amino peptide
linkages in same direction



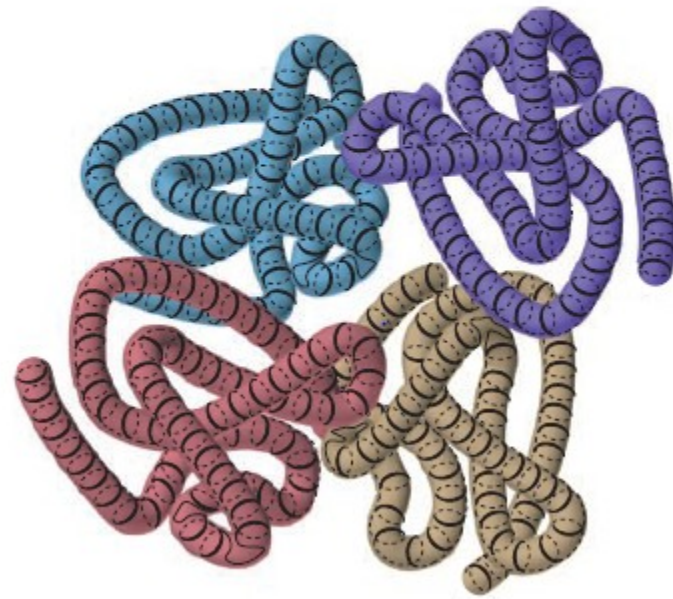
– Stabilizing interactions???



- Summary of protein structure



tertiary structure



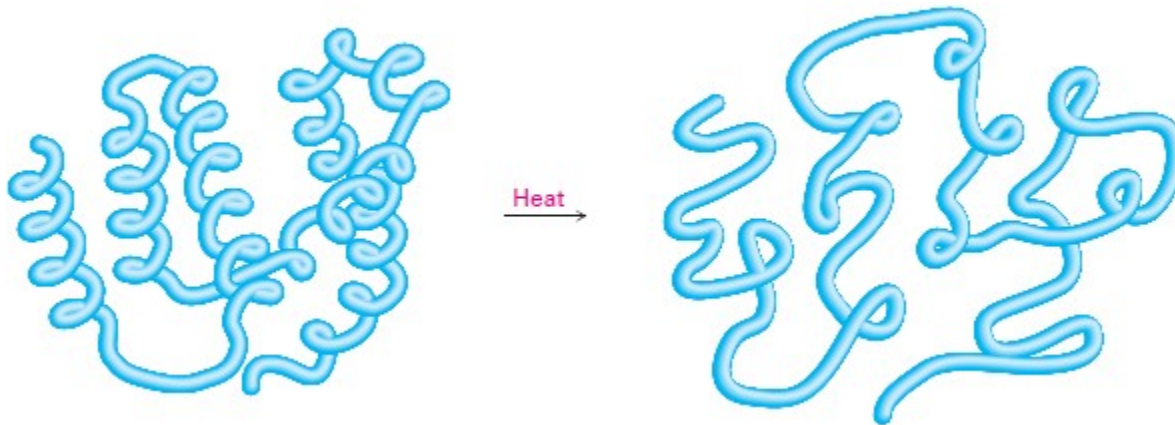
quaternary structure

Protein denaturation

- **Protein denaturation**

“Destroying the highly organized tertiary structure of a protein”

- Loss of biological activity plus
- Changes in *physical properties*



Exercises

- **Exercise 1**

- When a nonapeptide undergoes partial hydrolysis, it forms dipeptide, tripeptide and a tetrapeptide whose amino acid compositions are shown below. Reaction of the intact nonapeptide with Edman's reagent releases **PTH-Leu**. *What is the sequence of the nonapeptide?*

1. Pro, Ser
Leu

4. Gly, Ala

7. Met,

2. Gly, Glu

5. Glu, Ser, Val, Pro

8. His, Val

3. Met, Ala, Leu

6. Glu, Pro, Gly

- **Solution 1**

Leu-Met

Leu-Met-Ala

Ala-Gly

Gly-Glu

Gly-Glu-Pro

Pro-Ser

Glu-Pro-Ser-Val

Val-His

Complete structure

Leu-Met-Ala-Gly-Glu-Pro-Ser-Val-His

- **Exercise 2**

Complete hydrolysis of a heptapeptide showed that it has the following constituent amino acids: 2A, E, L, K, F, V

Deduce the amino acid sequence of this heptapeptide from the ff data.

1. Treatment of the heptapeptide with 2,4-dinitrofluorobenzene followed by incomplete hydrolysis gave, among other products: valine labeled at the α -amino group, lysine labeled at the ϵ -amino group, and a dipeptide, DNP VL (DNP 2,4-dinitrophenyl-).
2. Hydrolysis of the heptapeptide with carboxypeptidase gave an initial high concentration of alanine, followed by a rising concentration of glutamic acid.
3. Partial enzymatic hydrolysis of the heptapeptide gave a dipeptide (A) and a tripeptide (B).
 - a. Treatment of A with 2,4-dinitrofluorobenzene followed by hydrolysis gave DNP-labeled leucine and lysine labeled only at the ϵ -amino group.
 - b. Complete hydrolysis of B gave phenylalanine, glutamic acid, and alanine. When B was allowed to react with carboxypeptidase, the solution showed an initial high concentration of glutamic acid. Treatment of B with 2,4-dinitrofluorobenzene followed by hydrolysis gave labeled phenylalanine.

Solution 2

Val-Leu-Lys-Phe-Ala-Glu-Ala

- **Exercise 3**

- In determining the primary structure of insulin, what would lead you to conclude that insulin had more than one polypeptide chain?

Solution 3

Edman's or Sanger's method release two N-terminus amino acids

Other options for sequencing peptides

- Mass spectrometer
- NMR
- Nucleotide sequencing
 - (for DNA-encoded peptides)

Amino acid

sequence (protein)

Gln – Tyr – Pro – Thr – Ile – Trp

DNA sequence (gene)

CAGTATCCTACGATTTCG

Correspondence of DNA and amino acid sequences



Investigating proteins with mass spectrometry

